Fundamentals of Population and Conservation Genetics

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1 Preface

This book has been assembled from the notes for my colisted upper-level undergrad and graduate student course on conservation genetics (BIOB480/BIO548) at Montana State University. It is a rough draft and a work in progress. Over the next few years I hope to expand on the material within, particularly by adding practice questions and additional examples. It is complemented by a set of Jupyter Notebooks with simple data analysis activities in Python. I encourage you to use any and all of the material within, given appropriate attribution (especially when I am referencing external work).

Why another population and conservation genetics text? My reasons are typical. First, teaching from a much longer book is difficult, and students appreciate a concise guide to what they will be evaluated on. Assembling these notes (first in pen and ink, then in LaTeX, and now in Quarto) has deepened my own understanding of the material. Many formal textbooks skip aspects of derivations; I found this intolerable, and had to work through steps others more gifted in algebra may have taken for granted. Finally, I appreciate the flexibility of shifting emphasis, and creating my own figures. My bias is towards foundational theory, with brief discussions of empirical examples from the conservation biology literature (or failing that, nonthreatened wild organisms). This reflects my own interests, my sense of what is important at this stage in a student's education, and the fact that MSU's standalone population genetics class is no more.

I am indebted to numerous other resources. First and foremost is Frankham, Ballou, and Briscoe (2018)'s text *Introduction to Conservation Genetics*, the traditional resource for BIOB480. I have drawn heavily on their presentation of mathematical theory and delimitation of conservation genetics as a discipline, paraphrasing points and derivations throughout. Second—likely to increase in importance to these notes in the coming years—is Allendorf et al. (2022)'s *Conservation and the Genomics of Populations*, useful for its treatment of genomics methods and wealth of modern case studies. Third is Joe Felsenstein's *Theoretical Evolutionary Genetics* (Felsenstein (2005)), the DIY text I worked through as a mediocre student in his course in 2015, and which planted the idea that preparing one's own course materials could be worthwhile. Fourth is John Gillespie's *Population Genetics*, a slim and elegant introduction to the mathematical basis of the field that I returned to in 2018 when I found my memory of Joe's class fading. Lastly is Graham Coop's *Population and Quantitative Genetics* (Coop (2013))—a direct inspiration in form and spirit.

2 A Brief History of Conservation Genetics

Conservation genetics is a relatively young discipline that merges population and evolutionary genetics with conservation biology. Perhaps more accurately described as subfield, its roots can be traced to Charles Darwin's (1809-1882) attempts to identify a source of the bountiful variation he observed in the natural world—variation essential for his theory of evolution by natural selection. Darwin was ultimately unsuccessful in this endeavor, and remained aware it was a major weakness until the end of his life. Inspired by plant and animal breeding, his observations of sudden discontinuous variation in livestock traits (such as the short legs of Ancon or "Otter" sheep) initially led him to emphasize the role of "sports", or dramatic mutations.

Yet his acceptance of a mode of inheritance that blended parental traits together proved incompatible with the maintenance of sports from generation to generation. By 1868 he had shifted to a model of "gemmules"—atomic units of inheritance that individuals acquired through changes in their environment and passed on to their young. In doing so, he reasoned, adaptations could be maintained through time. Ultimately unconvincing to his peers, he died unaware that the monk Gregor Mendel had landed on a solution to his problem three years early through intensive study of pea plants: the inheritance of traits was particulate, and thus irreducible. Some genetic variants are dominant, some are recessive; genetic variants segregate during reproduction, and assort independently.

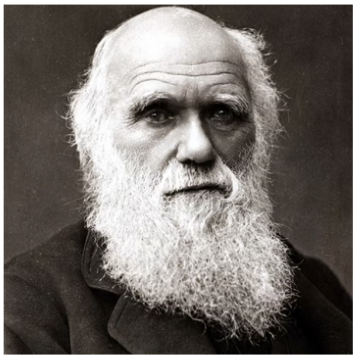




Figure 2.1: *Darwin and Mendel*

Mendel's framework lay the foundation for the work of the founding fathers of population

genetics: Sewell Wright (1889 - 1988), Ronald Fisher (1890 - 1962), and J.B.S. Haldane (1892 - 1964). Wright, the sole North American of the bunch, and relatively well-behaved compared to his compatriots, studied animal breeding and made major contributions to our understanding of inbreeding and path analysis (now common in causal inference statistics). Fisher was a British polymath—arguably a genius—whose footprints on genetics and statistics are pervasive. Politically conservative, he was an outspoken eugenicist and by all accounts unpleasant to those close to him. J.B.S. Haldane, the most charismatic of the bunch, was also a widely read Brit. The son of a Scottish physician who conducted physiological experiments on his family, Haldane earned a degree in Classics, fought in World War I, become an avid communist who never disavowed Stalin, and eventually migrated to India. His influence is particularly strong on the mathematics of natural selection, as well as linkage and gene mapping.

Wright, Fisher, and Haldane were important actors in the Modern Synthesis of evolutionary biology. A term coined by British evolutionary biologist, science popularizer (and, again, eugenicist) Julian Huxley, its better known figures include the German American Zoologist Ernst Mayr, the Russian geneticist Theodosius Dobzhansky, and the paleontologist George Gaylord Simpson. Together, the group pioneered much of our current understanding of how evolution occurs within and across lineages. Their work was followed in the 1960s and 70s by scientists who formalized population genetics into the diverse and active field it is today. Important figures in this era included James Crow (of University of Wisconsin) and Motoo Kimura (of Kyoto Imperial University), who co-developed the neutral theory of molecular evolution; Richard Lewontin, who generated some of the first empricial genetic data on human populations, showing variation was far greater within ethnic groups than among them, and who used his expertise in service of deeply felt leftist ideals; and Brian and Deborah Charlesworth, a married couple from the UK with strong interests in recombination.

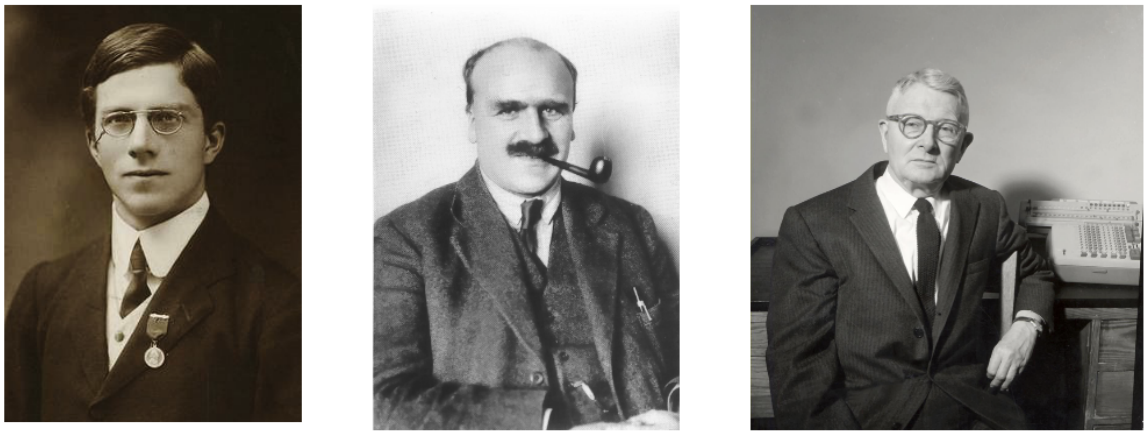


Figure 2.2: *Fisher, Haldane, & Wright.*

Conservation genetics *sensu strictu* emerged from this legacy in the late 1970s. Catalyzing events included a series of papers by Australian plant geneticist O.H. Frankel—who provacatively argued preserving genetic diversity was a moral responsibility and necessity to ensure continued evolution—his co-authored book with Michael Soulé entitled Conservation and Evolution, and Soulé's collaboration with Bruce Wilcox entitled *Conservation Biology: An Evolutionary-Ecological Perspective.* Soulé's involvement—a UC Santa Cruz-based ecologist and evolutionary biologist who can be given an enormous amount of credit for the development of the field of conservation biology more broadly—should indicate the close association between these applied, "crisis-driven" disciplines.

Since the 1980s, conservation genetics has benefited from astounding advances in our ability to collect marker data from across the genomes of individuals in wild populations, and concurrent advances in computational and statistical tools. For our purposes, it is interesting to note that several major figures in the field have been based in the Northwest or Montana. Joe Felsenstein—a theoretical population geneticist and phylogeneticist—trained with both Lewontin and Crow before ending up at University of Washington, where he eventually taught the author of this document courses in both subject areas. A much more notable Felsenstein student was Fred Allendorf, now emeritus faculty at University of Montana. Allendorf has made myriad contributions to conservation genetics, with a particular focus on salmonids; not coincidentally, Missoula and UM have become hubs for conservation genetics research, with Gordon Luikart and Mike Schwartz (of the US Forest Service's Rocky Mountain Research Station) running active and impactful labs. Back in Seattle, Robin Waples of NOAA has had a fruitful career using fisheries data to clarify outstanding issues in population genetics theory and practice.

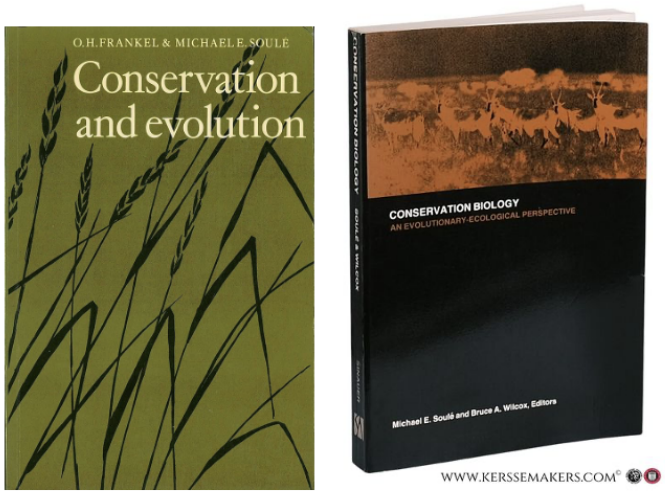


Figure 2.3: *Foundational texts.*

In its current iteration, conservation genetics aims to delimit the basic units of biological conservation and understand how their genetic composition is affected by habitat loss, exploitation,

and environmental change. There is a strong emphasis on what is unique about evolutionary processes in small populations; as genetic and genomic tools become easier to use, they are ever more frequently applied to non-evolutionary conservation questions (such as community metabarcoding). In this course, we will cover the following non-exhaustive list of topics:

- How evolutionary processes impact genotype frequencies and allele frequencies;
- How inbreeding is measured, and why it is harmful for reproduction and survival;
- How genetic diversity is lost and why it affects evolutionary potential;
- How fragmenting populations and habitat impacts gene flow, and what this means for diversity;
- How small populations end up evolving more by random processes (genetic drift) than by deterministic processes (natural selection);
- How mutations are accumulated and lost, and what their likely impacts are;
- How we determine taxonomy and management units using genetic data;
- How outcrossing (mating with distant relatives) can impact the fitness of populations.

As discussed in the Preface, my bias is towards foundational evolutionary genetics theory, with conservation-minded empirical examples. Much of the start of Frankham, Ballou, and Briscoe's *Introduction to Conservation Genetics* lays out the broader justification for conservation biology. Note that much of this field is shaped by values, ethics, and policy——not just science. The field is mission-driven, but can't tell us what we must do, just provide evidence of most effective way to pursue a particular societally desired goal.

Foundational genetics vocabulary and concepts

Mendel's Law of Dominance and Uniformity
Some alleles are dominant while others are recessive; an organism with at least one dominant allele will display the effect of the dominant allele.
Mendel's Law of Segregation
During gamete formation, the alleles for each gene segregate from each other so that each gamete carries only one allele for each gene.
Mendel's Law of Independent Assortment
Genes of different traits can segregate independently during the formation of gametes.
Darwin's Theory of Natural Selection
Variation of traits, both genotypic and phenotypic, exists within all populations of organisms. However, some traits are more likely to facilitate survival and reproductive success. Thus, these traits are passed onto the next generation. These traits can also become more common within a population if the environment that favors these traits remains fixed.
Gene
The basic unit of heredity *or* a sequence of nucleotides that is transcribed.
Locus
A specific, fixed position on a chromosome.

Allele
A variant in the sequence of nucleotides at a particular locus.
Genotype
The complete set of alleles an organism carries.
Genome
All the genetic information of an organism.
Phenotype
The set of observable characteristics or traits of an organism.
Ploidy
The number of complete sets of chromosomes in a cell.
Diploid
The state of having two homologous copies of each chromosome.
Homozygote
The state of having identical alleles at the same locus on homologous chromosomes.
Heterozygote
The state of having different alleles at the same locus on homologous chromosomes.
Dominant
An allele whose effect masks or overrides the effect of a different allele at the same locus.
Recessive
An allele whose effect is masked by a dominant allele.
Chromosome
A package of DNA with some or all of the genetic material in an organism. In most chromosomes, thin DNA fibers are coated with packaging proteins to form nucleosomes (histones in eukaryotes), which bind and condense DNA to protect it, generating a complex 3D structure.
DNA
A polymer composed of two polynucleotide chains that coil around each other to form a double helix. The polymer carries genetic instructions for the development, functioning, growth, and reproduction of all known organisms and many viruses.
Central Dogma
"DNA makes RNA, and RNA makes protein." DNA is transcribed to RNA by RNA polymerase, and RNA is translated into protein by ribosomes.
Nucleotide
An organic molecule composed of a nitrogenous base, a pentose sugar, and a phosphate. Nucleotides are the monomeric units of nucleic acids—DNA and RNA.
A, C, T, G
Adenine, cytosine, thymine, and guanine. Adenine and guanine are purines (two rings); cytosine and thymine are pyrimidines (one ring). A pairs with T, and C pairs with G.
Protein
A biomolecule composed of one or more amino acids.
Amino acid

An organic compound containing both amino and carboxylic acid functional groups; incorporated into proteins.

Enzyme

A protein that acts as a biological catalyst.

Active site

The region of an enzyme where substrate molecules bind and undergo a chemical reaction.

Allele frequency

The proportion of chromosomes with a particular allele at a particular locus in a population.

Genotype frequency

The proportion of individuals with a particular set of alleles at a particular locus in a population.

Monomorphic

Only one form; invariant.

Polymorphic

Many forms; variable.

Diagnostic locus

A locus at which all individuals of a particular population or species carry the same allele; can be used for identification.

Foundational statistics vocabulary and concepts

Parameter

A quantity that summarizes or describes an aspect of a *population*, such as a mean or standard deviation.

Statistic

A quantity computed from sample data for a statistical purpose; describes a *sample*.

Confidence interval

An interval that is expected to contain the true population parameter with a specified level of confidence.

Null hypothesis

The hypothesis that no relationship exists between variables being analyzed.

p-value

The probability of obtaining test results at least as extreme as the observed result, assuming the null hypothesis is true. A very small p-value indicates that the observed outcome would be unlikely under the null hypothesis (but does *not* mean the alternative hypothesis is correct).

Normal distribution

A continuous probability distribution in which values are symmetrically distributed around the mean.

R^2

The proportion of variation in the dependent variable that is predictable from the

independent variable(s).

Degrees of freedom

The number of values in the final calculation of a statistic that are free to vary.

3 How do we study genetic diversity?

i Paper Reading Strategies

Understanding both the intellectual history and contemporary practice of conservation genetics requires moving beyond textbooks to read the primary literature. In this class we'll do so on a weekly basis, engaging with a selection of a dozen or so scientific articles in person and through written assignments. In general, the papers I've picked are not unduly technical, represent different types of research and scholarship in conservation genetics, and emphasize local species when possible. (There is a vertebrate bias; I apologize.)
The following questions are helpful for efficiently grasping scientific articles:

- What type of paper is it? (review, perspective, empirical…)
- What is the main question or goal?
- How did the authors set out to answer that question?
- What did they find?
- How do they interpret their findings?
- Critique: strengths? limitations? stronger evidence?
- What does each figure say? Where and when was it published?

Empirical population and conservation genetics requires assessing genetic variation, either directly (through DNA sequences) or indirectly (through the property of phenotypes that reveal underlying sequence variation). Though seemingly trivial—"genetics" is in the name of both fields, after all—this emphasis on molecular variation distinguishes our topic from other ways biodiversity is studied (e.g. morphometrics). Consider a scenario in which you have both DNA sequence data and data from a meristic (countable) trait in the same set of populations of three-spined stickleback (*Gasterosteus aculeatus*). The trait in question—the number of lateral plates on each individual—ranges from n=20 to n=30. You are interested in the evolutionary relationships among populations. At first pass, inferring a close relationship among fish in ponds where all stickleback have either 30, 27, 23, or 20 plates would seem to make sense. Yet the sequence data from the same individuals show that there are multiple genetic variants that can give rise to the exact same number of plates! For example, 30-plated fish in pond A have haplotype $CTGA$ at the plate-determining locus, while 30-plated fish in pond B have haplotype $GTAA$. Given the vast number of variable sites in the genome, it is more probable that that the differences in sequence reflect the "true" relatedness (or relative evolutionary distance) than their identical number of plates. **Though repeated mutations and other**

factors can mask the evolutionary signal in genetic data, it is nonetheless a much more faithful signal of ancestral relationships than phenotype.

Assessing genetic variation has only become possible at scale over the last 50 or so years. Indeed, one of the most impressive aspects of population genetic theory is how much of it was developed before Watson, Crick, and Franklin determined the structure of DNA in 1953! Early empirical work focused on simple traits that behaved in Mendelian ways, such as eye color in *Drosophila* fruit flies; complex traits such height gave rise to a separate body of theory called quantitative genetics, which circumvented the question of underlying variants by focusing on the properties of statistical distributions. (More on this later in the course.) Beginning in the 1960s, *protein electrophoresis*—a method in which sequence variation in proteins is revealed by their different electrical charges in an agarose gel medium—allowed scientists to directly quantify genetic diversity in a wide range of organisms. Particular proteins—often polymorphic enzymes known as allozymes—became known as *genetic markers*, as they "marked" differences in underlying genes. The term is still widely applied to any method that characterizes alleles by means other than direct sequencing.

Protein electrophoresis confirmed many predictions of population genetic theory in natural populations. Yet it also brought surprises. For example, Hubby and Lewontin (Hubby and Lewontin (1966), Richard C. Lewontin and Hubby (1966), Charlesworth et al. (2016)) showed that there was significantly more allelic variation in *Drosophila* than expected, with differences in the electrical charge of proteins that were assumed to be "monomorphic" (i.e., had the same form). Six years later, Lewontin (Richard C. Lewontin (1972),) showed that there is far more allelic variation within than across ethnic groups, refuting widespread ideas about the biological reality of races.

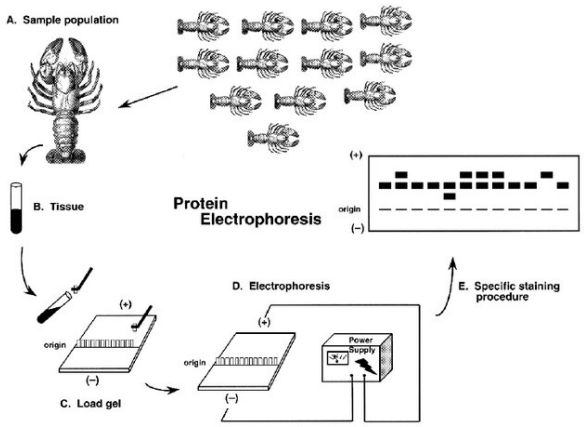


Figure 3.1: *A diagram of protein electrophoresis from Fetzner Jr and Crandall (2001).*

Despite its usefulness, protein electrophoresis only revealed variation at functional (i.e., protein coding) sites. The subsequent two decades lead to a suite of new genetic markers, including restriction fragment length polymorphisms (RFLPs; variants in restriction enzymes that

would result in different patterns of DNA fragmentation on a gel), single strand conformation polymorphisms (SSCPs; variants impact the shape of a single strand of DNA, affecting its mobility in a gel), and minisatellites and microsatellites (methods that detect alleles as repetitive DNA motifs, again indicated through gel electrophoresis). The polymerase chain reaction—a method of creating millions of copies of a DNA sequence of interest with the help of a heat-tolerant enzyme discovered in thermophilic bacteria from Yellowstone—further enhanced our ability to assay variation across the genome. It became particularly important to the first widely adopted approach to directly "sequencing" (determining the specific composition and order of nucleotides) DNA. Known as Sanger sequencing, this method is still used today, and involves creating a synthetic DNA strand of color-dyed bases to match the sample of interest; colored bases are then read by a lazer in a capillary gel.

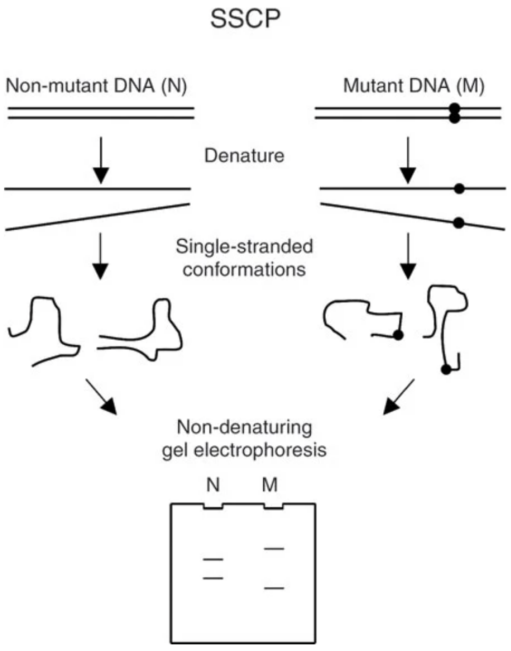


Figure 3.2: *The basis of single-stand conformation polymorphisms (Gasser et al. (2006)).*

After a moonshot endeavor to sequence the entire human genome in the early 2000s, *genomics*—a term that means the study of the genome-wide genetic variation—began to be applied to nonmodel organisms. Initially, this was largely done with so-called "reduced representation approaches" such as restriction-site associated DNA sequencing (RADseq; a relative of RFLPs, where restriction enzymes randomly fragment the genome and sections of a certain size are amplified and sequenced) and target capture (where previously identified regions of the genome are joined to synthetic probes to be amplified and sequenced). Reduced representation approaches leveraged high-throughput, short-read sequencing technology on platforms such as Illumina; this method works by joining millions of short DNA sequences to a platform

known as a flow cell, where they are amplified and tagged with dye (shown in a nice video here). Inevitably, Illumina and similar technologies were used to sequence the entire genome of individuals. Confusingly, such whole-genome sequencing (WGS) is sometimes known as whole-genome resequencing, with the implication that there is a "first" genome for any given species. Such "reference" genomes often involve more intensive "sequencing effort" (more "reads", or sequences generated, for each stretch of DNA). These days, they are generated with long-read sequencing technology, a method that attempts to sequence chromsome-scale stretches of the genome in a single go.

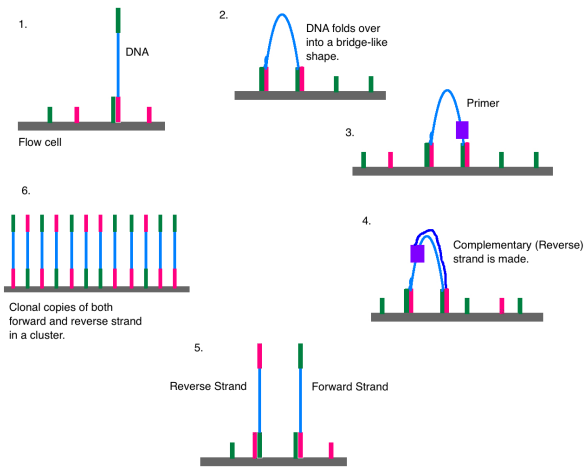


Figure 3.3: *Illumina sequencing technology, via Wikipedia.*

Everything from protein electrophoresis to WGS requires a source of genetic material: typically blood, muscle, heart, liver, scat, environmental traces. Obtaining this material can either be destructive (i.e., killing the animal) or nondestructive (a blood sample). A further distinction is invasive (catching the animal and bleeding it) or noninvasive (obtaining genetic material from scat or the environment). In diploid Eurkaryotes, most cells have 2 copies each of the nuclear genome and 100-1000 copies of the mitochondrial genome; chloroplast genomes are found in photosynthetic organelles called plastids (3-275+ copies each). This variation in copy number is no longer a determinant of what approaches are feasible, but is worth highlighting as a reminder that all empirical population genetics involves tangible biological structures.

i Probability Theory

A crucial assumption of population genetic theory is that alleles segregate (i.e., are passed on in gametes) by chance. As a consequence, population genetics relies heavily on probability theory. It is therefore in our interest to brush up the five Probability Rules, all of which will be used to solve problems this semester:

1. **Possible values** range from 0 (an impossible event) to 1 (a certain event).
2. **The sum of probabilities for all possible outcomes is 1.**
3. **Addition rule:** the probability that at least one of two events occurs:
   - **Mutually exclusive:**
     $P(\text{A or B}) = P(\text{A}) + P(\text{B})$
   - **Not mutually exclusive:**
     $P(\text{A or B}) = P(\text{A}) + P(\text{B}) - P(\text{A and B})$
4. **Multiplication rule:** the probability two events occur together:
   - **Independent:**
     $P(\text{A and B}) = P(\text{A}) \times P(\text{B})$
   - **Nonindependent:**
     $P(\text{A and B}) = P(\text{A}) \times P(\text{B} \mid \text{A})$
5. **Conditional probability:** the probability an event happens given another event occurs:
   - **Independent:**
     $P(\text{A} \mid \text{B}) = P(\text{A})$
     Example: If two coins are flipped, knowing one coin is tails gives no information about the other; $P(\text{heads}) = 0.5$.
   - **Nonindependent:**
     $P(\text{A} \mid \text{B}) = \dfrac{P(\text{A and B})}{P(\text{B})}$
     Example: A bag has 6 red, 3 blue, and 1 green marble.
     $P(\text{A and B}) = \frac{6}{10} = 0.6$,
     $P(\text{B}) = \frac{9}{10} = 0.9$,
     so $P(\text{A} \mid \text{B}) = 0.6/0.9 = 0.66$.

i Genetic polymorphisms in *Aplonis* starlings

In an early application of protein electrophoresis to wild birds, Kendall Corbin, Charles Sibley, and Andrew Ferguson (Corbin et al. 1974) assessed the extent of protein polymorphism in two species of starling in the genus *Aplonis*. The species, found on New Guinea and its offshore colonies, differed in their basic biology: Metallic Starling *Aplonis metallica* nests in large colonies and is tolerant of human disturbance, while Singing Starling *Aplonis cantoroides* is a solitary rainforest dweller.
New Guinea is geographically complex, and the authors hypothesized that its topography would restrict gene flow and lead to unique, geographically localized genetic variants.

Isolating variants at the lactase dehydrogenase-H (LDH-H) and esterase-1 loci, the authors found contrasting patterns. LDH-H showed variation but similar allele frequencies across *A. metallica*'s sampled range, while esterase-1 showed geographically correlated variation in allele frequencies (Figure 1). In *A. cantoroides*, LDH-H was essentially monomorphic, while esterase-1 showed some limited variation.
Allele frequencies were not significantly different from expectations under Hardy–Weinberg Equilibrium, suggesting there was not a strong influence of mutation, migration, or natural selection. The authors compare their findings to data from other birds, finding the reported polymorphism lower than captive populations but comparable to wild species.





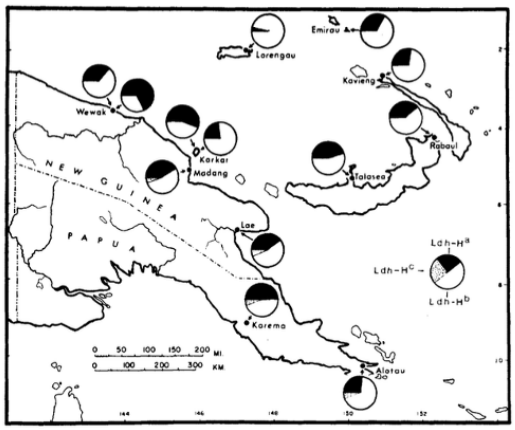


FIGURE 1. The geographic variation in the allelic frequencies at the lactate dehydrogenase-H locus in populations of *Aplonis metallica*. The frequency of an allele in a population is proportional to the area of the circle occupied by its symbol. Allelic frequencies at Rabaul localities 1 and 2 are essentially identical for the purpose of illustration. The same is true of Karkar localities 2 and 3 (see table 1). The Lorengau locality is on Manus Island.

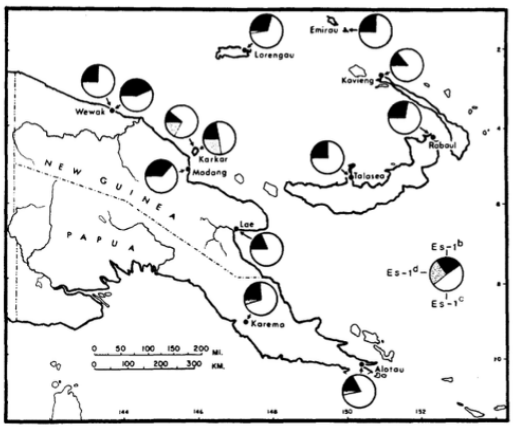


FIGURE 2. The geographic variation in the allelic frequencies at the esterase-1 locus in populations of *Aplonis metallica*. The frequency of an allele in a population is proportional to the area of the circle occupied by its symbol. Allelic frequencies at Karkar Island localities 2 and 3 are essentially identical for the purpose of illustration (see table 2).

Figure 3.4: *Figure 1. Protein polymorphism in A. metallica*

3.1 Quantifying genetic diversity

In the previous class we surveyed a range of methods for assaying genetic variation dating from the 1970s to the present day. Though technologies have varied across time, both genetic marker-based approaches (such as microsatellites or protein electrophorsis) and direct DNA sequencing (whether WGS or Sanger sequencing) result in a collection of individual genotypes

from a sample or set of samples. The unique genotypes present in these samples constitute *alleles*, or distinct genetic variants at a particular locus.

Quantifying genetic diversity involves the use of *summary statistics* to capture the various ways

Proportion of polymorphic loci:

$$P = \frac{p}{n} = \frac{\text{no. polymorphic loci}}{\text{no. loci}}$$

Average heterozygosity:

$$H = \frac{1}{n} \sum_{i=1}^{n} H_i$$

where H_i is the average heterozygosity (=proportion of individuals that are heterozygous) at locus i and n is the total number of loci. (This is just a fancy way to write an arithmetic mean.)

Allelic diversity:

$$A = \frac{1}{n} \sum_{i=1}^{n} A_i$$

where A_i is the number of alleles at locus i and n is the total number of loci.

4 Hardy-Weinberg Principle (or equilibirum, or proportions...)

To develop intuition about the expected frequency of different allelic combinations (or *genotypes*) in a sexually reproducing, diploid population, we'll start with a common analogy: coin flips. Imagine you have two quarters, each with a standard 50% probability of landing heads and 50% probability of landing tails. You are interested in how often you will get 0, 1, or 2 heads when flipping both coins simultaneously. More generally, this scenario can be thought of as the probability of 0, 1, or 2 independent "successes", where the probability of success is 50% ($p(\text{heads}) = 0.50$) and the probability of failure is 50% ($p(\text{tails}) = 0.50$). (This is an example of something called as **Bernoulli Trial**, which we will return to in several weeks during our discussion of genetic drift.)

Recall from the previous class that the probability of two events occurring together is given by the product of their individual probabilities (i.e., the **Multiplication Rule**). We can therefore determine the probability of neither quarter landing heads up—in other words, the probabilty of two failures:

- $p(0 \text{ heads}) = p(\text{tails}) * p(\text{tails}) = 0.50 * 0.50 = 0.25$

The probability of one success and one failure is slightly more complicated, as there are two possible independent paths to this combination: one in which the first quarter is heads and the second quarter is tails, and one in which the first quarter is tails and the second quarter is heads. The **Addition Rule** states that the overall probability of at least once of two outcomes occurring is the sum of their probabilities:

- $p(1 \text{ heads}) = p(1 \text{ head}) * p(1 \text{ tail}) + p(1 \text{ tail}) * p(1 \text{ head})$
- $p(1 \text{ heads}) = 0.50 * 0.50 + 0.50 * 0.50 = 0.50$

The probability of two heads is simple the product of the probability of heads on each coin (two successes):

- $p(2 \text{ heads}) = p(1 \text{ head}) * p(1 \text{ head}) = 0.50 * 0.50 = 0.25$

(Note that all possible outcomes sum to 1—the first of our probability rules.)

The extension of this to possible genotypes is straightforward. At a diallelic locus (i.e., a locus with two alleles), the probability that an individual receives two copies of a given allele is the probability its mother passes on that allele multipled by the probability its father passes on

that allele. Expected offspring genotypes thus vary with parental genotypes. If the mother is homozygous for the dominant allele at locus A and the father is also homozygous dominant, the probabilty of a homozygous recessive offspring is $p(aa) = 0 * 0 = 0$, the probability of a heterozygote is $p(Aa) = 1 * 0 + 0 * 1$ (note again there are two paths to this combination!), and the probability of a homozygous dominant individual is $p(AA) = 1 * 1 = 1$. A cross between two heterozygotes is a close match to our coin flip scenario: $p(aa) = 0.5 * 0.5 = 0.25$; $p(Aa) = 0.5 * 0.5 + 0.5 * 0.5 = 0.50$; and $p(AA) = 0.5 * 0.5 = 0.25$.

In a *population* (defined as a group of individuals that mate randomly with respect to a given locus), the **expected frequency** of genotypes in the next generation is equivalent to the probability an offspring inherits that genotype. While any given individual will have its genotype determined by the rules in the paragraph above, the overall distribution of genotypes is determined by the number of parents that have each allele—the *allele frequency* in the parent generation. It can be helpful to consider sexual reproduction in a randomly mating population as reaching into a stocking to draw out marbles in one of two colors. Offspring are created by the combination of two colors; the probabilty of (say) a green and red "child" is simply the probability of drawing a green and red marble with replacement. If 7 of the 10 marbles in the stocking are green, that is $p(green) = 0.7$, while $p(red) = 0.3$; we again account for the two different orders in which marbles can be drawn to get the total probabilty of a green red combination: $p(\text{green and red}) = 0.7 * 0.3 + 0.3 * 0.7 = 0.42$.

More formaly, a diallelic locus where allele A_1 occurs at frequency $f(A_1) = p$ and allele A_2 occurs at frequency $f(A_2) = q$, we expect the following genotype frequencies following one generation of random mating:

$$p * p + p * q + q * p + q * q = p^2 + 2pq + q^2 = 1$$

This simple equation is known by names, but most helpfully as **Hardy-Weinberg Proportions** or the **Hardy-Weinberg Principle** (for its two co-discoverers, G. H. Hardy and Wilhelm Weinberg).

In addition to random mating, HWP assumes a complete absence of the four evolutionary mechanisms capable of changing allele frequencies from generation to generation:

1) No natural selection;
2) No mutation;
3) No migration;
4) No genetic drift (infinite population sizes).

For example, if $f(A_1) = 0.8$ and $f(A_2) = 0.2$, we expect genotype frequencies of $f(A_1A_1) = 0.8 * 0.8 = 0.64$, $f(A_1A_2) = 0.8 * 0.2 + 0.2 * 0.8 = 2 * 0.8 * 0.2 = 0.32$, and $f(A_2A_2) = 0.2 * 0.2 = 0.04$.

Hardy-Weinberg Proportions are important because they are *a null model for evolution*—what we expect genotype frequencies to be in the absence of evolutionary mechanisms. For this reason, it is frequently referred to as *Hardy-Weinberg Equilibrium*, with "equilibrium" here

indicating a population that is not evolving (i.e., where allele frequencies are the same from generation to generation).

We can test whether observed deviations from Hardy-Weinberg Proportions are statistically significant with a Chi-Squared test:

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

The statistic measures the squared difference between the number of individuals actually observed with each possible genotype and the expected number of individuals with that genotype based on Hardy-Weinberg proportions and assumptions, divded by the expected number of genotypes. For a diallelic locus, the sum indicates you repeat the operation on the right-hand side three times (for A_1A_1, A_1A_2, and A_2A_2). This statistic is then compared to a table where statistical signficance is inferred given a particular value for the degrees of freedom (number of free parameters). Importantly, it only works with *count* data—not frequencies!

For example, if we imagine 100 chicks are born with genotype counts of $f(A_1A_1) = 20$, $f(A_1A_2) = 20$, and $f(A_2A_2) = 60$, we first determine $f(A_1) = \frac{2*20+20}{200} = 0.3$ and $f(A_2) = 1 - f(A_1) = 0.7$. Based on these values, we would expect $\#A_1A_1 = 100 * p^2 = 100 * 0.3^2 = 9$, $\#A_1A_2 = 100 * 2pq = 100 * 2 * 0.3 * 0.7 = 42$, and $\#A_2A_2 = 100 * q^2 = 100 * 0.7^2 = 49$

Our Chi-Squared statistic is then:

$$\chi^2 = \frac{(20-9)^2}{9} + \frac{(20-42)^2}{42} + \frac{(60-49)^2}{49} = 24.92$$

The value of 24.92 is our "test statistic". Since we are working with a diallelic locus where $p + q = 1$, we only have a single degree of freedom—the value p depends on q, and vice versa. We thus look at the row $df = 1$ in a table like this one and find our value. 24.92 is much greater than the test statistic value of 3.841 required to reach statistical significance at $p = 0.05$, so we can conclude the differences between the counts of observed and expected genotypes are unlikely to be due to random sampling error.

(More on the chi-squared distribution here—in the example above, we are looking at where on the line for $k = df = 1$ our statistic falls, which is far past the right-hand side of the plot, meaning the vast majority of the probability distribution is weighted towards less extreme differences.)

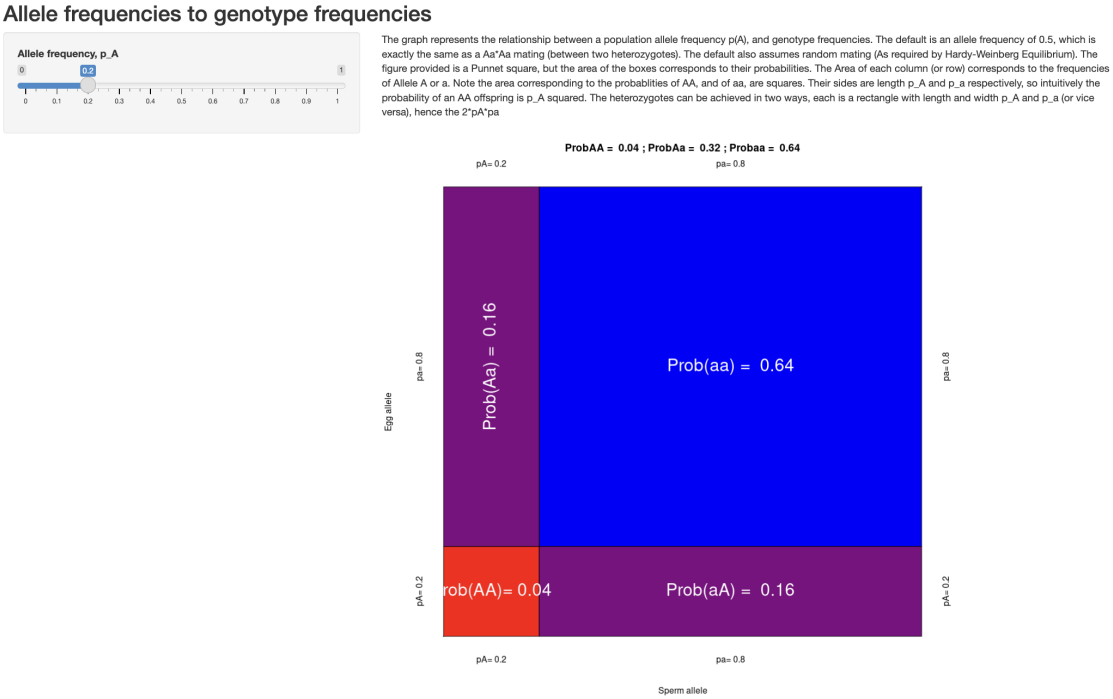
The idea of expected heterozygosity under Hardy-Weinberg proportions is an important one. We can more broadly define H_e for n loci as:

$$H_e = 1 - \sum_{i=1}^{n} p_i^2$$

In other words, the expected frequency of heterozygotes is what you have left over (i.e. the *complement*) after accounting for the expected frequency of all homozygotes. In a diallelic system, this is $1-p^2-q^2(=2pq)$; in a triallelic system, this is $1-p^2-q^2-r^2(=2pq+2pr+2qr)$, etc.

Hardy Weinberg Proportions App

Dan Bolnick has a useful app for visualizing Hardy-Weinberg proportions.



Open the app in your browser and consider the following two questions:

1) Under what allele frequencies is the frequency of heterozygotes maximized?

2) Why are the axes labeled with "sperm" and "egg"?

5 Linkage Disequilibrium

Hardy-Weinberg Proportions give us expectations for the frequency of genotypes at a locus. We now know this is useful and interesting because it provides a null model for patterns in the absence of a variety of biological processes of interest: mutation, selection, gene flow, and the stochastic consequences of finite population sizes. But what should we expect between different alleles at two independently assorting loci? In other words, if two loci are located on different chromosomes, or so far apart on the same chromosome that recombination during meiosis is highly probable, what is the expected frequency distribution of pairs of alleles in the next generation?

This question is the basis of concept of **linkage disequilibrium**, typically measured by the metric D. Consider diallelic loci A and B, with alleles A_1 and A_2 and B_1 and B_2. How often should we expect the gamete (or haplotype) A_1B_1 versus A_1B_2 versus A_2B_2 versus A_1B_1? If these loci are completely independent—say, on different chromosomes—we expect a given combination to be the product of its frequencies:

$$f(A_1B_1) = f(A_1) * f(B_1)$$

To measure D, we compare this expected, *unlinked* haplotype frequency to the actual data. For example, imagine we survey a population of N=8 individuals and count the following haplotypes (at 2N = 16 chromosomes): 6 A_1B_1s (= 6/16 = 0.375), 2 A_1B_2s (= 2/16 = 0.125), 6 A_2B_2s (= 6/16 = 0.375), 2 A_2B_1s (= 2/16 = 0.125). Based on these data, we know that the overall frequency of A_1 is $f(A_1) = 0.5$. Necessarily, $f(A_2) = 0.5$. Likewise, the frequency of B_1 is 8/16=0.5 and $f(B_2)$=0.5.

D is simply the difference between the observed frequency of a particular haplotype and our expectation of it:

$$D(A_1B_1) = f(A_1B_1) - f(A_1)f(B_1) = 0.375 - (0.5 * 0.5) = 0.125$$

Contrast this with a hypothetical situation in which all four possible haplotypes are found at equal frequencies (i.e. $f(A_1B_1) = f(A_1B_2) = f(A_2B_2) = f(A_2B_1) = 0.25$):

$$D(A_1B_1) = f(A_1B_1) - f(A_1)f(B_1) = 0.25 - (0.5 * 0.5) = 0$$

This comparison tells us two things. First, a positive D value means that the two alleles at different loci used to calculate its value are occuring together *more often* than we would expect if assortment is truly independent. A negative value would therefore indicate two alleles occur together *less often* than we would expect. Second, a value of $D = 0$ is our expectation if there is no linkage disequilibrium.

Importantly, we will get the same *absolute* value of D no matter which pair of alleles we choose:

$$D(A_1B_1) = f(A_1B_1) - f(A_1)f(B_1) = 0.375 - (0.5 * 0.5) = 0.125$$
$$D(A_1B_2) = f(A_1B_2) - f(A_1)f(B_2) = 0.125 - (0.5 * 0.5) = -0.125$$
$$D(A_2B_2) = f(A_2B_2) - f(A_2)f(B_2) = 0.375 - (0.5 * 0.5) = 0.125$$
$$D(A_2B_1) = f(A_2B_1) - f(A_2)f(B_1) = 0.125 - (0.5 * 0.5) = -0.125$$

It follows that the maximum value of D is $|0.25|$ (when a pair of alleles are *always* or *never* inherited together).

So what causes this non-random assortment of alleles? Physical linkage—the close proximity of two loci on the same chromosome—is one possibility. But LD can also be caused by interactions between genes, natural selection, population genetic structure, and demographic history. For example, speciation can be caused by incompatible allele combinations (a phenomenon known as negative epistasis). If inheriting A and b leads to fertility and health, but inheriting A and B results in infertility and death, the allelic combination A and B will disappear from the population. Natural selection can similarly favor particular allele combinations at different loci that impact traits that are functionally integrated, increasing their frequency above random expectations. And when all members of a species are not equally likely to interbreed, but instead randomly mate within distinct populations with distinct allele frequencies, assaying genotypes from multiple demes at once will produce elevated D values even in the absence of other LD-generating processes.

It should be clear this is a bad name for a complicated phenomenon (in a field full of bad names)——neither solely due to linkage, nor really a disequilibrium (frequencies of haplotypes can be stable through time). The term **gametic disequilibrium** has been proposed as an alternative, but has yet to reach a critical mass of use in the field.

5.0.1 LD Decay

Recombination between loci will break down LD through time in most cases, a relationship described by $D^t = D_0(1 - c)^t$, where D^t is linkage disequilibrium at generation t, D_0 is the initial value of linkage disequilibrium, and c is the recombination rate. (c is usually expressed as centimorgans per megabase, where 1 centimorgan is equivalent a 1% chance two loci on

a chromosome will become separated from one another as a result of recombination during meiosis. A megabase is simply one million nucleotide bases.)

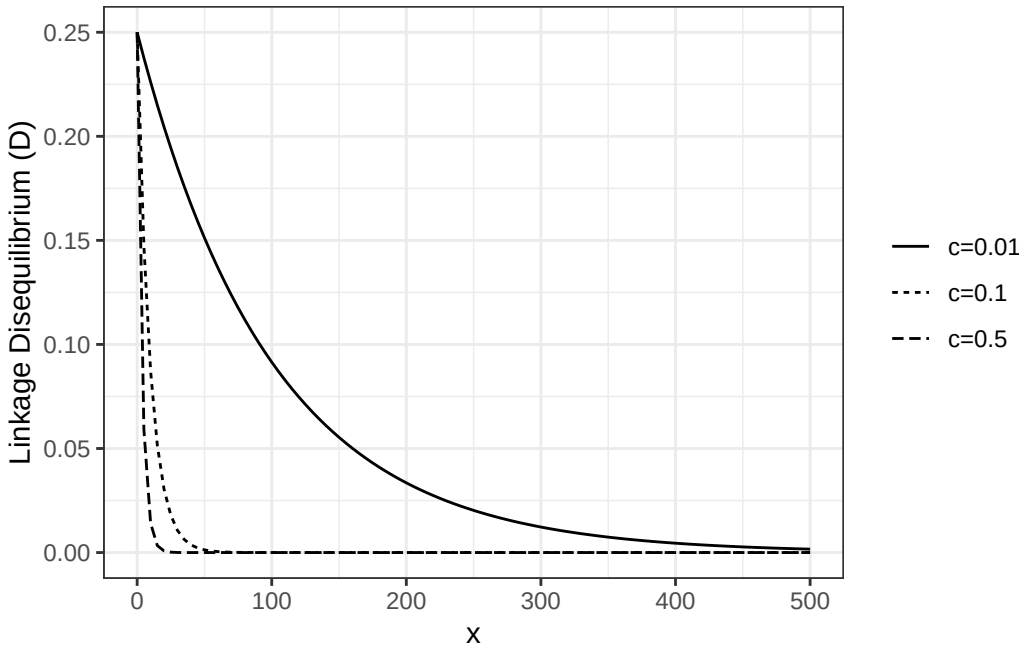
We can plot this relationship for a set of arbitrary recombination rate values as follows:

```
library(ggplot2)

# assign different values of c and D_0
c1 <- 0.01
c2 <- 0.10
c3 <- 0.25
D_0 = 0.25

# assign functions for each value of c
ld_1 <- function(x) D_0*(1 - c1)^x
ld_2 <- function(x) D_0*(1 - c2)^x
ld_3 <- function(x) D_0*(1 - c3)^x

# generate plot
p1 <- ggplot() +
  theme_bw() +
  theme(legend.title = element_blank())+
  stat_function(fun = ld_1, aes(linetype="c=0.01")) +
  stat_function(fun = ld_2, aes(linetype="c=0.1")) +
  stat_function(fun = ld_3, aes(linetype="c=0.5")) +
  scale_linetype_discrete() +
  ylim(0,0.25) +
  xlim(0, 500) +
  ylab("Linkage Disequilibrium (D)")
p1
```



Linkage Disequilibrium in Scandinavian Wolves

Gray wolves (*Canis familiaris*) in Scandinavia underwent a severe population bottleneck: after local extirpation in the middle of the 20th century, a single pair of immigrant wolves from the Finnish / Russian border eventually resettled central Sweden and began to reproduce. This pair and a third immigrant male arriving a decade or so later were the sole founders of a population that was estimated at 150 individuals in 18 packs in 2009. Because nearly all mating events in subsequent generations involved close relatives, Scandinavian wolves are highly inbred. Hagenblad et al. (Hagenblad et al. 2009) genotyped wolves from this population at 250 microsatellite markers with a known location in the canine genome. They found linkage disequilibrium values (measured by D') extended long physical distances—exactly what you would expect if genetic drift as a result of extremely small population sizes led to nonrandom associations among loci.



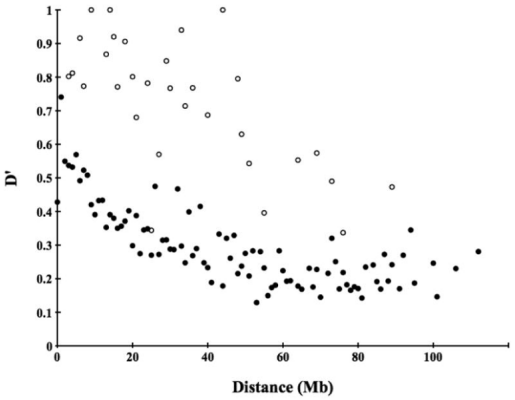
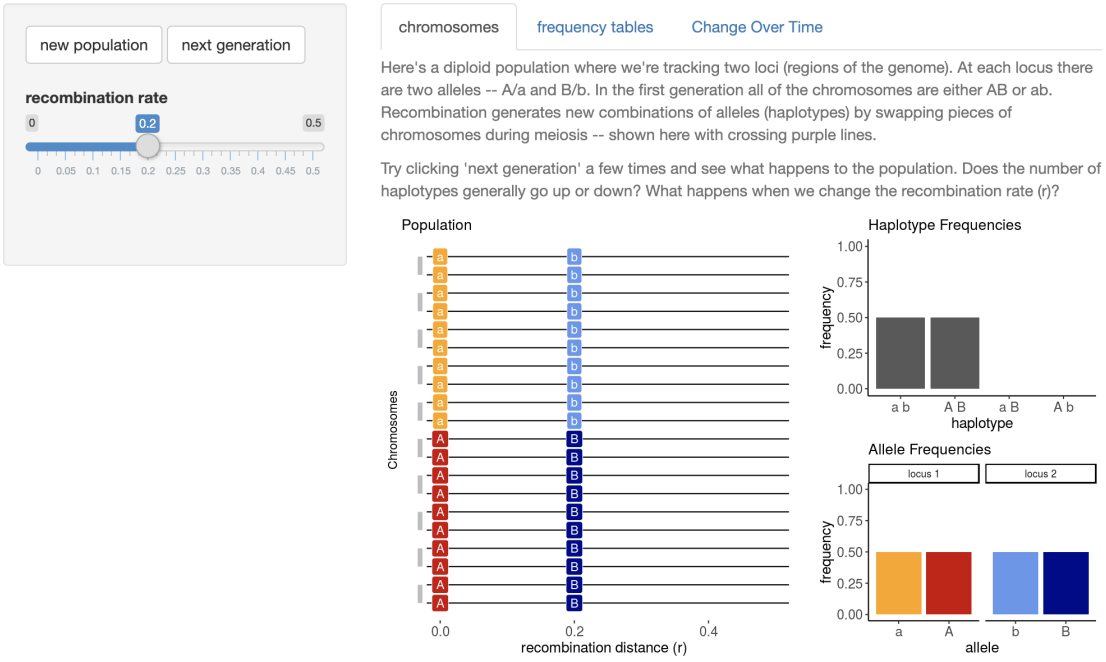


Figure 5.1: *A gray wolf in Finland* (https://www.inaturalist.org/observations/97159791) *and Figure 2 from Hagenblad et al. "Mb" refers to physical distance along chromosomes in megabases*

i LDSim Activity

CJ Battey's LDSim is a useful application to understand linakge disequilibrium more intuitively.



Open the app in your browser and consider the following two questions:

1) Under the "Chromosomes" tab, look at the plot that appears when the website loads. What is the x axis? What does each horizontal line represent? What do the gray bars pairing horizontal lines represent? How about the colors?

2) Using the same "Chromosomes" tab, move the "recombination rate" slider to 0, then click "Next Generation" several times. What happens to haplotype and allele frequencies? Click "New Population" to reset, and change the recombination rate to a higher value and repeat the simulation. How does your answer change?

3) Navigate to the "Change Over Time" tab and click "Run". What does the graph that appears below the sliders show? Describe all features.

4) Under the "Change Over Time" tab, run simulations using different combinations of recombination rate and population size. Which values lead to the quickest loss (or "decay") in LD? Which values lead to the slowest?

5) Develop a question that can be answered using this app.

6 Quantitative Genetics

The majority of this class focuses on theory and data relative to one or two loci. In some cases (such as plumage polymorphisms in birds), a single locus is responsible for all phenotypic variation present in a population. However, many traits we are interested in are *quantitative*, i.e. vary continuously, not categorically. (Human height is perhaps the most obvious example.) This observation suggests they are influenced by many genes of small effect (are *polygenic*). To study quantitative traits, we will need some introduction to the quantitative genetics, a body of work that developed in parallel to population genetics in the 20th century. Quantitative genetics is distinctive in treating the genetic variants responsible for phenotypes as essentially unknown and unobservable, something that has only recently begun to change with massive genomic datasets over hundreds of individuals. Perhaps the most important concept in the field is that of heritability.

We define heritability two ways. First, we can refer to *broad-sense heritability*, or the proportion of total phenotypic variance (V_P) that is attributable to genetic variation, meaning variation that parents can pass on to offspring ($H^2 = \frac{V_G}{V_P}$). In contrast, *narrow-sense* heritability is the proportion of phenotypic variance attributable to additive genetic variation ($h^2 = \frac{V_A}{V_P}$). This difference is relevant because parents pass on alleles, not genotypes, to offspring, and additive inheritance will maintain the resemblance between a heterozyogte and either homozygote, while dominance will not. (In both cases, the "exponent" is not actually an exponent, but simply a convention of the field.) Heritability is important because phenotypic variation also includes environmental influences:

$$\text{Phenotype} = \text{Genotype} + \text{Environment}$$

What is environmental influence? Height, for example, has a clear genetic basis. But it is also determined by nutrition during childhood—a non-genetic contribution of the environment a particular genotype finds itself in (and the resources available to it).

An important concept for estimating heritability is the definition of variance, or spread of data around the mean, which we borrow from statistics:

$$V_P = \frac{\sum_{i=1}^{n}(P_i - \bar{P})^2}{n-1}$$

where P_i is an individual's phenotypic value, $\bar{P}$ is the mean phenotypic value for the sample, and n is the sample size.

The standard deviation of a phenotype (also a measure of the spread around the mean) is the square root of its variance:

$$\sigma_P = \sqrt{V_P}$$

The covariance between offspring & parents is the sum of the products of the difference between individual values and the population (meaning all parents or all offspring) mean:

$$COV_{PO} = \frac{\sum_{i=1}^{n}(P_i - \bar{P})(O_i - \bar{O})}{n-1}$$

Building off this, the correlation between trait values for parents and offspring is:

$$r_{PO} = \frac{COV_{PO}}{\sqrt{(V_P * V_O)}}$$

where V_P is parental trait variance and V_O is offspring trait variance.

Heritability (h^2) can be directly estimated as the slope of a regression of offspring on average parent trait values, which is simply the covariance of parent and offspring trait values over total phenotypic variance ($\frac{COV_{PO}}{V_P}$).

To make these concepts more concrete, consider the following data:

Average Parent Height (cm)	Child Height (cm)
160	163
170	169
164	162
180	171
161	173

We first calculate V_P for the parents using a mean ($\bar{P}$) of (160+170+164+180+161)/5 = 167:

$$V_P = \frac{\sum_{i=1}^{n}(P_i - \bar{P})^2}{n-1}$$

$$V_P = \frac{(160-167)^2 + (170-167)^2 + (164-167)^2 + (180-167)^2 + (161-167)^2}{5-1} = 68 \text{ cm}$$

This gives us a standard deviation (σ_{po}) of $\sqrt{68} = 8.24$.

Following the same procedure, we calculate variance in offspring phenotypes, which requires mean offspring height ($\bar{O}$ = (163+169+162+171+173)/5=167.6): :

$$V_O = \frac{\sum_{i=1}^{n}(O_i - \bar{O})^2}{n-1}$$

$$V_O = \frac{(163-167.6)^2 + (169-167.6)^2 + (162-167.6)^2 + (171-167.6)^2 + (173-167.6)^2}{5-1}$$

$$V_O = 23.8 \text{ cm}$$

Next, we calculate the covariance between parent and offspring phenotypes:

$$COV_{PO} = \frac{\sum_{i=1}^{n}(P_i - \bar{P})(O_i - \bar{O})}{n-1}$$

$$= \frac{(160-167)(163-167.6) + ... + (161-167)(173-167.6)}{5-1} = 16.25 \text{ cm}$$

(Here "..." simply indicates the other values of P_i and O_i that don't fit on the page.)

With COV_{PO} we can calculate r_{PO}:

$$r_{PO} = \frac{COV_{PO}}{\sqrt{(V_P * V_O)}} = \frac{16.25}{\sqrt{(68 * 23.8)}} = 0.403934$$

...and heritability:

$$h^2 = \frac{COV_{PO}}{V_P} = \frac{16.25}{68} = 0.2389706$$

This is exactly the same as the slope of a linear regression of offspring on parent height values:

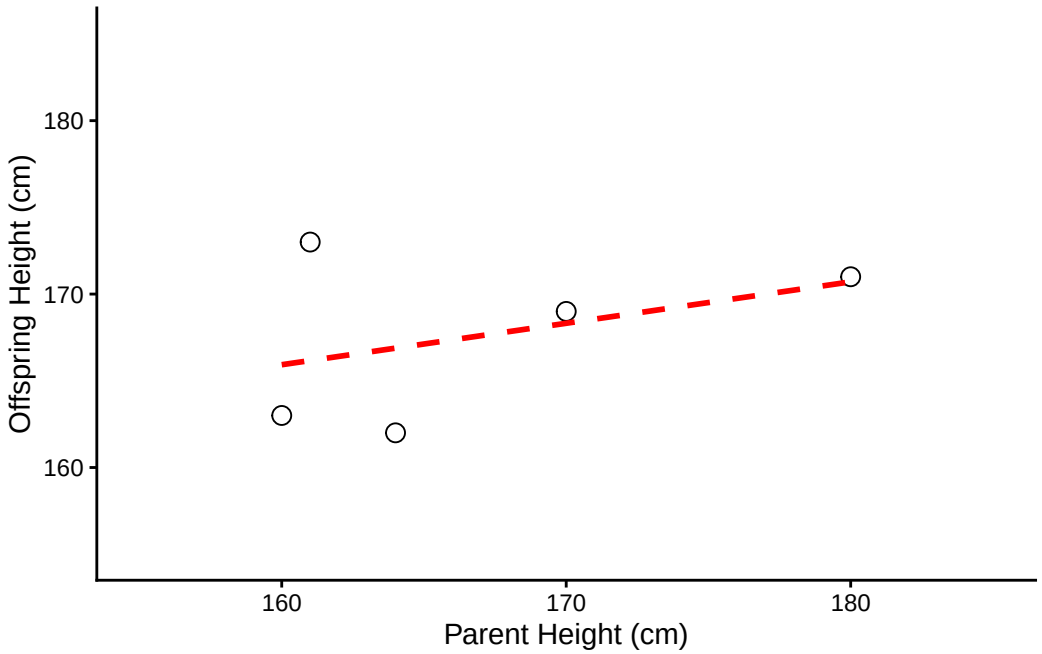
```
library(tidyverse)
p_i <- c(160, 170, 164, 180, 161)
o_i <- c(163, 169, 162, 171, 173)
df <- cbind(p_i, o_i) %>% as.data.frame()
model <- lm(o_i ~ p_i)
model$coefficients[2]
```

```
     p_i
0.2389706
```

(Beyond quantitative genetics, the slope of *any* regression can be calculated as Cov(x, y)/Var(x)).

Though not particularly impressive with so few data, the relationship is obvious when plotted:

```
ggplot(df, aes(x=p_i, y=o_i)) +
  theme_classic() +
  geom_point(pch=21, size=3) +
  xlim(155, 185) +
  ylim(155, 185) +
  geom_smooth(method='lm', se = FALSE, linetype="dashed", color="red") +
  xlab("Parent Height (cm)") +
  ylab("Offspring Height (cm)")
```



We can now also define total phenotypic variance (which we will also call V_P, sorry) as:

$$V_P = V_G + V_E + 2COV_{GE}$$

where V_G is phenotypic genetic variance, or the proportion of phenotypic variance that is rooted in heritable genetic variation; V_E is phenotypic environmental variance, and $2COV_{GE}$ is the covariance between genetics and the environment, which is negligible when individuals are under the same environmental conditions.

We can further partition V_G into effects from **additive**, **dominant**, or **partially dominant** loci:

$$V_G = V_A + V_D + V_{PD}$$

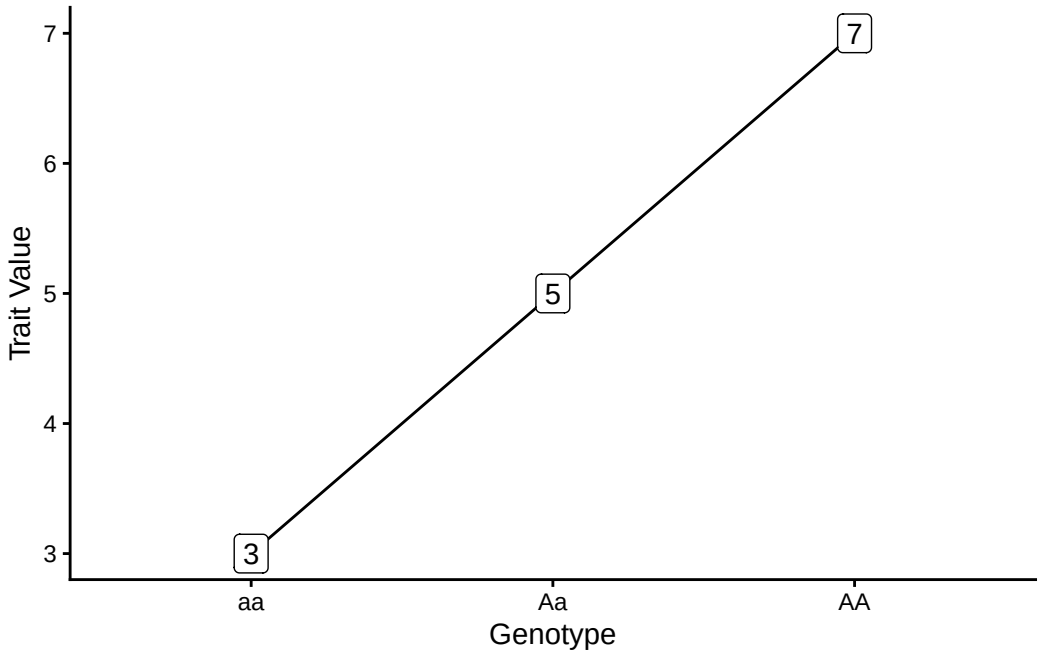
To determine which locus exhibits which mode of inheritance, we look at the relationship between genotypes and quantitative trait values.

First, we assign genotypes different values, which we represent algebraically with the letters a and d:

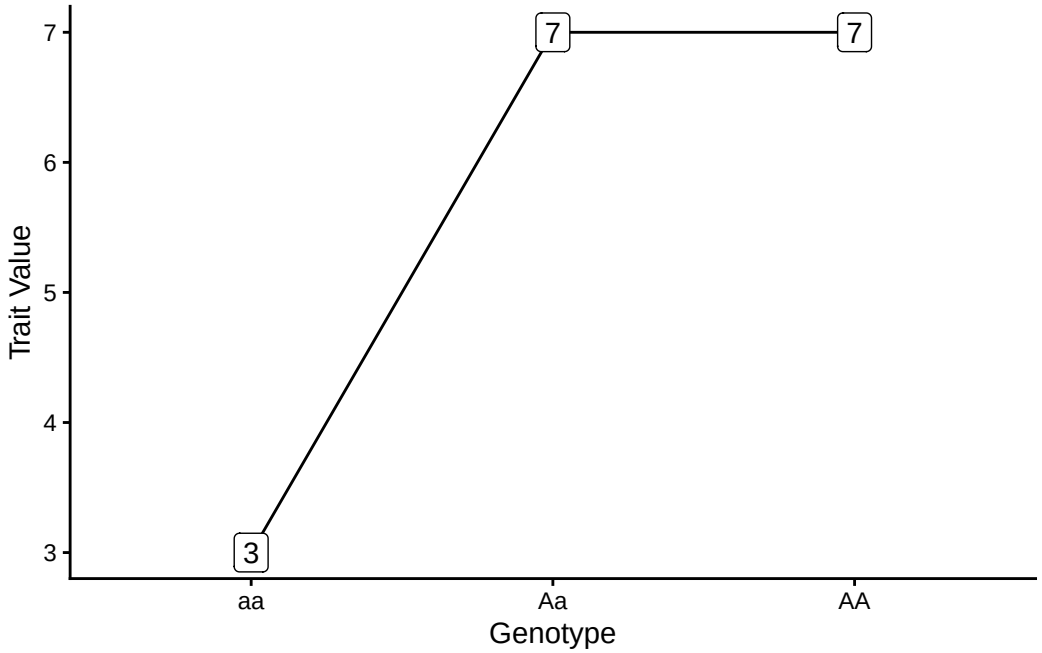
genotypes: AA, Aa, aa
genotypic values: $+a$, d, $-a$

In other words, $+a$ is the deviation of the trait value for AA away from the mean of the two homozygotes, d is the deviation of the trait value for the heterozygote away from the mean of the two homozygotes, and $-a$ is deviation of the trait value for aa away from the mean of the two homozygotes. (You can thus think of the "mean" trait value of the two homozygotes as 0, as $(+a + -a)/2 = 0$) Homozygotes differ in genotypic value by $2a$, and heterozygotes differ from the mean of the two homozygotes (again, 0) by d.

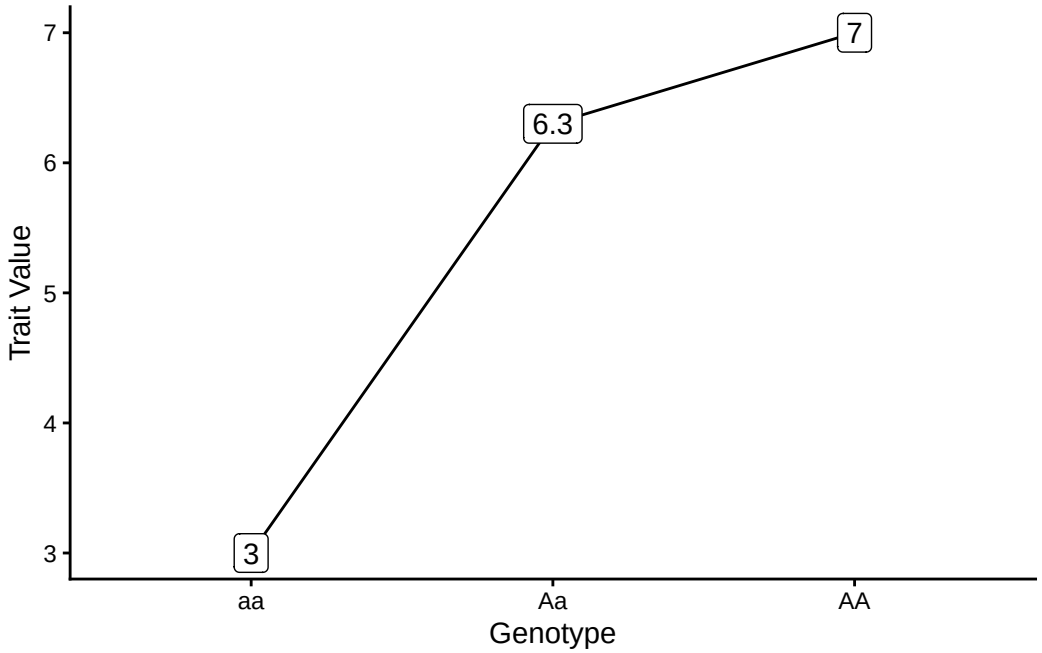
Different relationships between genotypes and genotypic values result from different modes of inheritance. In **additive genetic variance**, each allele contributes equally to the value of a phenotype, and the heterozygote's trait values are the average of the two homozygotes $(+a = 2; -a = 2; d = 0)$.



In contast, under **dominant genetic variance**, one copy of an allele is sufficient to attain the maximum value of a phenotype ($d = +a = 2$)



With **partially dominant genetic variance** one copy of an allele is not sufficient to attain the maximum value of a phenotype, but the genotypic value of the heterozygote is greater than the average value of the two homozygotes ($+a = 2; -a = 2; d = 1.3$)



As a shortcut, you can find a by dividing the difference between the trait values of the two homozygotes by 2:

$$a = \frac{\text{(AA value - aa value)}}{2} = \frac{(7-3)}{2} = 2$$

You can find d by first finding the midpoint (arithmetic mean) of the two homozygote trait values, and then subtracting that from the heterozygote trait value:

$$\text{midpoint} = \frac{\text{(AA value + aa value)}}{2} = \frac{(7+3)}{2} = 5$$
$$d = 6.3 - 5 = 1.3$$

Heritability of Migration Timing in Barn Swallows

Barn Swallows (*Hirundo rustica*) are cosmopolitan songbirds that are migratory across the majority of their range. An important migratory phenotype is *arrival date*, which is (unsurprisingly) simply the date a given individual arrives on breeding grounds. Because this has impacts on resource and mating partner availability, it is potentially under strong natural selection. Because arrival date varies continuously, studying any genetic basis it might have requires tools from quantitative genetics. Anders Møller (Pape Moller (2001)) used field data on the arrival date of 98 father / son pairs of barn swallows to estimate heritability with the regression approach described above. This deceptively simple method revealed $h^2 = 0.536$. This heritability was comparable to that of tail length, and indicates the trait may respond to selection.

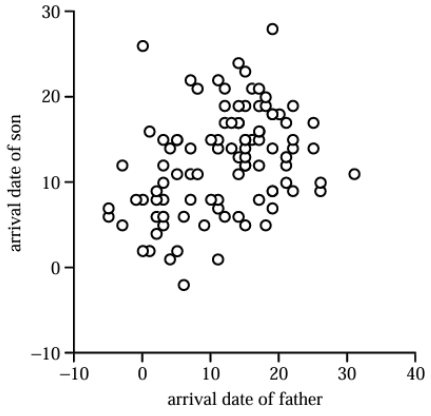


Figure 1. Arrival date of sons in relation to arrival date of their fathers in barn swallows. Date 1, 1 May; $n = 98$.

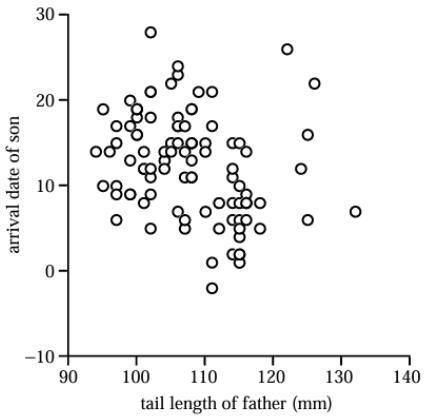
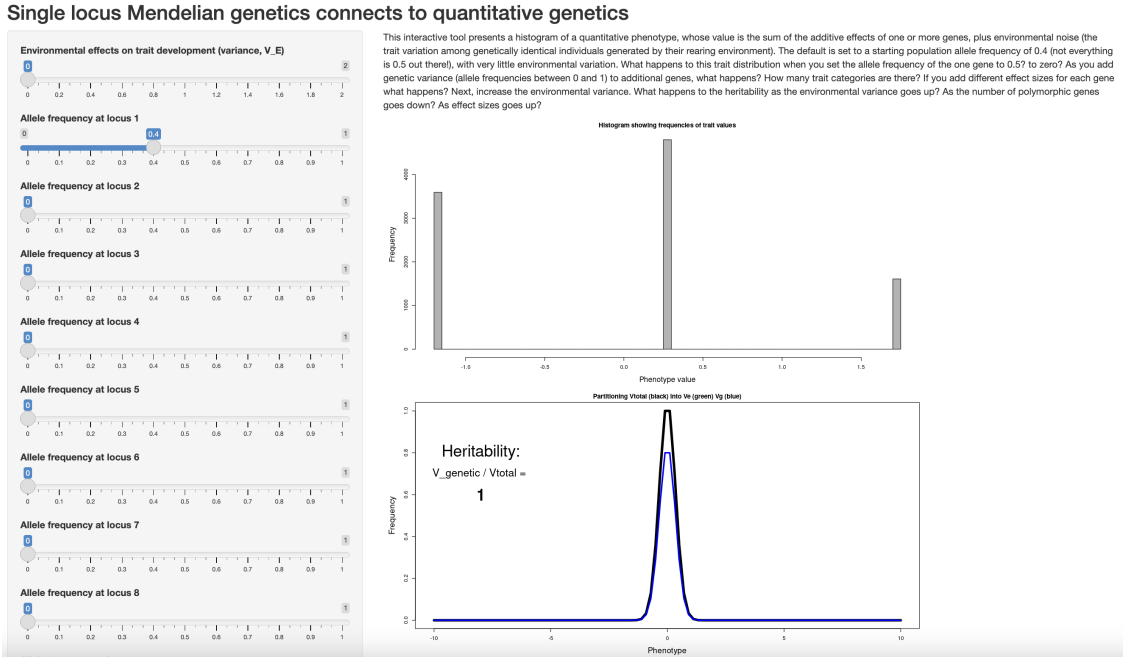


Figure 2. Arrival date of sons in relation to tail length (mm) of their fathers in barn swallows. Date 1, 1 May; $n = 98$.

Figure 6.1: *Figures 1 and 2 from Møller 2001*

Quant Gen App

Dan Bolnick has another app that helps connect this material to Mendelian single locus genetics.



Open the app in your browser and consider the questions:

1) One by one, change the allele frequencies of additional loci to intermediate values (i.e., not 0 or 1). What happens to estimates of heritability and the distribution of phenotypes?

2) Boost the environmental effect on trait variance slider. How do heritability and the distribution of phenotypes change?

7 Natural selection

The expected genotype frequencies for a diallelic locus under Hardy-Weinberg Proportions assumes random mating, infinite population sizes, no mutation, no migration, and no natural selection. The next section of the class will focus on developing our understanding of how genotype and allele frequencies change across generations when each of these assumptions is violated (both independently and in some cases, jointly). We will begin with the last assumption on our list: natural selection. As population and conservation genetics are at their core the study of the most fundamental type of microevolutionary change, it is appropriate to begin with the most famous evolutionary mechanism—environmental factors that lead to differential survival and reproduction across genotypes.

We will now develop a single locus model for allele frequency changes in response to natural selection. Let's begin with an extreme example—a lethal recessive disease. The following table lays out phenotypes, genotypes, and frequencies under HWP. It also details relative fitness (in which the most-fit genotype is 1, and others are scaled proportionally.)

phenotype	normal	normal	dead	total
genotype	AA	Aa	aa	1
relative fitness	1	1	0	1
freqs	p^2	$2pq$	q^2	1
freqs after selection	$p^2 * 1$	$2pq * 1$	$q^2 * 0$	$1 - q^2$
adjusted freqs	$\frac{p^2}{1-q^2}$	$\frac{2pq}{1-q^2}$	0	1

The frequency of the recessive allele (q) in the next generation is therefore the number recessive homozygotes (now 0) + 1/2 the number of heterozygotes divided by the total number of individuals minus the recessive homozgygotes that perished:

$$\frac{0 + pq}{1 - q^2} = \frac{q(1-q)}{(1-q)(1+q)} = \frac{q}{1+q}$$

(Note that since $p + q = 1$, $p = 1 - q$).

The change in the frequency of q (Δq) will be the difference between the new frequency of q (q_1) and its initial frequency (q):

$$\Delta q = q_1 - q = \frac{q}{1+q} - q = \frac{q}{1+q} - \frac{q(1+q)}{1+q} = \frac{q - q(1+q)}{1+q} = \frac{q - q - q^2}{1+q} = \frac{-q^2}{1+q}$$

We can expand this model for nonlethal cases where homozygous recessive genotypes have lower relative fitness by a quantity s, which we will refer to as the *selection coefficient*:

phenotype	normal	normal	sickly	total
genotype	AA	Aa	aa	1
relative fitness	1	1	1 - s	1
freqs	p^2	$2pq$	q^2	1
freqs after selection	$p^2 * 1$	$2pq * 1$	$q^2 * (1 - s) = q^2 - sq^2$	$1 - sq^2$
adjusted freqs	$\frac{p^2}{1-sq^2}$	$\frac{2pq}{1-sq^2}$	$\frac{q^2-sq^2}{1-sq^2}$	1

What, then, will happen to the frequency of allele p given a q has a fitness of $1 - s$? We know that its frequency in the next generation will be the adjusted frequency of homozygotes plus 1/2 the adjusted frequency of heterozygotes:

$$p_1 = \frac{p^2}{1-sq^2} + \frac{1}{2}(\frac{2pq}{1-sq^2}) = \frac{p^2 + pq}{1-sq^2} = \frac{p(p+q)}{1-sq^2} = \frac{p}{1-sq^2}$$

This gives us p_1, or the frequency of p in the next generation. (Note again that one step above involves substituting 1 for $p + q$.) The change (Δ) in p between generations is simply $\Delta p = p_1 - p$:

$$\Delta p = \frac{p}{1-sq^2} - p = \frac{p}{1-sq^2} - \frac{p(1-sq^2)}{1-sq^2}$$

$$\Delta p = \frac{p - p(1-sq^2)}{1-sq^2} = \frac{p - p + spq^2}{1-sq^2} = \frac{spq^2}{1-sq^2}$$

This tells us three things: that p will increase in frequency, and that its degree of increase depends on the initial frequency of q and the value of the selection coefficient, s.

The last case we will address is partial dominance, which we will model with the addition of a dominance coefficient, h. (Under complete dominance, $h = 0$; under additive inheritence, $h = 0.5$.)

phenotype	normal	iffy	sickly	total
genotype	AA	Aa	aa	1
relative fitness	1	1 - hs	1 - s	1
freqs	p^2	$2pq$	q^2	1

freqs after selection	p^2	$2pq(1-hs) = 2pq - 2hspq$	$q^2(1-s) = q^2 - sq^2$	$1 - 2hspq - sq^2$
adjusted freqs	$\frac{p^2}{1-2hspq-sq^2}$	$\frac{2pq-2hspq}{1-2hspq-sq^2}$	$\frac{q^2-sq^2}{1-2hspq-sq^2}$	1

Turning this into an expression for Δp is a little gnarly:

$$\Delta p = p_1 - p = \frac{p^2}{1-2hspq-sq^2} + \frac{1}{2}\frac{2pq-2hspq}{1-2hspq-sq^2} - p$$

$$\Delta p = \frac{p^2+pq-hspq}{1-2hspq-sq^2} - p$$

$$\Delta p = \frac{p^2+pq-hspq}{1-2hspq-sq^2} - \frac{p(1-2hspq-sq^2)}{1-2hspq-sq^2} = \frac{(p^2+pq-hspq)-p(1-2hspq-sq^2)}{1-2hspq-sq^2}$$

$$\Delta p = \frac{p((p+q-hsq)-(1+2hspq+sq^2))}{1-2hspq-sq^2}$$

Once again substituting 1 for $p+q$:

$$\Delta p = \frac{p(1-hsq-1+2hspq+sq^2)}{1-2hspq-sq^2}$$

$$\Delta p = \frac{p(-hsq+2hspq+sq^2)}{1-2hspq-sq^2} = \frac{p((2pq-q)hs+sq^2)}{1-2hspq-sq^2} = \frac{spq((2p-1)h+q)}{1-2hspq-sq^2}$$

Because $2p-1 = p+p-1$ and $p = 1-q$, $2p-1 = p+1-q-1 = p-q$:

$$\Delta p = \frac{spq((p-q)h+q)}{1-2hspq-sq^2}$$

That's about as good as we can get it!

7.1 Summary

As the math in the course becomes more complicated, it can be easy to get lost in the weeds. The big takeaway here is that deterministic models of selection show that homozygous genotypes with low fitness relative to the homozygote for the alternate allele (not necessarily recessive!) will decrease in frequency in the population, reducing the frequency of the allele they are homozygous for in turn. The change in frequency is determined by the frequency of the focal allele (q) in the present generation and the strength of selection (**s**); in anything less than complete dominance, the largest single-generation shifts are possible when alternate alleles are at intermediate frequencies and s is close to 1. I repeat our important derivations:

1) The frequency of a recessive lethal allele in the next generation under complete dominance;

$$q_1 = \frac{q}{1+q}$$

2) The *change* in frequency of a recessive lethal allele and the alternate allele under complete dominance;

$$\Delta p = \frac{q^2}{1+q};\ \Delta q = \frac{-q^2}{1+q}$$

3) Alelle frequency changes under complete dominance and selection against q; and

$$\Delta p = \frac{spq^2}{1-sq^2};\ \Delta q = \frac{-spq^2}{1-sq^2}$$

4) Alelle frequency changes under partial dominance and selection against q:

$$\Delta p = \frac{spq((p-q)h+q)}{1-2hspq-sq^2};\ \Delta q = \frac{-spq((p-q)h+q)}{1-2hspq-sq^2}$$

These models let us predict allele frequency changes under scenarios where selection is the only evolutionary force acting on a given population. For example, consider the mice studied by Nachman et al. (Nachman, Hoekstra, and D'Agostino (2003)) Coat color appears to show complete dominance: Individuals with the genotypes DD and Dd at Mc1r are dark, while those with the genotype dd are light. If $f(D) = 0.3$ and $f(d) = 0.7$, and the relative fitness of white mice on lava rocks is 0.3 ($W(dd) = 0.3$, which means a selection coefficient of $s = 1-0.3 = 0.7$), how much will allele frequencies in that habitat change in one generation? We can solve for this with simple substitution:

$$\Delta p = \frac{spq^2}{1-sq^2};\ \Delta q = \frac{-spq^2}{1-sq^2}$$
$$\Delta p = \frac{0.7*0.3*0.7^2}{1-0.7*0.7^2};\ \Delta q = \frac{-0.7*0.3*0.7^2}{1-0.7*0.7^2}$$
$$\Delta p = 0.1566;\ \Delta q = -0.1566$$

7.2 Time required for a given allele frequency change

As discussed above, the frequency of a lethal deleterious allele in the next generation is $q_1 = \frac{q}{1+q}$. It follows that its frequency in the generation *after that* is $q_2 = \frac{q_1}{1+q_1}$. Since we know what q_1 is in terms of q (the original allele frequency), we can substitute it and solve for a general relationship:

$$q_2 = \frac{\frac{q_0}{1+q_0}}{1+\frac{q_0}{1+q_0}} = \frac{q_0}{1+q_0} \div 1 + \frac{q_0}{1+q_0}$$

$$q_2 = \frac{q_0}{1+q_0} \div \frac{1+q_0}{1+q_0} + \frac{q_0}{1+q_0} = \frac{q_0}{1+q_0} \div \frac{1+q_0+q_0}{1+q_0}$$

$$q_2 = \frac{q_0}{1+q_0} \div \frac{1+2q_0}{1+q_0} = \frac{q_0}{1+q_0} * \frac{1+q_0}{1+2q_0} = \frac{q_0(1+q_0)}{(1+q_0)(1+2q_0)} = \frac{q_0}{1+2q_0}$$

By extension, $q_t = \frac{q_0}{1+tq_0}$.

We can rearrange this to get the number of generations (t) required to change the allele frequency from q_0 to q_t:

$$q_t = \frac{q_0}{1+tq_0};\ q_t(1+tq_0) = q_0$$

$$q_t + tq_0q_t = q_0;\ tq_0q_t = q_0 - q_t$$

$$t = \frac{(q_0 - q_t)}{q_0q_t} = \frac{1}{q_t} - \frac{1}{q_0}$$

We can apply this to determine how long it would take for a recessive lethal allele to drop from a frequency of $f(q_0) = 0.7$ to $f(q_t) = 0.2$:

$$t = \frac{1}{0.2} - \frac{1}{0.7} = 3.57 \text{ generations}$$

Selection on Armor in Sticklebacks

Freshwater threespine stickleback (*Gasterosteus aculeatus*) on the coast of British Columbia are derived from heavily-armored marine ancestors, but have repeatedly evolved a reduced number of lateral plates. Hypothesizing this transition was due to developmental advantages associated with avoiding investment in plate development, Rowan Barrett and colleagues (Barrett, Rogers, and Schluter (2008)) transplanted marine sticklebacks to freshwater ponds and tracked the frequency of an allele associated with fewer lateral plates through time. They observed increases in multiple replicate populations, indicating selection consistent with their growth advantage hypothesis (though oscillations in the

fitness of genotypes throughout each fish's life cycle suggest the story is more complicated).

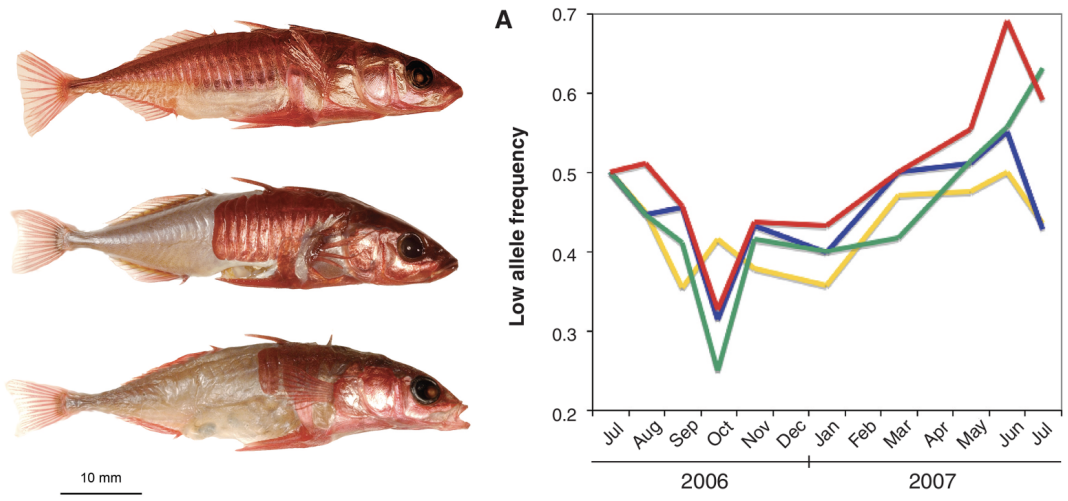
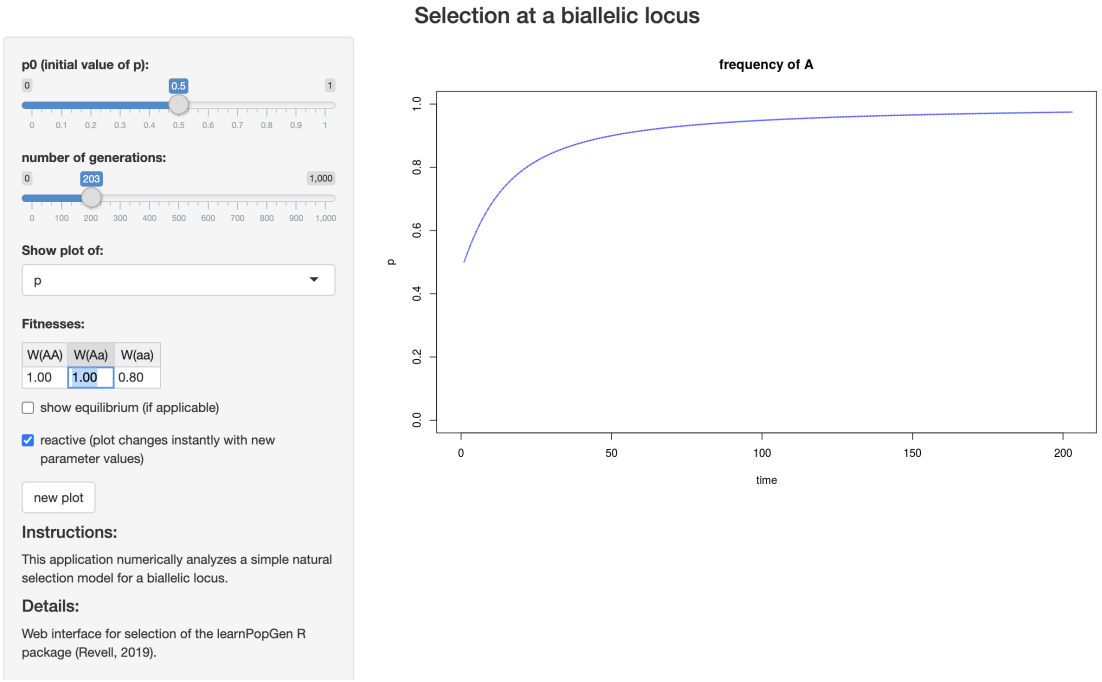


Figure 7.1: *Figures 1 and 2a from Barrett et al. 2008. Left: Plate morphs in threespine sticklebacks, with an individual homozygous for the "low" allele at the bottom of the image. Right: the change in frequency of the "low" morph allele through time in replicate freshwater ponds*

i Selection App

Liam Revell has an app to analytically solve for allele frequencies under different selection and dominance coefficients: https://phytools.shinyapps.io/selection/



Open the app in your browser and work through the following questions.

1) Set fitnesses to $WW(AA) = 1.0$, $W(Aa) = 1.0$, and $W(aa) = 0.8$. Set $p0$ to 0.01, and make sure you are showing a plot of p. What mode of inheritance is occuring at this locus? What is the value of the selection coefficient? How many generations does it take to reach $p = 0.6$?

2) Change $W(Aa)$ to 0.9. What mode of inheritance are you modeling now? Does the time it take to reach $p = 0.6$ change?

3) Switch to showing a plot of q without changing any other settings. What can you conclude about the relationship between p and q through time?

4) Develop a question and address it using this app. Be prepared to report back to the class.

8 Mutation, migration, and selection

The next violation of HWP assumptions we will examine is mutation. Mutations are a sudden genetic change in an allele or chromosome. They can be caused by the impacts of oxygen, radiation, chemicals, or cellular processes. DNA repair fixes many mutations but is imperfect. Importantly, mutations are the origin of all genetic diversity.

Common types of mutations include:

- Base substitutions
- Additions or deletions in DNA sequences
- Gene duplications
- Insertion of mobile DNA elements (**transposons**)

While we often think of mutations as inherently negative, they can have a variety of fitness effects:

- **silent substitutions**: a mutation in a coding region that does not result in an amino acid substitution in a protein
- **neutral mutations**: mutations with little to no impact on fitness
- **deleterious mutations**: mutations that negatively impact fitness
- **beneficial mutations**: mutations that positively impact fitness

Importantly, some mutations may be beneficial in some circumstances but not others!

A key parameter for a given mutation type is its rate, typically denoted by u or μ ("myu"). Mutation rates describe the probability of mutation per unit per generation. Different genetic markers have different rates. As a set of benchmarks:

- 1 x 10^{-8} to 1 x 10^{-9} per base per year: DNA nucleotides
- mammalian mtDNA typically has 5 - 10x the nuclear DNA mutation rate
- 1 x 10^{-4} per locus: mammalian microsatellites
- 1 x 10^{-5} per locus: morphological mutations in fruit flies

For example, a per nucleotide rate of 10^{-5} (equivalent to $\frac{1}{100000}$) leads us to expect a single mutation in a stretch of 100,000 bases each generation (because the independent probabilities of mutation at each base are summed). If instead our mutation rate is per *gene* and we know there are roughly 20,000 genes in the human genome, we would expect $10^{-5} * 20,000 = 0.2$ mutations per individual (so, roughly every 5 individuals we'll find one new mutation in a coding region).

The probability of an individual NOT having a mutation in one of their 20,000 genes is a little more complicated. To determine it, we need to account for both the probability of no mutation in their maternally-inherited chromosomes and the probability of no mutation in their paternally inherited chromosomes. Each haploid genome requires multiplying the independent probability of NO mutation ($= 1-\mu$) by itself 20,000 times, as these independently probabilities must occur concurrently. We then must multiply the probabilities of no mutation in each haploid genome by each other, as these two events must also occur together:

$$p(\text{no mutation}) = (1-\mu)^{20000} + (1-\mu)^{20000} = (1-\mu)^{40000} = (1-\frac{1}{100000})^{40000} \sim 0.67$$

Conversely, to find the probability of at least one new mutation in a coding region in a human's diploid genome, we can either take the complement of no mutation (i.e., $1 - 0.67 = 0.23$).

But what does this mean for allele frequencies? We start with a simple model where allele A_1 (occuring at frequency p_0) mutates to become A_2 (occuring at frequency q_0) at a rate u / μ. (For now, we assume mutation is only one-dimensional.)

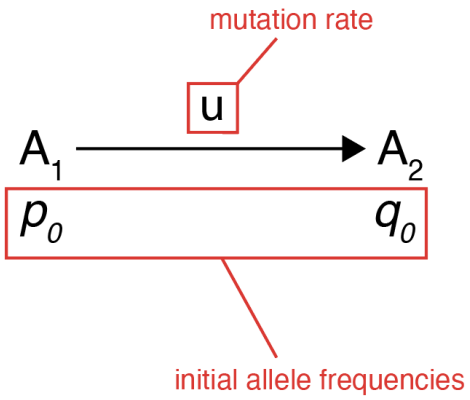


Figure 8.1: *A simple model of one-directional mutation*

The frequency of A_1 in the next generation (p_1) will be its initial frequency multiplied by the complement of the mutation rate, i.e. the expected proportion of alleles that do NOT mutate:

$$p_1 = p_0(1-\mu)$$

We can use this formula in the following example:

Q: What is the frequency of allele A_1 in the next generation if its mutation rate u is 2.5 x 10^{-5} and it occurs at an initial frequency of 0.7?

A:

$$p_1 = p_0(1-u);$$
$$p_1 = 0.7(1-0.000025);$$

$$p_1 = 0.6999825$$

This shows us that among other things, mutation is a *slow* way to change allele frequencies and restore genetic diversity!

This same model helps us determine the change in A_1 from generation to generation: it is the difference between p_1 and p_0, which simplifies to the negative product of the mutation rate and initial frequency of A_1:

$$\Delta p = p_1 - p_0 = p_0(1-\mu) - p_0 = p_0 - \mu p_0 - p_0 = -\mu p_0$$

We can also use this relationship to build a model of how mutation changes allele frequencies t generations in the future. If $p_1 = p_0(1-\mu)$, then $p_2 = p_1(1-\mu) = p_0(1-\mu)*(1-\mu) = p_0(1-\mu)^2$. Therefore, $p_t = p_0(1-\mu)^t$, which means that $p_t \sim p_0 e^{-\mu t}$. (This equation is an approximation of exponential decay.)

An immediate application of our model is to address the question of how long it takes for mutation to restore lost genetic diversity. We can do this by isolating t in the equation above:

$$p_t \sim p_0 e^{-\mu t}$$
$$\frac{p_t}{p_0} = e^{-\mu t}$$
$$ln(\frac{p_t}{p_0}) = -\mu t$$
$$\frac{ln(\frac{p_t}{p_0})}{-\mu} = \frac{ln(p_t) - ln(p_0)}{-\mu} = \frac{-ln(p_0) - ln(p_t)}{\mu} = t$$

We can apply this equation if (for example) we have a mutation rate of 10^{-6} and want to know how long it takes for p to go from a frequency of 1 to 0.9:

$$\frac{-ln(1) - ln(0.9)}{10^{-6}} = t = 105,360 \text{ generations}$$

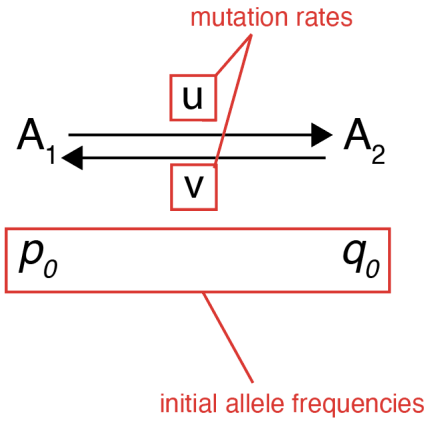


Figure 8.2: *A simple model of one-directional mutation*

But what happens if we have mutation in two directions? In this model, allele A_1 (occuring at frequency p_0) still mutates to become A_2 (occuring at frequency q_0) at rate u, but we also permit A_2 to mutate back to A_1 at rate v:

In this scenario, we would expect the equilibrium frequency of each allele to be determined by the relative strength of mutation in each direction:

$$\hat{p} = \frac{v}{\mu + v};\ \hat{q} = \frac{\mu}{u + v}$$

For example: what is the equilibrium frequency of A_2 if $\mu = 10^{-5}$ and $v = 3 * 10^{-6}$?

$$\hat{q} = \frac{10^{-5}}{10^{-5} + (3 * 10^{-6})} = 0.77$$

Next, we incorporate mutation. Selection more or less inevitably acts on the new mutations that are being continuously added to populations as the majority are deleterious (it's much easier to break something than improve it!). Intuitively, the overall change in the frequency of q will be equal to its change due to mutation and its change due to selection–a phenomenon known as **mutation-selection equilibrium**, or the balance between the addition of deleterious alleles by mutation and their removal by selection. For a deleterious recessive allele, this is: :

$$\Delta q = \Delta q_{mutation} + \Delta q_{selection}$$

Assuming a model of simple dominance, we know that deleterious mutations increase as a product of their forward mutation rate (u) and frequency (p), and decrease at rate determined by the one locus selection models we covered last week ($\frac{-spq^2}{1-spq^2}$). Therefore:

$$\Delta q = \mu p - \frac{spq^2}{1 - sq^2}$$

Since sq^2 is an incredible small quantity for rare alleles, the denominator is essentially 1, so

$$\Delta q = \mu p - spq^2$$

By definition, an equilibrium is when inputs (mutations) match outputs (alleles to selection) ($\Delta q = 0$). When this is true, $up \sim spq^2$. We can divide both sides of the similarity by sp to isolate q^2:

$$0 \sim \mu p + spq^2;\ \mu p \sim spq^2;\ \frac{\mu p}{sp} = \frac{\mu}{s} \sim q^2$$

We then take the square root of the right side of the equation to derive the equilibrium frequency of A_2:

$$\hat{q} \sim \sqrt{\frac{\mu}{s}}$$

In plain English, the equilibrium frequency of q will be the square root of the ratio of mutation from A_1 to A_2 to mutation from A_2 to A_1. We further know that:

- The equilibrium only depends on the mutation rate and the selection coefficient
- Increased mutation rates will increase the frequency of A_2
- Increased strength of selection against it will decrease it.

Here are two simple examples of the use of this equation:

Q: What is the mutation-selection equilibrium for a recessive lethal allele if the forward mutation rate is 3 x 10^{-5}?

A:

$$q \sim \sqrt{\frac{3x10^{-5}}{1}} \sim 0.0054$$

Q: What is the mutation-selection equilibrium for a recessive deleterious allele if the forward mutation rate is 1 x 10^{-5} and the selection coefficient is 0.1?

A:

$$q \sim \sqrt{\frac{3x10^{-5}}{0.1}} \sim 0.017$$

In the same vein, we can model the counteracting forces of migration and selection to predict allele frequencies at **migration-selection equilibria**. (Because mutation is rare and only changes allele frequencies very slowly, this will be more important for most scenarios discussed in class). The change in the frequency of allele q is going to be determined by the proportion of alleles in a population of interest that are contributed by migrants, a variable we call m, multiplied by the difference in the frequency of q in the migrant population and the local population:

$$\Delta q = m(q_m - q_0)$$

For example, if migrants are fixed (homogenous) for an allele that is absent from the focal population, and contribute 20% of alleles to the focal population, we expect a change in q of $+0.2$ in a single generation:

$$\Delta q = 0.2(1 - 0) = 0.2$$

By similar logic, the frequency of q in the generation 1 (q_1) will be the sum of the product of m and q_m and the initial frequency of q and the proportion of alleles that are NOT migrants, which simplifies to the intial allele frequency plus the migration rate multiplied by the difference between the frequency of q in the migrant population and focal population:

$$q_1 = (1 - m)q_0 + mq_m = q_0 - mq_0 + mq_m = q_0 + m(q_m - q_0)$$

This gives us the basis for the derivation of Δq above:

$$\Delta q = q_1 - q_0 = (q_0 + m(q_m - q_0)) - q_0 = m(q_m - q_0)$$

As an example of its application, imagine a scenario in which we wish to estimate the migration rate of domestic dog alleles into a wild dog population. We will consider a locus that is *diagnostic* (fixed in one species or population and not the other), treat the new, hybrid population as q_0, q_1 as the initial allele frequency of "pure" wild dogs, and q_m as the allele frequency of domestic dogs:

wild dog	q_0	1
hybrid	q_1	0.78
dog	q_m	0

To do this need to isolate m from $q_1 = q_0 + m(q_m - q_0)$:

$$
\begin{aligned}
q_1 &= q_0 + m(q_m - q_0) \\
q_1 - q_0 &= m(q_m - q_0) \\
\frac{q_1 - q_0}{q_m - q_0} &= m \\
m &= \frac{0.78 - 1}{0 - 1} = 0.22
\end{aligned}
$$

If migration and selection form an equilibrium allele frequency, what is it? This question is relevant to the analysis of clines and hybrid zones. We again consider the overall change in q to be the sum of the change in q due to opposing forces:

$$\Delta q = \Delta q_{selection} + \Delta q_{migration}$$

i Migration-Selection Equilibrium in *Zonotrichia capensis*

Populations that are continuously distributed across environmental gradients are likely to experience migration-selection equilibrium because adjacent individuals—that are otherwise free to interbreed—may be under selection from different abiotic or biotic factors. The predictable reduction in the partial pressure of oxygen (PO_2) with increasing elevation exerts strong selection on respiratory function in endotherms, leading to a suite of adaptations to hypoxia in high mountains. Zac Cheviron and Rob Brumfield (Cheviron and Brumfield (2009)) asked how gene flow and selection interacted to shape allele frequencies in mitochondrial and nuclear DNA in the widespread Rufous-collared Sparrow (*Zonotrichia capensis*). Sampling across multiple elevational gradients and a horizontal "control" transect, they found a clinal change in the frequency of a haplogroup (= group of similar haploid DNA sequences) that reached fixation at high elevations. They found no detectable pattern at nuclear loci, suggesting gene flow homogenizes allele frequencies at most loci across the gradients, while selection acts on proteins encoded in the mitochondria across elevation (but not within elevational bands).

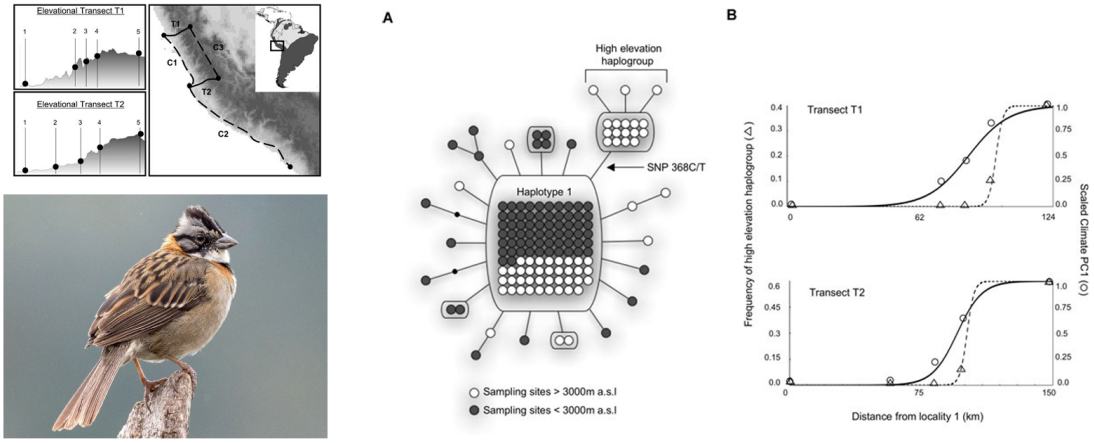


Figure 8.3: *Zonotrichia capensis, inset with Figures 1 and 2 from Cheviron et al. 2009. Note the horizontal control transect on the sampling map; pane B shows the change in frequency of the high altitude haplotype across the two experimental transects.*

9 Genetic Drift

The next assumption of HWP we will violate is that of infinite population sizes. When populations are sufficient large in size, the allele frequency dynamics are more or less deterministically driven by mutation, migration, and selection. With finite populations (and particularly with small populations), chance events begin to take on greater importance. This is of particular interest to us because conservation genetics is, in essence, the population genetics of small populations of conservation concern. In Montana, for instance, there only ~350 Black Footed Ferrets remaining, only ~500 Kootenai sturgeon, and only ~320 whooping cranes. The International Union for the Conservation of Nature (IUCN) considers a species critically endangered if there are fewer than 50 individuals, endangered if there are fewer than 250, and vulnerable if there are fewer than 1000.

The primary mechanism for the shift away from deterministic dynamics is sampling error of gametes. Consider two crosses between sexually reproducing diploid individuals at a single triallelic locus: an A_1A_1 x A_1A_2 cross, and an A_1A_3 x A_1A_2 cross. In this parental generation, allele frequencies are $p = 0.5$, $q = 0.375$, and $r = 0.125$. The first pair then has three offspring, formed by each parent randomly contributing one of their two alleles. The genotypes of these offspring are A_1A_1, A_1A_2, and A_1A_2. The second pair has a single child, with genotype A_1A_1. While this new generation has the same population size and total number of alleles or chromosomes, allele frequencies have changed to $p = 0.75$, $q = 0.25$, and $r = 0$.

You will notice that since $r = 0$, the allele has disappeared from the population. We describe this as its *loss*: the alternative scenario, in which an allele reaches a frequency of 1.0, is termed *fixation*. To develop intuition about the relationship between population size and the probability of fixation or loss of an allele, using a simulation tool (like CJ Battey's driftR) is helpful. More formally, we can we model the expected distribution of allele frequencies for a given number of gametes using the *binomial distribution*.

Let's start by revisiting Hardy-Weinberg proportions. As usual, we have two alleles (A_1 and A_2) at frequencies p and q. What is the probability that a single individual ($N = 1$) will have a particular genotype? We can determine this by the binomial expansion of $(p+q)^{2N} = (p+q)^2$. The reason is that we have two chances (one allele from both mom and dad, i.e. the exponent) at one of two possible outcomes (receiving either allele A_1, which will occur with a probability proportional to its frequency, p, or A_2, which will occur with a probability proportional to q):

$$(p+q)^2 = (p+q)(p+q) = p^2 + pq + qp + q^2$$

We see our for outcomes: p^2, pq, qp, and q^2. (Typically, $pq + qp$ are lumped into the more familiar $2pq$.) For a single individual, the probability (which you can think of as "expected frequency if there were only one outcome") of having each genotype is therefore $\frac{1}{4} = 0.25$. But what if we have 2 individuals instead? Our binomial expansion will then be raised to a power of $2N = 4$:

$$(p+q)^4 = (p+q)(p+q)(p+q)(p+q) = p^4 + 4p^3q + 6p^2q^2 + 4pq^3 + q^4$$

You can see how this quickly gets complicated. The probability of getting 4 copies of A_1 (out of 4 total alleles in 2 individuals) is p^4, the probability of getting 3 ps and 1 q is $4p^3q$, etc. But how can we generalize this for larger population sizes without the prohibitively tedious process of manually solving the binomial expansion? The formula for the binomial distribution provides us with a shortcut:

$$\binom{2N}{K} p^K q^{2N-K}$$

This formula gives you the probability of there being exactly K copies of an allele with frequency p given $2N$ gametes. Here, $\binom{2N}{K}$ is read "2N choose K" and is equivalent to $\frac{2N!}{K!(2N-K)!}$. q is the frequency of the alternate allele (the binomial expansion only works for diallelic loci). We can return to our example above to demonstrate its use. What, then is the probability of getting 3 copies of allele A_1 given $2N = 4$ and allele frequencies of $p = 0.5$ and $q = 0.5$?

$$\binom{4}{3} 0.5^3 0.5^{4-3} = \frac{4!}{3!(4-3)!} 0.5^3 0.5^1 = \frac{4*3*2*1}{3*2*1*(1)} 0.125 * 0.5 = 0.25$$

Importantly, this is exactly equal to the term in the binomial expansion demonstrated above: $4p^3q = 4 * 0.5^3 * 0.5 = 0.25$.

The variance of the distribution is $\sigma = \frac{p_0 q_0}{2N}$, or the initial allele frequencies over the number of alleles in the population. It is maximized when $p_0 = q_0$ and $2N = 2$, which tells us genetic drift (random change in alelle frequencies due to sampling error) is strongest in small populations with relatively equal frequencies at diallelic alleles. You can build intuition about the binomial distribution by playing around with frequencies ("probability of success") and number of alleles ("number of Bernoulli trials") using this app: https://istats.shinyapps.io/BinomialDist/.

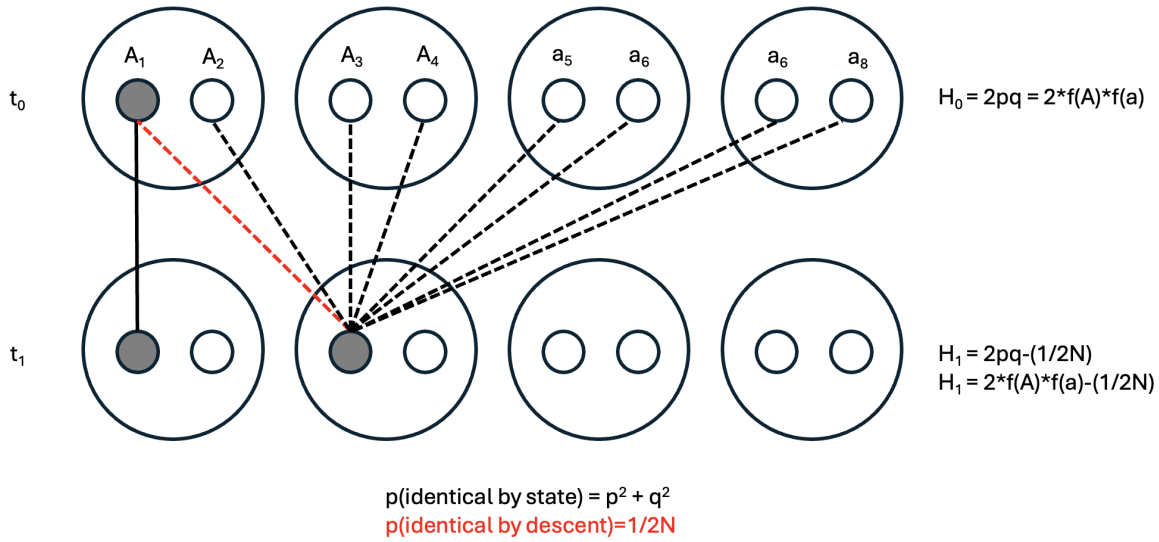
A simple extension of the binomial distribution is that the probability of fixation of an allele is given by its frequency to the $2N$ power, i.e. p^{2N}. Conversely, the probability of its loss is its complement to the $2N$ power, ie. $(1-p)^{2N}$.

A different way to arrive at this result using the building blocks of an area of population genetics known as **coalescent theory**, which we will discuss in greater length shortly. First, let's (re)define familiar terms. In a single individual, homozygosity and heterozygosity in an

refer to the state of having identical or different alleles at a given locus (respectively). In a randomly mating population, however, homozygosity refers to the probability two alleles drawn at random are identical ($p^2 + q^2$) or different ($2pq$). As indicated in the parentheses, these probabilities are given by terms from Hardy-Weinberg proportions.

Given this reminder, consider a single-generation population bottleneck of size N, where a formerly large population is reduced to small numbers (Figure 1). The subsequent generation $(t + 1)$ will be formed from the $2N$ alleles present during the bottleneck. Selecting any two alleles in different individuals at random in generation $t + 1$, the probability that they came from the same parental copy—in other words, are **identical by descent**—is simply 1 out of the total number of parental alleles, i.e. $\frac{1}{2N}$. (To make this more intuitive, select an allele in $t + 1$ and its ancestor in t, then select a second allele in $t + 1$. That second allele has $2N$ possible ancestors, with a $1/2N$ probability of the same ancestor as your first allele.)

Of course, other alleles may be **identical by state** (e.g., both be $A_1 s$) but come from different parents. These will result in homozygotes, but not change their overall frequency.



i Drift in *Lycaeides idas* butterflies

Drift constantly shapes allele frequencies in finite natural popualtions, but can be difficult to detect without temporal comparisons. Zach Gompert and colleagues (Gompert et al. (2021)) used population genomic time series data from from ten *Lycaeides idas* butterfly subpopulations in the Greater Yellowstone Ecosystem to test the hypothesis that gene flow over larger timescales maintained genetic diversity but did not impact generation to generation evolution. In doing so, they documented significant ($\sim$) changes in the frequencies of genetic variants across the gene—drift in action.

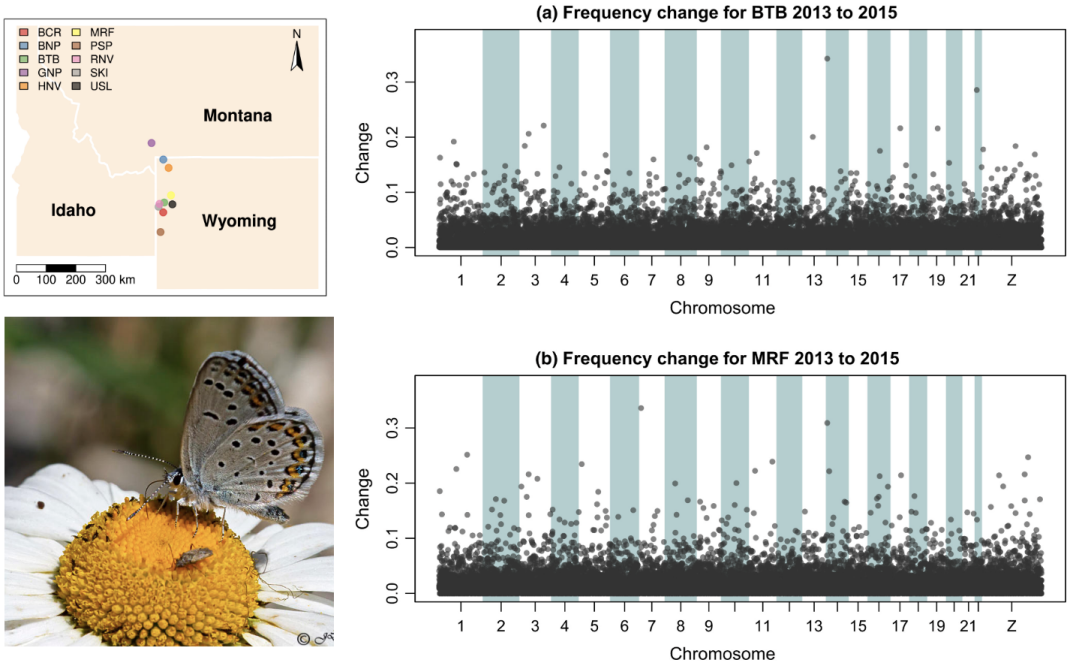


Figure 9.1: *Figures 1 and 2 from Gompert et al. 2021. Left: A map of population sampling localities, and a photo of Lycaeides idas (https://www.butterfliesandmoths.org/species/Plebejus-idas). Right: Changes in variant frequency in two subpopulations over time, plotted across chromosomes.*

9.1 Drift and Heterozygosity

To understand how genetic drift impacts genetic diversity, it's helpful to consider an extreme scenario we will return to several times: a population bottleneck, or sudden reduction in population size to a limited number of individuals. The most severe population bottleneck possible for sexually reproducing organisms is 2. Assuming they are diploid, the following table gives us the expected frequencies and heterozygosities of different allelic combinations. (H_e is calculated as $2pq$ given $f(A_1)$ and $f(A_2)$ from the Allele combination column.)

Allele combination	Frequency (f)	Heterozygosity (H_e)	f x H_e
4 A_1s	p^4	0	0
3 A_1s; 1 A_2	$4p^3q$	0.375	$1.5p^3q$
2 A_1s; 2 A_2	$6p^2q^2$	0.5	$3p^2q^2$
1 A_1s; 3 A_2	$4pq^3$	0.375	$1.5pq^3$
4 A_2s	q^4	0	0

Allele combination	Frequency (f)	Heterozygosity (H_e)	f x H_e
Total	1.0		$1.5pq$*

*Derivation: Total $= 1.5p^3q + 3p^2q^2 + 1.5pq^3 = 1.5pq(p^2 + 2pq + q^2) = 1.5pq(1) = 1.5pq$

Mean heterozygosity in the bottlenecked population (H_1) will be a proportion of the heterozygosity in the original population (H_0). In this case, $\frac{H_1}{H_0}$ will be $\frac{1.5pq}{2pq} = \frac{3}{4}$ or $1 - \frac{1}{4}$. Since 2N = 4, $\frac{H_1}{H_0} = 1 - \frac{1}{2N}$. Therefore, a single pair bottleneck will reduce H_0 by 25%; a bottleneck of 10 individuals will only reduce it by 5. We will return to this idea from a different perspective in a later lecture.

The Mauritius Kestrel Bottleneck

Mauritius Kestrel (*Falco punctatus*), the last surviving endemic raptor on the island of Mauritius, underwent a short-lived but severe population bottleneck due to forest loss and pesticide use, reaching a single known breeding pair in 1973. An intensive conservation breeding program halted and reversed the decline, leading to a current population of >400 individuals. Richard Nichols and colleagues (Nichols, Bruford, and Groombridge (2001)) used microsatellite data from contemporary tissue samples and historic museum specimens to describe changes in genetic diversity through time, showing that the restored population has significantly lower observed heterozygosity than either the ancestral (pre-bottleneck) population or mainland populations of other kestrel species.

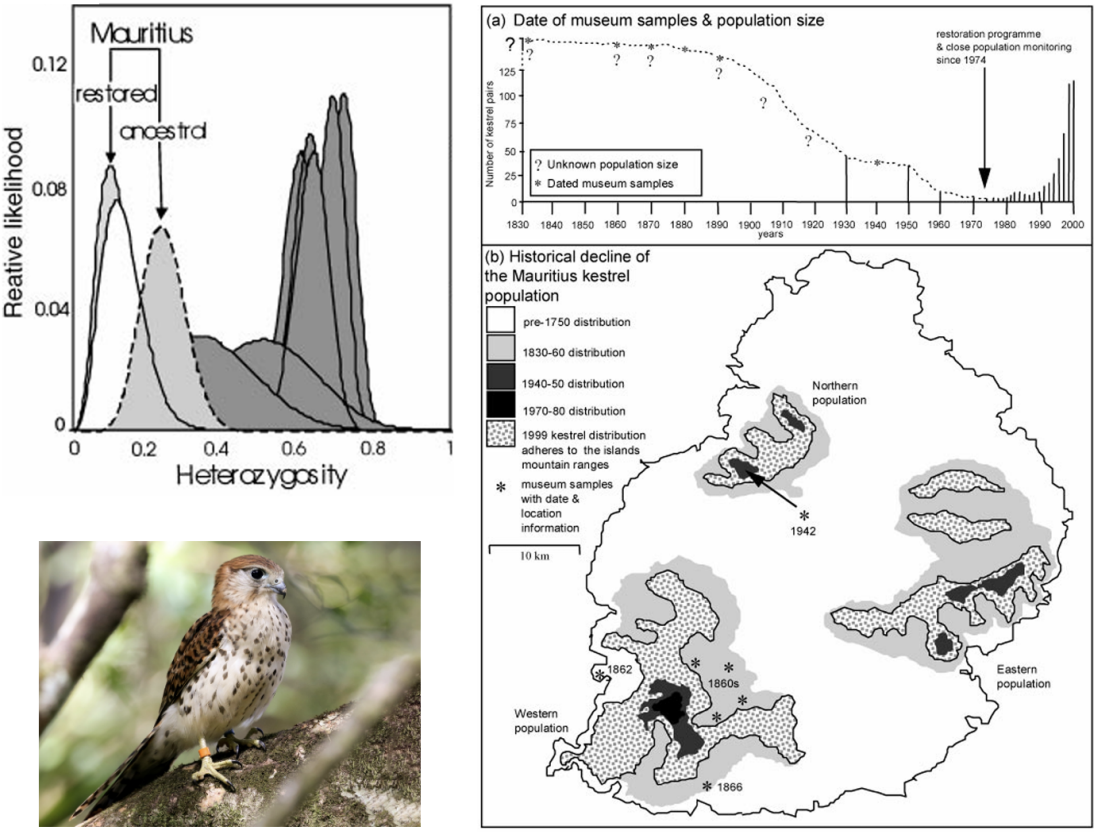
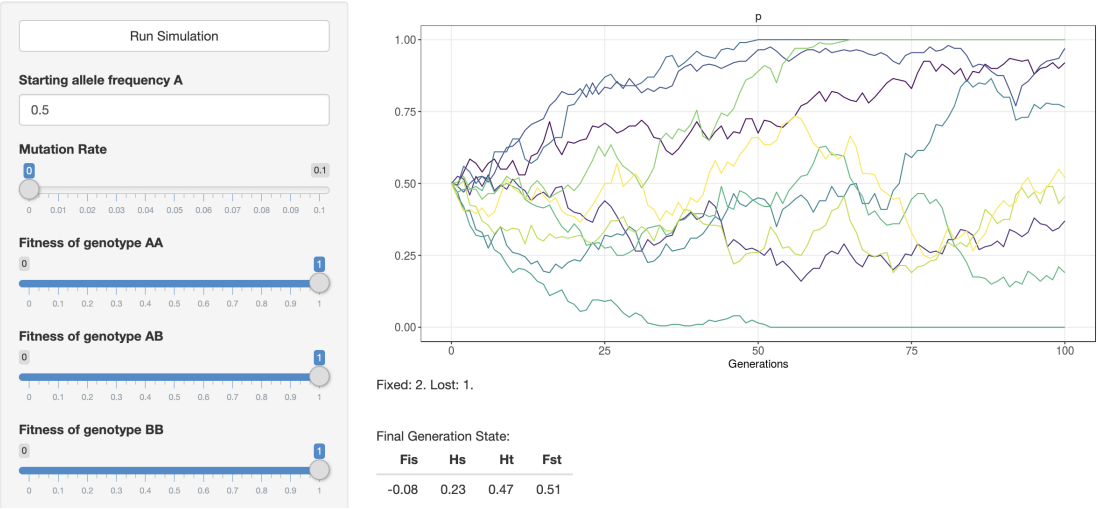


Figure 9.2: *Figures 1 and 2 from Nichols et al. 2001. Left:* H_o across microsatellite loci in extant Mauritius Kestrels, the ancestral population of Mauritius Kestrels, and other non-insular kestrel species. A photo of a banded Mauritius Kestrel is inset below (https://macaulaylibrary.org/asset/250861081). Right: Inferred distributional and population size changes in the species

i Genetic Drift Activity

CJ Battey wrote an app to simulate allele frequency change due to drift and selection: https://phytools.shinyapps.io/selection/

driftR: Population Genetic Simulations in R



Open the app in your browser and work through the following questions.

1) Orient yourself. What are the axes? What variables can you change? What does the plot display when you click "Run Simulation"?

2) Set "Starting allele frequency A" to 0.5 (instead of two values, as is default). Set all fitnesses to 1, the Migration Rate to 0, the Number of Populations to 10, and Number of Generations to 100. Make sure you are plotting "p" alone. Run simulations for population sizes of 10, 100, 1000, and 10,000. What pattern do you observe? What does it mean when a colored line hits the top of the graph? What does it mean when a colored line hits the bottom of the graph?

3) Repeat the process in 2 but setting the fitness of one of the homozygotes to a value between 0.01 and 0.5. What do you observe?

4) Develop a question to address with the driftR.

10 Effective Population Size

One of the most important (and confusing) topics in population genetics, effective population size (denoted N_e) refers to the size of an ideal population that loses heterozygosity at the same rate as the observed population. Effective population size differs from census population size (N, i.e., the number of individuals you can count) due to fluctuations in population size, variation in family size, and overlapping generations. N_e/N ratios are typically very low, with 0.1 being a good estimate in the absence of any other information. (Up until this point, all our models have been implicitly using N_e! We'll be careful to clarify this in the future.)

Because in a randomly-mating population, heterozygosity (technically expected heterozygosity, or H_e) is the probability two alleles are different, this population-wide increase in identity-by-descent (i.e., homozygosity) of $1/2N$ each generation reduces H proportionally:

$$H_1 = (1 - \frac{1}{2N})H_0$$

(Here, H_0 is the population's initial expected heterozygosity, while H_1 is the heterozygosity of the next generation.) Though we will not focus on it in this class, the loss of genetic diversity (as measure by H_e, or the probability two randomly drawn alleles are different) will vary across modes of inheritance: $\frac{1}{N_e}$ for haploid organisms, $\frac{1}{4N_e}$ for tetraploids, $\frac{1}{1.5N_e}$ for sex chromosomes, $\frac{1}{N_{ef}}$ for mitochondrial DNA (which is maternally inherited—thus the subscript f). This tells us rates of genetic diversity loss will be greater in haploid than tetraploid organims, and in mtDNA than in sex chromosomes.

We can use the general relationship for diploid organisms above to predict the expected heterozygosity t generations in the future, which should take on a familiar form:

$$H_1 = (1 - \frac{1}{2N})H_0$$

$$H_2 = (1 - \frac{1}{2N})H_1 = (1 - \frac{1}{2N}) * (1 - \frac{1}{2N})H_0$$

$$H_2 = (1 - \frac{1}{2N})^2 H_0$$

$$\frac{H_2}{H_0} = (1 - \frac{1}{2N})^2$$

$$\frac{H_t}{H_0} = (1 - \frac{1}{2N})^t$$

$$H_t \sim e^{\frac{-t}{2N}}$$

Heterozygosity thus decays exponentially at a rate determined by the effective population size.

We can apply this equation to determine the remaining heterozygosity at a particular time given initial heterozygosity and the (effective) number of individuals in the population. For example, if initial heterozygosity is 0.6, our population size is 50, and we are interested in the heterozgosity remaining at generation $t = 20$:

$$H_t = 0.6e^{\frac{-20}{2*50}} = 0.4912$$

To determine the effective population size required to maintain a particular level of heterozygosity in the population, we can isolate the variable N using the natural logarithm. Let's say we want to figure out the minimum population size required to maintain 40% of heterozygosity over 10 years:

$$0.40 = \frac{H_t}{H_0} = (1 - \frac{1}{2N_e})^{10} \sim e^{\frac{-10}{2N_e}}$$

$$ln(0.40) = \frac{-10}{2N_e}$$

$$ln(0.40) = \frac{-10}{2N_e}$$

$$ln(0.40) * 2N_e = -10$$

$$2N_e = \frac{-10}{ln(0.40)}; Ne = \frac{-5}{ln(0.30)} = 5.45$$

As mentioned above, effective population size will be impacted by factors like variation in the size of successive generations, fluctuating population size, and unequal sex ratios. The amount of heterozygosity retained over t generations will be the product of the effective population size in each:

$$\frac{H_t}{H_0} = \prod_i^t (1 - \frac{1}{2N_{ei}})$$

For three populations of size 10, 500, and 200, this would be:

$$\frac{H_t}{H_0} = \prod_i^t (1 - \frac{1}{2N_{e_i}}) = (1 - \frac{1}{20})(1 - \frac{1}{1000})(1 - \frac{1}{400}) = 0.947$$

For fluctuating population sizes, N_e is calculated as a the harmonic mean of the population size at each of t timesteps:

$$N_e = \frac{1}{\frac{1}{t}\sum_i^t \frac{1}{N_i}}$$

Because the harmonic mean is more heavily impacted by small quantities than the arithmetic mean, N_e will be shaped by the minimum population sizes through time (which makes sense, given these miniature bottlenecks will lead to the biggest losses of heterozygosity).

It's worth looking at the derivation of effective population size under unequal sex ratios in detail. We'll start by considering the probability two alleles from different females are IBD from a female ancestor. This is equivalent to determining the loss of heterozygosity in females alone, and should be $\frac{1}{4}\frac{1}{2N_f}$. Here, $\frac{1}{4}$ refers to the probability (FF out of the possible combinations FF, FM, MF, and MM) both of selected individuals are females, while $\frac{1}{2N_f}$ refers to the probability the alleles are IBD in a female ancestor (this the numerator invoves the number of females—N_f—not overall population size). The loss of heterozygosity in males alone will have an identical form, giving us an overall expected loss of heterozygosity per generation given an unequal sex ratio:

$$\frac{1}{2N_e} = \frac{1}{4}\frac{1}{2N_f} + \frac{1}{4}\frac{1}{2N_e}$$

Isolating N_e is now just a matter of algebra:

$$\frac{1}{2N_e} = \frac{1}{8N_f} + \frac{1}{8N_m}$$

$$\frac{1}{2N_e} = (\frac{N_m}{N_m})\frac{1}{8N_f} + \frac{1}{8N_m}(\frac{N_f}{N_f}) = \frac{N_m}{8N_fN_m} + \frac{N_f}{8N_mN_f} = \frac{N_m + N_f}{8N_mN_f}$$

$$2N_e = \frac{8N_mN_f}{N_m + N_f}$$

$$N_e = \frac{4N_mN_f}{N_m + N_f}$$

For example, if we have 6 males and 2 females ($N = 8$ overall), N_e will be:

$$N_e = \frac{4 * 6 * 2}{6 + 2} = 6$$

i.e., 2 smaller than the census population size.

N_e in Conservation Biology

Because effective population size (N_e) reflects both future evolutionary potential and the risk of inbreeding depression, it is an important metric for estimating extinction risk in small populations. In a 2008 review paper, Friso Palstra and Daniel Rizzante (Palstra and Ruzzante (2008)) reviewed 83 studies that reported N_e from wild populations, finding an average value of $N_e = 260$ and an average effective to census population size ratio of $N_e/N = 0.14$. Estimates tended to be smaller in species of conservation concern, and in those with limited gene flow from other populations. These findings provide an empirical confirmation of N_e's value for conservation biology.

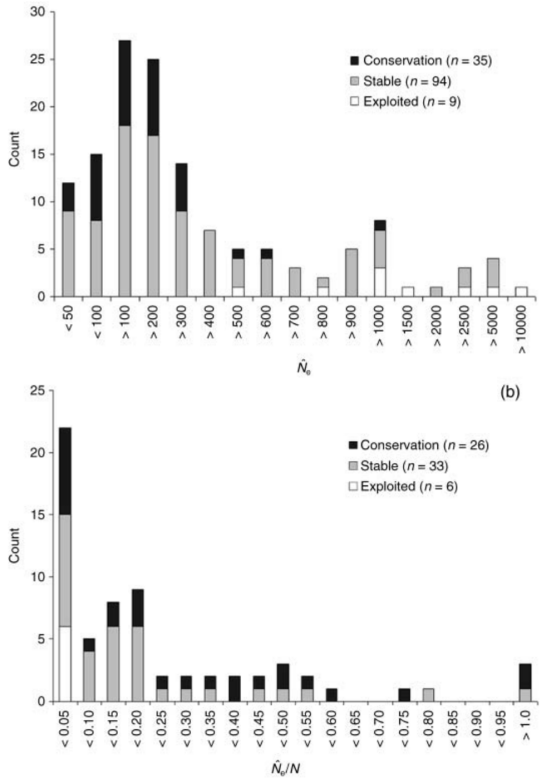


Figure 10.1: *Figures 2 from Palstra and Ruzzante, showing the distribution of binned N_e* an N_e/N estimates for species of different conservation status

11 The Coalescent

To conclude our unit on N_e, we will now briefly address how it is estimated with empirical data. A large number of methods exist, relying on data on the loss of heterozygosity in subsequent generations (discussed previously), the rate of decay in linkage disequilibrium, the loss of allelic diversity, and estimates of drift / mutation equilibrium (for long temporal time scales). Perhaps most useful is Watterson's estimator, which relies on a parameter θ ("theta"), defined as $\theta = 4N_e\mu$.

θ is something known as the population mutation parameter, which gives us the expected number of segregating (=different) alleles in a population of a given size and given mutation rate. To understand this, recall that any two alleles in a single generation are identical by descent in the previous generation at probability $\frac{1}{2Ne}$. This observation is the basis of a body of work known as coalescent theory, which is concerned with finding the time to the most recent common ancestor ("MRCA") of a pair of alleles. We can expand the probability of IBD to get at the probability any two alleles *coalesce* in t generations, which is simply the probability they *don't* coalesce in the previous generation multiplied by the probability of coalescence in the generation of interest:

$$P_{coalesce}(t) = (1 - \frac{1}{2N_e})^{t-1}(\frac{1}{2N_e})$$

This form should look familiar, as it can be approximated by the exponential distribution when N_e is sufficiently large:

$$P_{coalesce}(t) = \frac{1}{2N_e}e^{-\frac{t-1}{2N_e}}$$

This is helped because the expected value (think weighted average) of the exponential distribution is $\frac{1}{\lambda}$, where λ is the rate parameter of the distribution. In this case, that's $\frac{1}{2N_e}$, which means that the expected time to coalescence for any two alleles is $1 \div \frac{1}{2N_e} = 2N_e$ generations. The plot below shows the expected time to coalescence and probability of coalescence in each generation for populations with effective sizes $N_e = 50$, $N_e = 100$, and $N_e = 500$:

```r
library(ggplot2)
library(dplyr)

# define functions for different effective population sizes
coalesce_50 <- function(x){(1/(2*50))*exp(-(x-1)/(2*50))}
coalesce_100 <- function(x){(1/(2*100))*exp(-(x-1)/(2*100))}
coalesce_200 <- function(x){(1/(2*200))*exp(-(x-1)/(2*200))}

# create a data frame with values from all functions
x_vals <- seq(0, 500, by = 1)

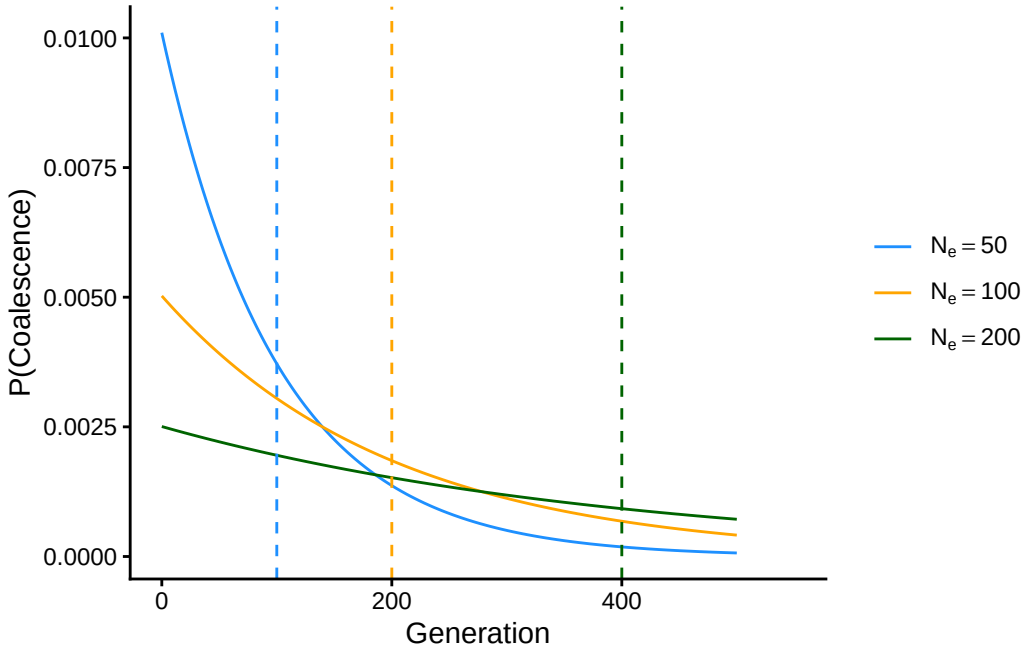
labels <- c("Ne=50", "Ne=100", "Ne=200")
levels_ordered <- factor(labels, levels = labels)

# create function data frame
df <- data.frame(
  Generation = rep(x_vals, 3),
  Probability = c(coalesce_50(x_vals), coalesce_100(x_vals), coalesce_200(x_vals)),
  Function = factor(rep(labels, each = length(x_vals)), levels = labels)
)

# create expected values data frame
vline_df <- data.frame(
  Function = levels_ordered,
  xintercept = c(100, 200, 400)
)

# plot
ggplot(df, aes(x = Generation, y = Probability, color = Function)) +
  theme_classic() +
  xlim(c(0, 550)) +
  geom_line() +
  geom_vline(data = vline_df,
             aes(xintercept = xintercept, color = Function),
             linetype = "dashed", show.legend = FALSE) +
  xlab("Generation") +
  ylab("P(Coalescence)") +
  scale_color_manual(
    values = c("Ne=50" = "dodgerblue",
               "Ne=100" = "orange",
               "Ne=200" = "darkgreen"),
    labels = c(
```

```
      expression(N[e] == 50),
      expression(N[e] == 100),
      expression(N[e] == 200)
    ),
    name = NULL  # Removes legend title
  )
```



For diploid individuals, the expected number of mutations on these haplotypes (or the expected number of mutant alles out of all $2N_e$) is simply this value multiplied by 2μ—the 2 here indicating ploidy, as each copy of the allele has a chance of mutating. Put this all together, and the expected number of segregating sites in any two randomly selected nonrecombining haplotypes is $\theta = 4N_e\mu$. In practice, θ is usually estimated with a metric called π (or θ_π), which is calculated as the average number of nucleotide differences per site in all possible pairs of the sample population. Given this, effective population size can be estimated from $\theta/4\mu$.

As a measure of genetic diversity, θ is closely related to heterozygosity. Using coalescent theory, we can define expected (or mean) heterozygosity as the probability of a mutation in a given generation divided by the total probability of an "event" happening in that generation (either mutation or coalescence):

$$\bar{H} = \frac{2\mu}{2\mu + \frac{1}{2N_e}} = \frac{2\mu}{\frac{4\mu N_e + 1}{2N_e}} = \frac{2\mu}{1} \cdot \frac{2N_e}{4\mu N_e + 1} = \frac{4\mu N_e}{4\mu N_e + 1}$$

Since $\theta = 4\mu N_e$, $\bar{H} = \frac{\theta}{\theta+1}$!

Watterson's θ in Conservation Biology

As with other correlates of effective population size, the population mutation parameter θ is a concise summary of genetic diversity in a population, and may thus reflect its census populaton size and demographic trajectory—in other words, its conservation status. Jong Yoon and colleagues (Jeon et al. (2024)) tested this hypothesis using publicly available whole genome sequences from 82 mammal species and IUCN conservation categories (Data Deficient, Non Threatened, and Threatened). They found "Threatened" species had significantly lower θ estimates than "Non-Threatened" species, suggesting genomic data may be useful for establishing risk even in the absence of other sources of information on population health.

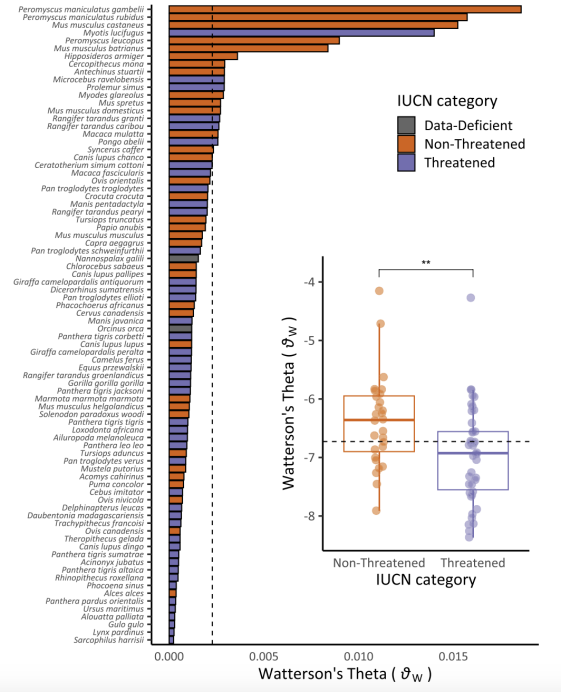


Figure 11.1: *Figure 2 from Jeon et al. 2024, showing estimates of Watterson's θ* across species, color-coded according to IUCN conservation status

12 Inbreeding and Inbreeding Depression

An individual is inbred when its parents share one or more common ancestors. Because inbreeding is unavoidable in small populations, it is a topic of major importance in conservation genetics. Inbreeding reduces reproductive fitness by increasing homozygosity (though more identical by descent pairings), which exposes deleterious recessive alleles. We measure inbreeding with the inbreeding coefficient F, which scales from 0 (completely outbred) to 1 (completely inbred).

We can start with a simple model of self fertilization in a diploid lineage, where a parent (with genotype A_1A_2) produces gametes in proportions $\frac{1}{2}A_1$ and $\frac{1}{2}A_2$. There is thus a $\frac{1}{2} * \frac{1}{2}$ probability its offspring will be homozygous for A_1, a $\frac{1}{2} * \frac{1}{2}$ probability it will be homozygous for A_2, and a $\frac{1}{2} * \frac{1}{2} + \frac{1}{2} * \frac{1}{2} = \frac{1}{2}$ probability it will be heterozygous (either A_1A_2 or A_2A_1). Its inbreeding coefficient is the summed probabilities both its alleles are IBD, which in this case is equal to the sum of the probabilities of both homozygous genotypes, as there is no other ancestral source of alleles: $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

We can expand this model to two unrelated grandparents producing two children who have a child with each other (so-called "full sib" mating). Labeling each grandparental allele independently regardless of state to indicate descent gives us genotypes A_1A_2 and A_3A_4. The child's inbreeding coefficient is the sum of the probability its two alleles are derived from the same grandparental copy. For example, there is a $\frac{1}{2}$ probability the first grandparent passes on A_1 to the first parent, and a $\frac{1}{2}$ probability that same copy is passed on to their child, which we multiply for a $\frac{1}{4}$ overall probability of inheriting the first copy of A_1. Similarly, the probability of the first grandparent passing A_1 on to the second parent and that second parent passing it on to their child is also $\frac{1}{4}$.

We thus end up with a $\frac{1}{16}$ the child of a full sib mating event to be homozygous for A_1 with alleles that are identical by descent. Their full inbreeding coefficient is the sum of all such possible IBD combinations, which in this case is $F = \frac{1}{16}(\text{ for } A_1A_1) + \frac{1}{16}(\text{ for } A_2A_2) + \frac{1}{16}(\text{ for } A_3A_3) + \frac{1}{16}(\text{ for } A_4A_4) = \frac{4}{16} = \frac{1}{4}$

The calculation above is based on finding a "loop" in a pedigree: a route where a single allele copy can be passed on through descendants before combining (to become identical by descent) in a common ancestor. This forms the basis for calculating the pedigree inbreeding coefficient, F_P (or F_X, where X is an individual on the pedigree). The inbreeding coefficient was derived by Sewell Wright (1889 - 1988), who you will recall was one of the three major founders of population genetics. One of Wright's major contributions is the path coefficient, which is a method for assessing the correlation between two nodes on a directional graph. In conservation

genetics, this lets us determine the proportion of ancestry passed on from an ancestor to an offspring n generations apart, while accounting for inbreeding (more on that in a second). Path coefficients can range from from $\frac{1}{2}$ (e.g., father / mother to child) to $\to 0$ (e.g., the common ancestor of all humanity to you):

$$P_{AO} = 2^{-n} \cdot \sqrt{\frac{1+f_A}{1+f_O}}$$

For example, grandparents contribute an average of 1/4th of their genes to their grandchildren ($n = 2$ generations apart), assuming no inbreeding:

$$P_{AH} = 2^{-2} \cdot \sqrt{\frac{1+0}{1+0}} = \frac{1}{4} \cdot 1 = \frac{1}{4}$$

Path coefficients can be used to determine the *coefficient of relationship*, or the proportion of genetic material two individuals B and C share given each common ancestor they share (here denoted by A_i). Here, we keep track of the number of ancestors between B and A with n, and C and B with m:

$$r_{BC} = \sum_i P_{A_iB} \cdot P_{A_iC} = \sum_i 2^{-n_i} \cdot \sqrt{\frac{1+f_{A_i}}{1+f_B}} \cdot 2^{-m_i} \cdot \sqrt{\frac{1+f_{A_i}}{1+f_C}}$$

$$r_{BC} = \sum_i (\frac{1}{2})^{(n_i+m_i)} \frac{1+f_{A_i}}{\sqrt{(1+f_{B_i})(1+f_{C_i})}}$$

For example, the relationship coefficient between two non-inbred siblings will be the product of the path coefficients between each child and their father plus the product of the path coefficients of each child and their mother. Since f is 0 in all cases, the fraction on the right side reduces to 1, and we only need to worry about the first half of the equation (i.e., $r_{BC} = \sum_i (\frac{1}{2})^{(n_i+m_i)}$):

$$r_{DE} = P_{AD} \cdot P_{AE} + P_{BD} \cdot P_{BE} = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$$

In his original paper, Wright (1922) defines the inbreeding coefficient as $f_x = \frac{1}{2} r_{SD} \sqrt{(1+f_S)(1+f_D)}$. Here, $\frac{1}{2}$ represents the probability an ancestor contributes IBD alleles to the next generation ($p(A_1A_1) = 0.5 * 0.5 = 0.25 + p(A_2A_2) = 0.5 * 0.5 = 0.25 = 0.5$)). Intuitively, this probabilty is then influenced by the relationship coefficient, r_{SD}, and the geometric mean of the inbreeding coefficients of the parents. In a selfing "population" of $n = 1$, this term reduces to $f_x = \frac{1}{2} * 1\sqrt{(1+1)(1+1)} = \frac{1}{2}$, i.e., exactly the probability of selecting alleles that are IBD in the parental generation. In a population of unrelated, sexually reproducing organisms, $f_x \sim 0$.

The probability two alleles are IBD decays as the two parents (focal individuals; Wright's "sire" and "dam", S and D) become more distantly related. If their common ancestor is not inbred, and they themselves are not inbred, the inbreeding coefficient of a full-sib mating event is equal to $f_x = \frac{1}{2} * \frac{1}{2} * \sqrt{(1+0)(1+0)} = \frac{1}{4}$

Here is a plot of how inbreeding increases along with the relationship coefficient for different levels of parental inbreeding:

```
library(ggplot2)
library(dplyr)

# define functions for different effective population sizes
r_low <- function(x){0.5*x*sqrt((1+0)*(1+0))}
r_mod <- function(x){0.5*x*sqrt((1+0.1)*(1+0.1))}
r_high <- function(x){0.5*x*sqrt((1+0.25)*(1+0.5))}

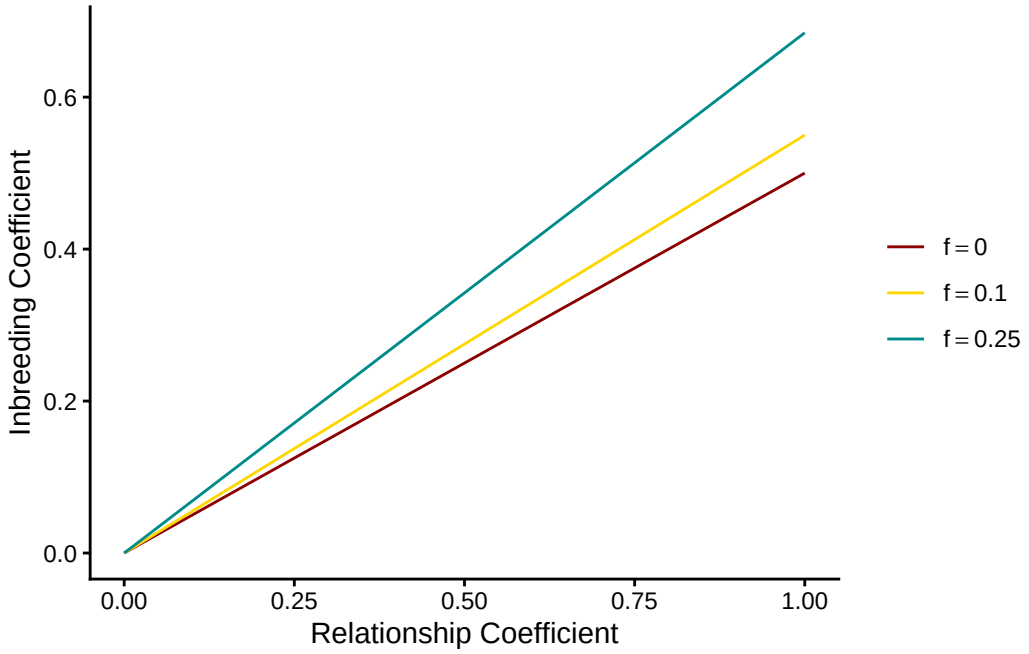
# create a data frame with values from all functions
x_vals <- seq(0, 1, by = 0.05)

labels <- c("f=0", "f=0.1", "f=0.25")
levels_ordered <- factor(labels, levels = labels)

# create function data frame
df <- data.frame(
  relationship = rep(x_vals, 3),
  inbreeding = c(r_low(x_vals), r_mod(x_vals), r_high(x_vals)),
  Function = factor(rep(labels, each = length(x_vals)), levels = labels)
)

# plot
ggplot(df, aes(x = relationship, y = inbreeding, color = Function)) +
  theme_classic() +
  xlim(c(0,1)) +
  geom_line() +
  xlab("Relationship Coefficient") +
  ylab("Inbreeding Coefficient") +
  scale_color_manual(
    values = c("f=0" = "red4",
               "f=0.1" = "gold1",
               "f=0.25" = "darkcyan"),
    labels = c(
      expression(f == 0),
      expression(f == 0.1),
```

```
      expression(f == 0.25)
    ),
    name = NULL  # Removes legend title
  )
```



(Note that increased inbreeding in the parents increases the slope of the relationship, i.e. the correlation between two alleles in a descendant after controlling for their own relationship.)

To derive a general expression for F that takes into account only the ancestry / path coefficients to each ancestor capable of contributing an allele that could be IBD in a descendent, Wright substitutes his expression for the relationship coefficient into the above equation:

$$F_X = \frac{1}{2} * \sum_i (\frac{1}{2})^{(n_i+m_i)} \frac{1+f_{A_i}}{\sqrt{(1+f_{B_i})(1+f_{C_i})}} * \sqrt{(1+f_{B_i})(1+f_{C_i})}$$

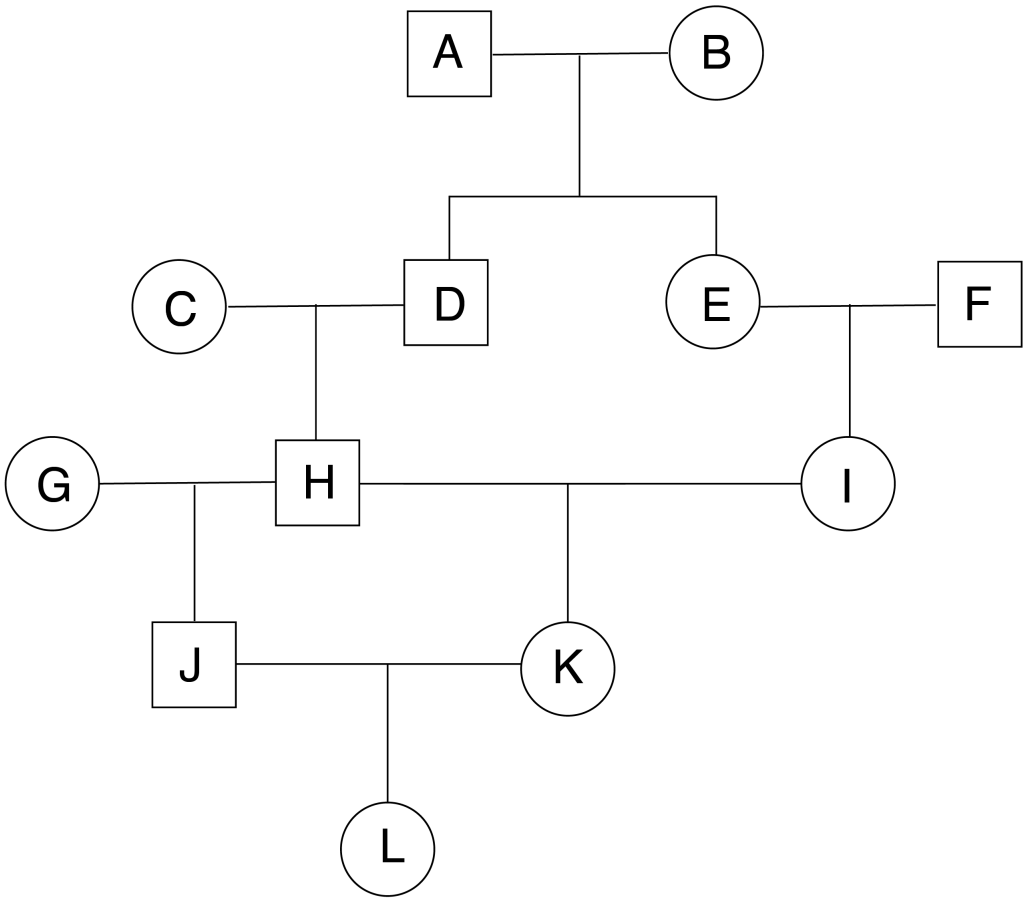
This unwieldy equation simplifies nicely:

$$F_X = \sum \frac{1}{2}^{n_i+m_i+1} \cdot (1+F_A)$$

In this equation, n_i refers to the number of generations between X and the common ancestor through one side of the pedigree, while m_i comes from the number of generations between X and the common ancestor through the *other* side of the pedigree. (The "+1" comes from

multiplying $\frac{1}{2}^{(n_i+m_i)} \cdot \frac{1}{2}^{(n_i+m_i)}$—this is the same as the increase in the exponent in $2^1 \cdot 2^2 = 2^3$. You may find it easier to think of the entire exponent as $n-1$, which is simply the number of links in the entire loop from X to a common ancestor and then back the other side. We'll use this notation for the rest of the class.)

This coefficient is then multiplied by 1 + the inbreeding coefficient of the common ancestor (F_A) and then summed over all loops by which an allele could be IBD. For example, consider the following pedigree, where squares indicate males, circles indicate females, and lines between males and females lead to offspring in the next generation (i.e., further down the figure):



We can calculate the inbreeding coefficient of individual K as follows:

$$F_K = \sum \frac{1}{2}^{n-1} \cdot (1 + F_A) = \frac{1}{2}^{5} \cdot (1 + 0) + \frac{1}{2}^{5} \cdot (1 + 0) = 0.0625$$

Note that there are two routes in which an allele can be IBD in K (highlight the common ancestor of the two copies with a bar): $HD\bar{A}EI$ and $HD\bar{B}EI$, accounting for an origin in her great-grandfather or great-grandmother, respectively. A slightly more complicated situation is calculating the inbreeding coefficeint for L:

$$F_L = \sum \frac{1}{2}^{n-1} \cdot (1 + f_A) = \frac{1}{2}^{7} \cdot (1 + 0) + \frac{1}{2}^{7} \cdot (1 + 0) + \frac{1}{2}^{3} \cdot (1 + 0) = 0.14063$$

Now we have loops $JHD\bar{A}EIK$, $JHD\bar{B}EIK$, and JHK. However, there is still no inbreeding among potential common ancestors, so F_A remains 0. It can helpful to organize these information into a table, especially as the number of loops grows:

Loop	$n-1$	F_{CA}
$JHD\bar{A}EIK$	7	0
$JHD\bar{B}EIK$	7	0
JHK	3	0

i Pedigree-based Inbreeding Coefficients in Arctic Fox *Vulpes lagopus*

Inbreeding coefficients originated in the field of animal breeding, where pedigrees are typically known with certainty. In all but a handful of species, however, such detailed kinship data are absent or difficult to collect. Assessments of impact of pedigree-based inbreeding coefficients on fitness in wild populations are thus rare and valuable confirmations of theory. In a 2016 paper in *Molecular Ecology*, Karin Norén and colleagues (Norén et al. (2016)) provide one such example from a 9-year study of a subpopulation of Scandinavian Arctic Foxes. They showed a 10-fold increase in the average inbreeding coefficient in the population. Highly inbred individuals were less likely to survive in low-resource years, and less likely to mate in high-resource years.

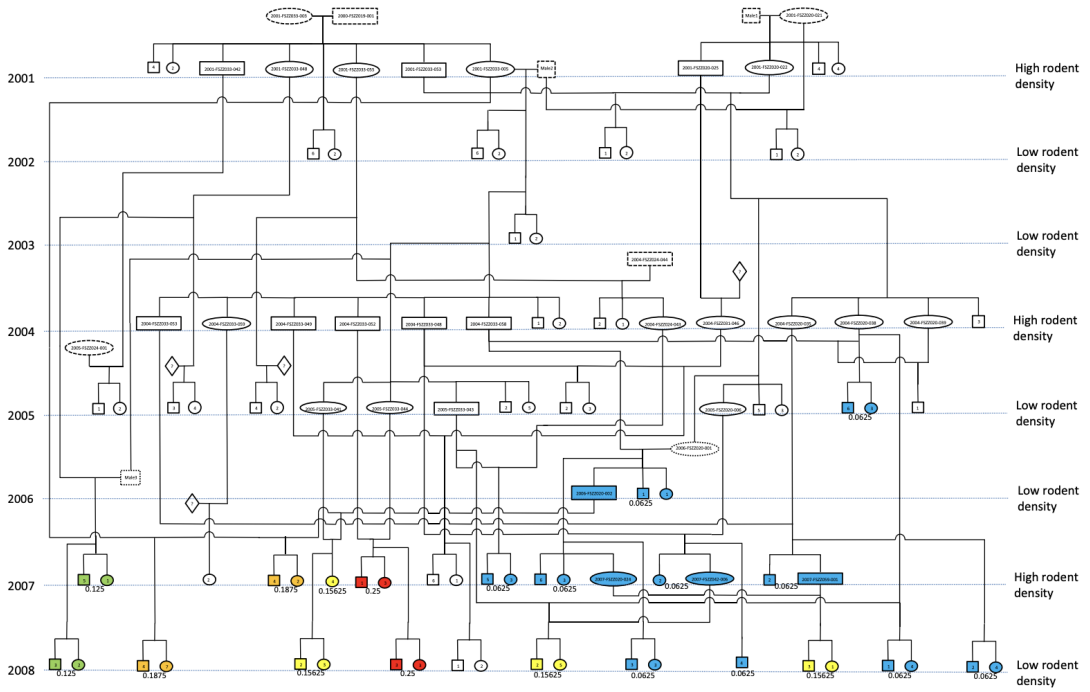


Figure 12.1: *Figure 1 from Norén et al. 2016. This pedigree, which was verified genetically, also shows resource abundance from 2001 to 2008. Individuals in 2007 and 2008 are color-coded by inbreeding coefficient (blue for low values, green for moderate values, orange or red for higher values).*

12.1 Inbreeding without pedigrees

In most cases we will have no pedigree to use to calculate F. We therefore need a way to infer inbreeding from genotype frequency data. Because F is equivalent to the probability two alleles are identical by descent, the genotype frequency of heterozygotes in a fully inbred population is 0 (0% * $2pq$), while the frequency of each homozygote is equivalent to its respective allele frequency, as the probability of an individual being homozygous is 100% once you account for the frequency of the allele in the population. A partially inbred population will see a reduction in heterozygosity proportional to F, leaving a frequency of $2pq$ heterozygotes multiplied by the complement of the inbreeding coefficient. The frequency of the two homozygotes will be a mixture of the predicted frequencies under random mating and those under full inbreeding, weighted by the inbreeding coefficient:

	F	A_1A_1	A_1A_2	A_2A_2
random mating	0	p^2	$2pq$	q^2
fully inbred	1	p	0	q
partially inbred	0<F<1	$p^2(1-F)+p*$ $1*F=p^2+Fpq$	$2pq(1-F)$	$q^2(1-F)+q*$ $1*F=q^2+Fqp$

(Note that I have shortened the derivations for $p^2(1-F)+p*1*F=p^2-Fp^2+pF=p^2+Fp(1-p)=p^2+Fpq$ and $q^2(1-F)+q*1*F=q^2(1-F)+q*1*F=q^2-Fq^2+qF=q^2+Fq(1-q)=q^2+Fqp$.)

We can therefore understand the reduction in heterozygosity of an inbred population as $\frac{H_{inbred}}{H_e}=\frac{2pq(1-F)}{2pq}=1-F$. Simplifying, we get:

$$\frac{H_{inbred}}{H_e}+F=1$$
$$F=1-\frac{H_{inbred}}{H_e}$$
$$F=1-\frac{H_o}{H_e}$$

In other words, the inbreeding coefficient is equal to 1 minus the ratio of observed to expected heterozygosity.

Lastly, we derive a model for relating the inbreeding coefficient to population size to predict the increase in F across generations. Recall that F is defined as the probability of sampling two alleles that are identical by descent. This can happen two ways: by new inbreeding (at rate $\frac{1}{2N}$), and by sampling alleles that are already IBD from previous inbreeding (at frequency $(1-\frac{1}{2N})F_{t-1}$, where F_{t-1} is the inbreeding coefficient of the parental generation):

$$F_t=\frac{1}{2N}+(1-\frac{1}{2N})F_{t-1}$$

This tells us that the increase in the inbreeding coefficient each generation is $\Delta F=\frac{1}{2N}$: equivalent to the *loss* of homozygosity each generation. From this, we can develop a general relationship between expected inbreeding at generation t and the initial inbreeding coefficient:

$$F_t=1-[(1-\frac{1}{2N})^t(1-F_0)]$$

You should once again notice our model for the exponential decay of heterozygosity through time embedded within, and recall that $1-F_0$ is the reduction in in heterozygosity below expected values under random mating caused by inbreeding. F therefore grows larger with smaller population sizes, greater initial inbreeding values, and a greater number of generations.

12.2 Inbreeding depression

In a vaccuum, inbreeding itself would not be particularly interesting. It is its effects on expected trait values—particularly fitness traits—that make it important for conservation biologists to consider. We now develop a model to understand its impacts on mean trait values using simple quantitative genetics. Remember that $+a$ is the deviation of the trait value for A_1A_1 away from the mean of the two homozygotes, d is the trait value deviation for the heterozygote, and $-a$ is deviation of the trait value for A_2A_2. The average trait value for the population will require weighting these deviations by the frequency of the genotype they are associated with. The table below lays out these weightings for different mating schemes:

	A_1A_1	A_1A_2	A_2A_2
random mating	p^2a	$2pqd$	$-q^2a$
fully inbred	pa	0	$-qa$
partially inbred	$p^2a + Fpqa$	$2pqd(1-F)$	$-q^2a - Fqpa$

From this, we can determine that the mean trait value deviation of a randomly mating population (M_0) from the midpoint of the two homozygotes is:

$$M_O = p^2a + 2pqd - q^2a = p^2a - q^2a + 2pqd = a(p^2 - q^2) + 2pqd$$
$$M_O = a(p+q)(p-q) + 2pqd = a(p-q) + 2pqd$$

(Note the substitution of 1 for $p + q$.)

Similarly, the mean trait value deviation of an inbred population (M_F) from the midpoint of the two homozygotes is:

$$M_F = p^2a + Fpqa + 2pqd(1-F) - q^2a - Fpqa = p^2a - q^2a + Fpqa - Fpqa + 2pqd - 2pqdF$$
$$M_F = a(p^2 - q^2) + 2pqd - 2pqdF = a(p+q)(p-q) + 2pqd - 2pqdF = a(p-q) + 2pqd - 2pqdF$$

Because $a(p-q)+2pqd = M_O$, $M_F = M_O - 2pqdF$. In this equation, the term $2pqdF$ represents inbreeding depression, which we denote as δ. Inbreeding depression is thus the degree to which the average trait value of the population is depressed below random mating expectations.

This tells us a few other things too:

1) If there is no dominance (i.e., $d = 0$, there can be no inbreeding depression;
2) If there is no inbreeding (i.e., $F = 0$), there can be no inbreeding depression;
3) The degree of inbreeding depression (δ) is a linear function of F;
4) We can estimate δ as $1 - \frac{\text{fitness of inbred offspring}}{\text{fitness of outbred offspring}}$.

H_o-based estimates of inbreeding correlate with sperm quality in mammals

The difficulty of obtaining detailed pedigrees for endangered species has resulted in a heavy reliance on heterozygosity reduction-based estimates of F and d in empirical conservation genetics. Yet variation in H_o across loci—and the limited number of loci in many studies, particularly before the current genomic era—has led to questions as to the reliability of these estimates. Fitzpatrick and Evans (Fitzpatrick and Evans (2009)) aggregated data on heterozygosity and sperm characteristics for 20 mammal species, 11 of which were threatened or endangered. They found the highest proportion of sperm abnormalities (and lowest sperm motility) in those species with the lowest heterozygosity. This result supports the reliability of $F = 1 - \frac{H_o}{H_e}$ for real-life data, and confirms that depressed sperm quality is a common form of inbreeding depression in mammals.

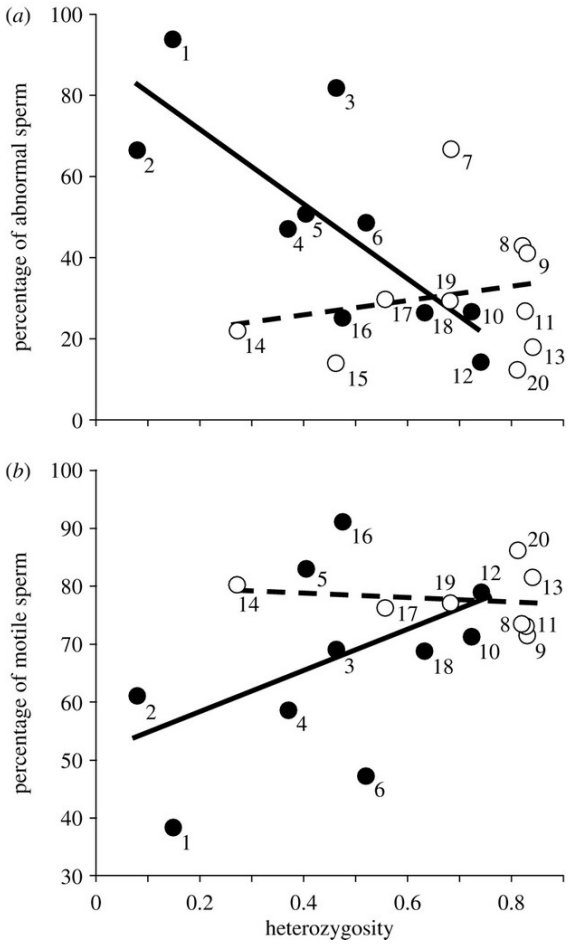


Figure 12.2: *Figure 1 from Fitzpatrick and Evans, showing the relationship between* H_o and sperm quality (a) or sperm mobility (b). Closed circles are non-endangered species; open circles are endangered or threatened. The dashed regression line reveals that the difference between threat categories—not among all individuals—drives the trend, something expected if heterozygosity loss was driven by population bottlenecks rather than naturally low N_e.

13 Population Subdivision

Up until this point, our models have focused on the dynamics of a group of individuals that mate randomly with respect to a locus of interest—a genetic **population**, or deme. Within species, however, geography, ecology, and happenstance frequently produce complex patterns of nonrandom mating among individuals. A simple natural example of this phenomenon might be flowering plants arranged on either side of a large river. While pollinators occasionally cross this barrier to exchange genes between geographically distant individuals, mating is far more likely among those found on the same bank. In conservation genetics, habitat fragmentation, road construction, and other sources of anthropogenic change can have similar impacts on mating probabilities and gene flow. The effect is to generate **population genetic structure**, also referred to as **population subdivision**.

We build off our understanding of inbreeding depression to model population subdivision. Consider two populations: in then first, we find 324 A_1A_1 individuals, 72 A_1A_2 individuals, and 4 A_2A_2 individuals. In the second, we find 36 A_1A_1 individuals, 168 A_1A_2 individuals, and 198 A_2A_2 individuals. In population 1, $f(A_1) = 720/800 = 0.9$ and $f(A_2) = 80/800 = 0.1$. In population 2, $f(A_1) = 240/800 = 0.3$ and $f(A_2) = 560/800 = 0.7$.

Are these populations in HWP? Yes: In population 1, expected frequencies are $f(A_1A_1) = 400{*}0.9^2 = 324$, $f(A_1A_2) = 400{*}2{*}0.9{*}0.1 = 72$, and $f(A_2A_2) = 400{*}0.1^2 = 4$. In population 2, expected frequencies are $f(A_1A_1) = 400 * 0.3^2 = 36$, $f(A_1A_2) = 400 * 2 * 0.3 * 0.7 = 168$, and $f(A_2A_2) = 400 * 0.7^2 = 196$. But the situation changes if we lump both together and treat them as a single, putatively randomly mating population. Now, $f(A_1) = 0.6$ and $f(A_2) = 0.4$ overall, leading us to expected frequencies of $f(A_1A_1) = 800 * 0.6^2 = 288$ (fewer than the 360 observed), $f(A_1A_2) = 800 * 2 * 0.6 * 0.4 = 384$ (greater than the 242 observed), and $f(A_2A_2) = 800 * 2 * 0.4 * 0.4 = 128$ (fewer than the 202 observed).

This observed excess of homozygotes and observed deficit of heterozygotes is known as the Wahlund Effect, and is an analog to the defecit of heterozygosity seen in inbred populations. In subdivided populations—the norm in conservation biology—we expect average genotype frequencies to diverge from HWP as they do under inbreeding. This leads us to a so-called "family" of *F*-statistics. The first we have already covered in depth: the inbreeding coefficient is defined as the ratio of observed hetrozygosity to expected heterozygosity:

$$F_{IS} = 1 - \frac{H_o}{H_e}$$

Next we have Wright's fixation index, or F_{ST}, which is the ratio of the expected heterozygosity of each subpopualtion averaged to the expected heterozygosity based on the average allele frequencies of all populations lumped together (the S in H_S stands for "subdivided", while the T in H_T stands for "total"):

$$F_{ST} = 1 - \frac{H_S}{H_T}$$

F_{ST} for our example above is therefore $1 - \frac{242}{384} = 0.3697$: "significant differentiation", according to the commonly applied benchmark of $F_{ST} = 0.15$.

(The third major statistic in the family, F_{IT}, is not covered in class.)

Interestingly, we can model the impacts of population subdivision on expected allele frequencies either as a consequence of inbreeding or as a consequence of drift (recall that $\frac{pq}{2N}$ is the variance of the binomial distribution):

genotype	frequency under inbreeding	frequency under drift
A_1A_1	$p^2 + Fpq$	$p^2 + \frac{pq}{2n}$
A_1A_2	$2pq(1-F)$	$2pq - 2(\frac{pq}{2n})$
A_2A_2	$q^2 + Fpq$	$q^2 + \frac{pq}{2n}$

Here, you should see that the change in homozygosity is $2 * pq * \frac{1}{2N}$—in other words, 2 times the product of allele frequencies multipled by the per-generation loss of heterozygosity. Correspondingly, the increase in frequency for both p^2 and q^2 leads to a *a loss* of homozygosity of $2 * pq * \frac{1}{2N}$*. ($\frac{pq}{2N}$ is the formula for the variance after a single generation; we can refer to the variance after any number of generations more generally by σ^2)

We can understand the relationship with a simple example. Given allele frequencies $f(A_1) = 0.7$ and $f(A_2) = 0.3$, $H_e = 2 * 0.7 * 0.3 = 0.42$ and an inbreeding coeficient of $F = 0.64$, heterozygosity is reduced to $2pq(1-F) = 0.42(1-0.64) = 0.15$. Approaching the same scenario through the lens of drift, we would first need to determine the number of generations required to achieve an inbreeding coefficient of 0.67 in our model of decay of heterozygosity through time. We will imagine there are $N = 10$ individuals:

$$F = 1 - (1 - \frac{1}{2N})^t$$

$$0.64 = 1 - (1 - \frac{1}{2 * 10})^t$$

$$0.64 = 1 - (0.95)^t$$

$$1 - 0.64 = 0.95^t;\ 0.358 = 0.95^t$$

$$log(0.358) = t * log(0.95)$$

$$\frac{log(0.358)}{log(0.95)} = t \sim 20 \text{ generations}$$

Variance in allele frequencies after 20 generations will therefore be $\sigma^2 = pq(1 - (\frac{1}{2N})^{20}) = 0.7 * 0.3 * (1 - (\frac{1}{20})^{20}) = 0.135$. The expected heterozygosity is thus $2pq - \sigma^2$: $2 * 0.7 * 0.3 - 2 * 0.135 = 0.15$. In other words, the same answer, regardless of how you model it!

In the absence of natural selection, population subdivision reflects a balance between genetic drift and migration. A second important F_{ST} estimator is derived on the basis of this observation. We begin by noting that the probability two alleles are identical by descent is equal to the product of the alleles are IBD in the focal population, weighted by the proportion of alleles that are not migrants ($(1-m)^2$, the exponent reflecting the probability of not drawing a migrant allele twice). Because migrant alleles will by definition not be IBD with alleles in the focal population, this will reduce F_t below expectations in the absence of migration:

$$F_t = [\frac{1}{2N} + (1 - \frac{1}{2N})F_{t-1}](1-m)^2$$

This simplifies to the following expression:

$$F_t = [\frac{1}{2N} + F_{t-1} - \frac{F_{t-1}}{2N}](1-m)^2$$

At an equilibrium between drift and migration, the probability of two alleles being IBD will not change between generations ($F_t = F_{t-1}$):

$$\hat{F} = [\frac{1}{2N} + \hat{F} - \frac{\hat{F}}{2N}](1-m)^2$$

We can now distribute $(1-m)^2$ across the terms in brackets:

$$\hat{F} = \frac{(1-m)^2}{2N} + \hat{F}(1-m)^2 - \frac{\hat{F}(1-m)^2}{2N}$$

We continue simplifying:

$$\hat{F} - \hat{F}(1-m)^2 + \frac{\hat{F}(1-m)^2}{2N} = \frac{(1-m)^2}{2N}$$

$$\hat{F}(1 - (1-m)^2 + \frac{(1-m)^2}{2N}) = \frac{(1-m)^2}{2N}$$

$$\hat{F}(1 - (1-m)^2(1 - \frac{1}{2N})) = \frac{(1-m)^2}{2N}$$

$$\hat{F} = \frac{(1-m)^2}{2N} \frac{1}{(1-(1-m)^2(1-\frac{1}{2N}))}$$

$$\hat{F} = \frac{(1-m)^2}{2N - 2N(1-m)^2(1-\frac{1}{2N})}$$

$$\hat{F} = \frac{(1-m)^2}{2N - (1-m)^2(2N-1)}$$

$$\hat{F} = \frac{(1-2m+m^2)}{2N - (1-m)^2(2N-1)}$$

$$\hat{F} = \frac{(1-2m+m^2)}{2N - (1-2m+m^2)(2N-1)}$$

Because m^2 is likely to be extremely small, we can ignore it in both the numerator and the denominator:

$$\hat{F} = \frac{(1-2m)}{2N - (1-2m)(2N-1)}$$

$$\hat{F} = \frac{(1-2m)}{2N - 2N + 4Nm + 1 - 2m}$$

By similar logic, we can assume $2m$ is very small and ignore it as well, which brings us to our final equation:

$$F_{ST} = \frac{1}{4Nm+1}$$

For example, if have a population of size 50 and a migration rate of 0.1, we expect an equilibirum level of differentiation of $F_{ST} = \frac{1}{4*50*0.1+1} = 0.047$.

i F_{ST} in alpine plants

The inverse relationship between F_{ST} and the product of N_e in the source population and the migration rate m leads to the expectation that populations that are more geographically isolated from each other—such as those greater distances apart—will have higher F_{ST} values. Ægisdóttir and colleagues (Ægisdóttir, Kuss, and Stöcklin (2009)) tested this prediction (of what is known as the "isolation by distance" hypothesis) in 32 populations of the alpine perennial *Campanula thyrsoides* across Switzerland. Genotyping 736 individuals from 32 populations at 5 microsatellite loci, they found that pairwise F_{ST} was indeed correlated with linear geographic distance.

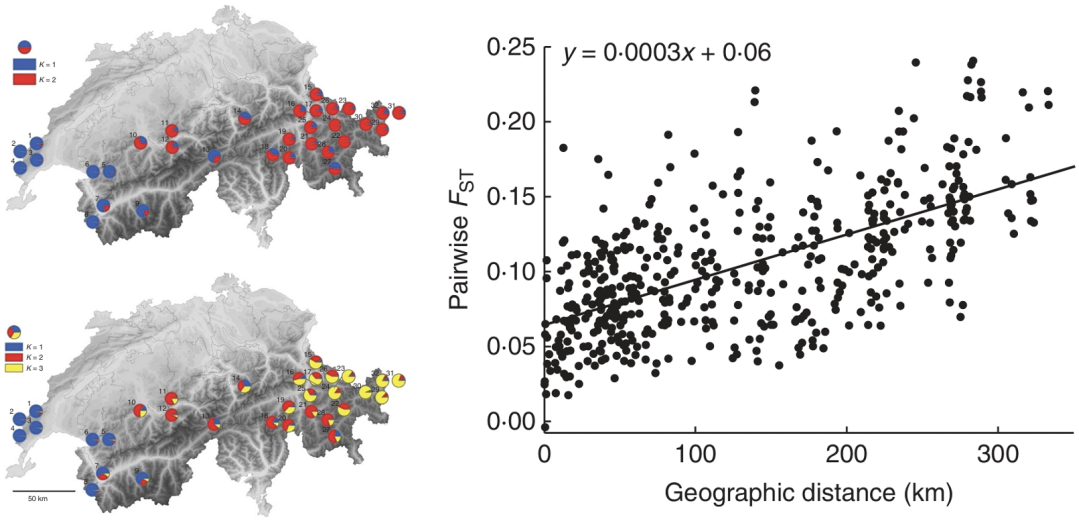


Figure 13.1: *Parts of Figure 1 and Figure 2 from Ægisdóttir et al. 2009. The lefthand pane shows the assignment of individuals to a model with two or three hypothetic populations; note the geographic coherence of either arrangement. The righthand pane shows the correlation between pairwise* F_{ST} and geographic distance in the same set of samples.

14 Genetic Viability

A fundamental question in conservation genetics is how big a population has to be to maintain viability into the future. The Australian geneticist I.R. Franklin took an early but influential stab at addressing it with his 50/500 Rule, which posits $N_e = 50$ is required to avoid significant fitness loss due to inbreeding depression in the short term (~5 generations), while $N_e = 500$ is required to maintain long-term evolutionary potential.

His derivation of $N_e = 50$ was fairly simple. Evidence from animal breeders suggested that keeping F below 0.05 would be sufficient to avoid impactful inbreeding depression. Franklin then solved for N_e using the equation relating inbreeding depression to population size and number of generations:

$$F = 1 - (1 - \frac{1}{2N_e})^t$$

$$0.05 = 1 - (1 - \frac{1}{2N_e})^5$$

$$0.95 = (1 - \frac{1}{2N_e})^5 \sim e^{\frac{-5}{2N_e}}$$

$$ln(0.95) \sim \frac{-5}{2N_e};\ 2N_e = \frac{-5}{ln(0.95)}$$

$$N_e = \frac{-2.5}{ln(0.95)} \sim 49$$

(50 was catchier!)

Franklin's derivation of 500 was slightly more complicated. He began with the assumption that to maintain evolutionary potential, the loss of additive genetic variance (the genetic variance most important for responding to selection) must be balanced out by its input by mutation:

$$\Delta V_A = V_M - (\frac{V_A}{2N_e})$$

Here, V_M represents the mutation rate, while the second term is the additive variance analog of $\frac{pq}{2N_e} = \sigma^2$ (i.e., the variance in allele frequencies due to drift). At equilibrium, $\Delta V_A = 0$, so $V_M = \frac{V_A}{2N_e}$. Therefore:

$$2N_eV_M = V_A$$
$$2N_e = \frac{V_A}{V_M}$$
$$N_e = \frac{V_A}{2V_M}$$

The required effective population size to maintain V_A therefore depends on the existing additive genetic variance and the rate at which it is replenished: the higher the latter, the lower the necessary V_M. At the time of the derivation, the best estimate of a mutation rate for V_M came from *Drosophila* bristles, and was $10^{-3} * V_E$, where V_E is the environmental variance (i.e., mutation adds roughly $\frac{1}{1000}$ the variance the environment does). This estimate gives us the following relationship:

$$N_e = \frac{V_A}{2 * 0.001 * V_E} = 500\frac{V_A}{V_E}$$

Next, Franklin took into account heritability (h^2), which is equal to the proportion of a phenotype that is due to additive genetic variance. Since phenotype is roughly equivalent the sum of the additive genetic variance that contributes to it and the envionmental variance that contributes to it, $h^2 \sim \frac{V_A}{V_A+V_E}$:

$$\frac{h^2}{1} \sim \frac{V_A}{V_A + V_E}$$
$$\frac{h^2}{1-h^2} \sim \frac{V_A}{V_E}$$

He then substituted this relationship into his previous work, and finally solved for N_e using an estimate of bristle heritability ($h^2 = 0.5$):

$$N_e = 500\frac{V_A}{V_E} = 500\frac{h^2}{1-h^2} = 500\frac{0.5}{1-0.5} = 500$$

Thus—very crudely—an effective population size of 500 is required to maintain long-term evolutionary potential.

How many species meet the 50 / 500 rule?

Despite its strong assumptions, the 50 / 500 rule sets relatively modest conservation targets. In a 2024 paper, Shannon Clarke and coauthors assembled a dataset of georeferenced N_e estimates from 3,829 populations across diverse taxonomic groups. They found that plant, mammal and amphibian populations had a <54% probability of meeting the N_e=50 rule,

and a <9% probability of meeting the N_e=500 rule; broadly speaking, lower N_e was correlated with a higher human footprint. However, the extent to which low N_e reflects contemporary threatened status—versus long-term patterns due to life history—remains unknown.

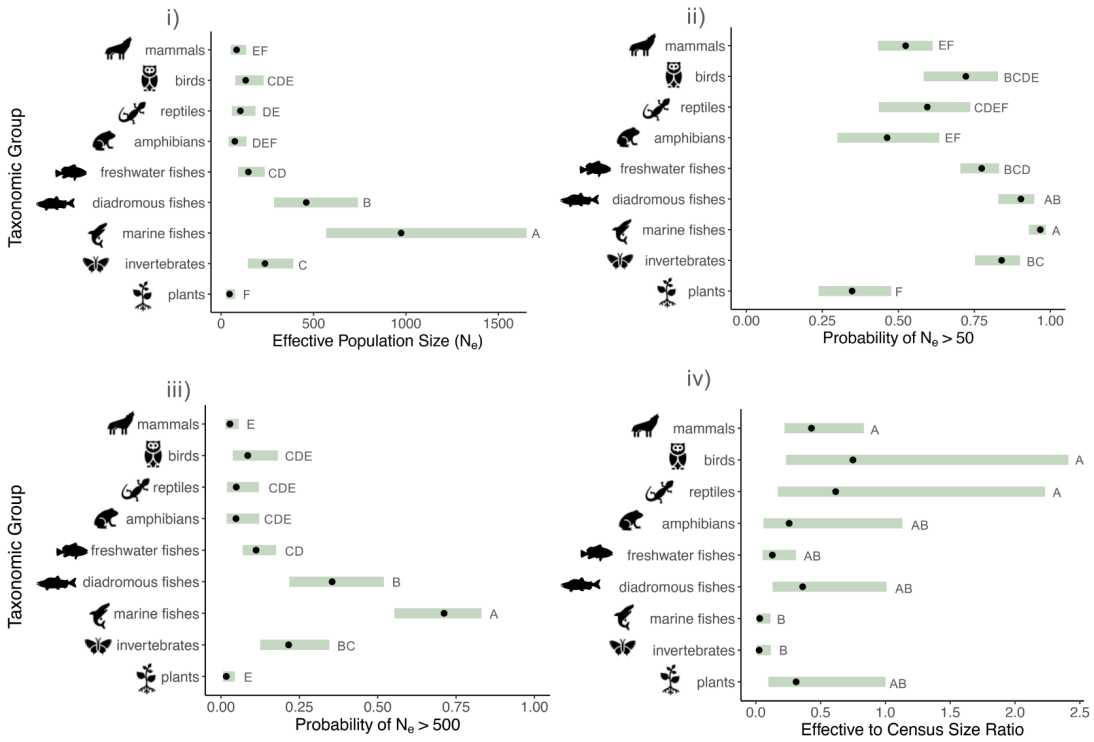


Figure 14.1: Figure 2 from Clarke et al. 2024, illustrating estimates of effective population size (i), probability of meeting the N_e>50 rule (ii), probability of meeting the N_e>500 rule (iii), and N_e/N_c ratio across taxonomic groups.

15 Taxonomy, Systematics, and Management Units

Conservation genetics encompasses the use of genetic tools to better understand the taxonomy and systematics of threatened species and populations. By some definitions, it may be the topic area that produces the single greatest number of scientific articles in the field. For example, in a recent review of conservation genetics papers aimed at describing global disparities in sequence effort and technology (Linck and Cadena (2024)), Daniel Cadena and I categorized 394 papers in leading conservation genetics journals by (crude) topic area. Of these, 28 (~7%) were in the category that explicitly included taxonomy and systematics. However, the category that included defining population genetic structure—a task at the core of identifying conservation units, and anecdotally the single most common one in conservation genetics papers—had a whopping 330 (83%) of the remaining studies. It would therefore be malpractice for me not to touch on the conceptual basis of this work in passing.

To start, some definitions are in order. Per the Convention on Biological Diversity, Taxonomy is "the science of naming, describing and classifying organisms and includes all plants, animals and microorganisms of the world". Systematics is often used interchangeably with taxonomy, but adds an evolutionary dimension: what are the *relationships* among taxonomic units?

So why does taxonomy matter for conservation? There are four key reasons:

1. Unrecognized endangered species may be allowed to go extinct;
2. Endangered species may be denied protection if considered a population of a common species;
3. Incorrectly diagnosed species may be hybridized with other species, potentially reducing fitness;
4. Populations that could be used to ameliorate inbreeding might be overlooked.

All four of these points circle around the need to define the basic units of biodiversity. Familiarity with **species concepts** is important to this goal. Species concepts are famously abundant, with and counting at the time of a review in the mid-90s (Mayden (1997)). Part of the reason for their proliferation is widespread confusion between conceptual basis and operational criteria (*definitions* versus *delimitation*). For anyone who is not a professional systematist or philospher of science, most will appear redundant or irrelevant. The box below defines six broad categories of species concepts; the **biological species concept**, **genotypic cluster species concept** and **phylogenetic species concept** are most relevant to modern conservation genetics.

Species Concept Categories

- **Biological species concept**: A species is a group of actively or potentially interbeeding individuals separated from other such groups by intrinsic or extrinsic reproductive isolation (not only geography)
- **Ecological species concept**: A species is a group of individuals that share the same ecological niche or adaptive zone.
- **Evolutionary species concept**: A species is a group of individuals with the same evolutionary role, tendencies, and historical fate.
- **Phylogenetic species concept**: A species is a group of individuals that conform to one or more phylogenetic criteria (reciprocal monophyly, coalescence of all gene trees).
- **Phenetic cluster species concept**: A species is a group of individuals that form a cluster in morphological space separated from other such groups (similar trait values for multiple features).
- **Genotypic cluster species concept**: A species is a group of individuals that form a cluster in genoptypic space separated from other such groups (similar allele frequencies at multiple loci).

For any given situation——say, a group of partially intergrading populations with uncertain limits and relationships—multiple studies will frequently arrive at conflicting conclusions about the number and identify of species, genetic demes, and / or conservation units. Methological sources of this conflict include the application of alternate species definitions, insufficient data, or varied sampling schemes. A common *biological* reason for ambiguity is that speciation is a continuum, and any two lineages diverging from a shared common ancestor may be at a different point on trajectories ending in complete reproductive isolation and valid biological species. As a result, some understanding of speciation as a process is useful for our goals.

In vertebrates, at least, speciation typically proceeds in **allopatry**, i.e. between geographically isolated populations. Rarely in vertebrates—though theoretically possible—does speciation proceeed in **sympatry** (populations with overlapping distributions) or *parapatry* (populations with non-overlapping but adjacent distributions). More important than geography *per se* are its impacts on gene flow. In its pure form—think flightless birds on different oceanic islands—allopatric speciation occurs when the migration rate between diverging subpopulations is $m = 0$. Sympatric speciation occurs when $m = 0.5$, i.e. the probability of receiving an allele from a member of the other, diverging lineage is equivalent to the probability of receiving one from a conseecific. Parapatric speciation occurs between these extremes: in other words, when $0 < m < 0.5$.

The current consensus is that most speciation events, regardless of geography or migration rate, are driven by natural selection, not drift. However, in some lineages—particularly plants———speciation can occur almost instantly with changes in ploidy (the number of chromosomes), even if those changes are not immediately adaptive. Nonetheless, the process can take a long time in many contexts, mediated by changes in gene flow, selection, and other factors that can halt or reverse divergence. It is thus reasonable to conclude that for some nontrivial number of edge cases, disagreements about what individuals consistute what population or species have no hard answer. Here, values and politics will be more useful than hard science.

The process of identifying the boundaries of species and assigning individuals to groups is known as **species delimitation**. Species delimitation requires *operational criteria* (which may or may not be independent from species concepts): How do we separate one species from another in practice? In the past, this work was delegated to taxonomic experts, who made often-subjective judgement calls about the significance of particular phenotypic characters, and assessed individuals and populations against their standards. This work gradually became more quantitative, requiring algorithims to assess phenotypic or genetic differences between populations.

The past two decades have seen the ongoing synthesis of phylogenetics and systematics, and populations and species are now routinely demarcated using methods that simultaneously infer evolutionary relationships among genetic demes using statistical models of DNA sequence evolution. This work builds off an earlier tradition of quantifying molecular divergence with statistics such as Nei's Genetic Distance (D_N):

$$D_N = -ln(\frac{\sum_{i=1}^{m}(p_{ix}p_{iy})}{\sqrt{(\sum_{i=1}^{m}p_{ix}^2)(\sum_{i=1}^{m}p_{iy}^2)}})$$

Nei's Genetic Distance has the useful property that if the substitution rate is constant per generation, D_N will increase proportionally to divergence time (not true of F_{ST}, for example). As a result, when $p_{ix} = p_{iy}$, $D_N = 0$; when no alleles shared, $D_N = \infty$. Assuming mutation rates across markers are equal, and that they cannot occur at same locus twice, D_N should faithfully summarize evolutionary divergence in a way that can be compared to benchmarks for particular taxa.

> **i** The Metapopulation Lineage Species Concept
>
> A paper that has come as close to ending species concept debates as anyone is ever likely is Kevin De Quieroz's 2007 *Systematic Biology* article entitled "Species Concepts and Species Delimitation". De Queiroz argues convincingly that nearly all evolutionary species concepts agree that a species should be considered a lineage of metapopulations, and that (as I echo above) disagreement stems from the fact that various operational criteria only apply at different time points in the speciation continuum.

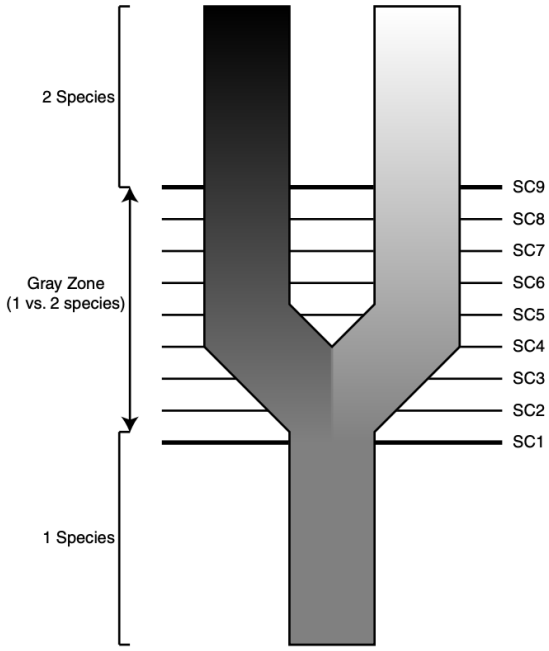


Figure 15.1: *De Quieroz demonstrating how different species concepts (indicated by the prefix SC below) will come to different conclusions in the "Gray Zone" of speciation.*

15.1 Management Units

Species delimitation is also important because conservation law requires categories to which to apply. The US Endangered Species Act permits listing of "any subspecies of fish or wildlife or plants, and any distinct population segment of any species of vertebrate fish or wildlife which interbeeds when mature". Unfortunately, the ESA does not define DPS, but FWS offers guidance: "a population segment's DPS status is determined by the *discreetness* and *significance* of the population segment compared with the species as a whole".

Grizzlies in the Greater Yellowstone Ecosystem (GYE) are a prime example of the difficulty of applying the ESA. In 2007, the US Fish & Wildlife Service (USFWS) decided bears in the region constituted a valid DPS that had sufficiently recovered from mid-20th century lows, delisting them as a result. Two years later, the US District Court for District of Montana determined USFWS had not adequately considered Whitebark Pine (*Pinus albicaulis*) decline in their decision, leading to relisting of the DPS. In 2017, USFWS declares that their research had indicated Whitebark Pine decline was *not* an existential threat to the DPS, again delisting it. The same year, the Crow Indian Tribe responded with a lawsuit to relist on the basis of

species-wide popualtion trends. This argument, though successful, muddied the waters about what was actually on the table for legal protection, as the basis for listing GYE grizzlies in the first place was their demographic (and perhaps evolutionary) independence.

DPS are a special case—in this case, a legal designation within a certain jurisdiction—of the idea that there are often important targets of conservation below the species level, which should reflect distinct population-level attributes. In an influential 1994 paper, Craig Moritz (Moritz (1994)) introduced the idea of evolutionarily significant units, more or less equivalent to intraspecific lineages and identifiable with phylogeographic methods. More generally, conservation units attempt to preserve demographic independence, locally adapted populations, and other forms of phenotypic and genetic variation (Funk et al. (2012)).

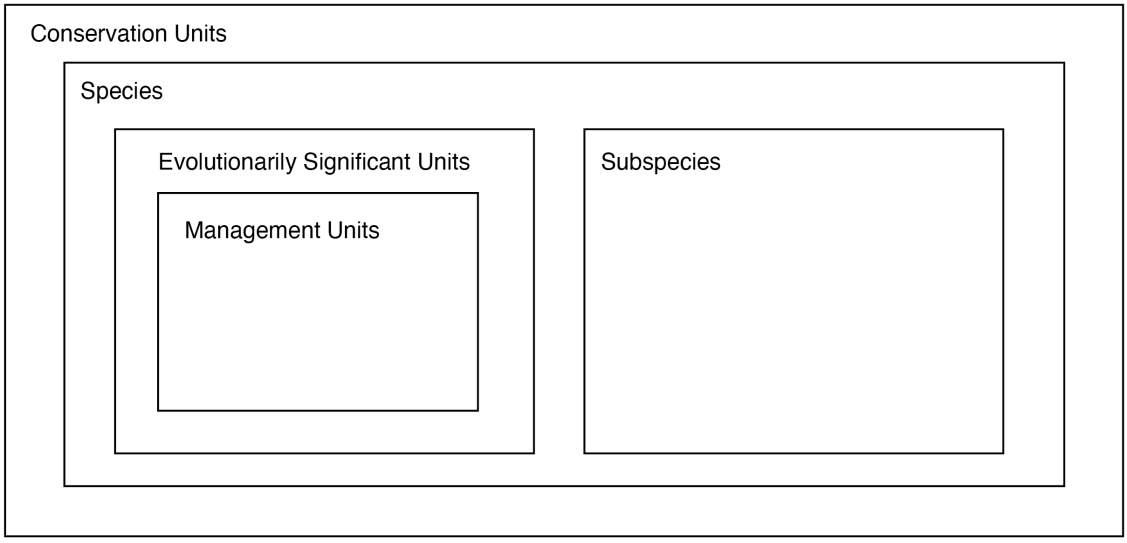


Figure 15.2: *A hierarchy of conservation units: species contain evolutionarily significant units (ESUs) and subspecies; management units are typically nested within ESUs, which may have no particular relationship to subspecies.*

i Subspecies Delimitation in California Gnatcatchers (*Polioptila californica*)

Because identification of conservation units can determine the political status—and ultimate survival—of specific populations of organisms, scientific debates over their distinctiveness or validity can become heated. One example of this can be found in a series of papers (Zink et al. (2013), McCormack and Maley (2015), Zink et al. (2016)) over the validity of two subspecies of the California Gnatcatcher. In the exchange, Bob Zink and colleagues suggest named subspecies have no validity as ESUs, pointing to phylogenetic

trees with poor statistical support and geographic incoherence. John McCormack and colleagues respond that their conclusion—which may permit development in coastal chapparal—was an artifact of using genetic markers with poor resolution.

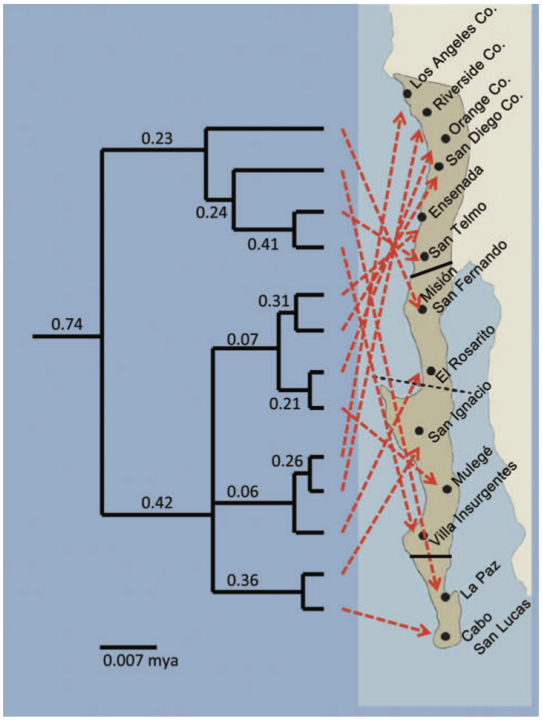


Figure 15.3: *Figure 1 from Zink et al. 2016, rehashing their earlier findings of poor statistical support for geographic subspecies, indicated by the dashed line crossing Baja California. Numbers on the phylogeny are posterior clade credibility values, none of which exceed 50%.*

15.2 Outbreeding Depression

A consequence of poor species delimitation and taxonomy is that managers may inadvertently generate **outbreeding depression** when translocating or crossing individuals from unexpectedly divergent lineages. Outbreeding depression may be caused for two primary reasons: The fact that populations may be adapted to **different environmental conditions**, and that they may have evolved **different coadapted gene complexes (positive epistasis)**. The former concern is more broadly applicable, as local adaptation is pervasive and poorly known in most wild species. But cryptic species level diversity is surprisingly widespread, particularly in non-vertebrates and subtropical and tropical lineages. Last-ditch conservation breeding

therefore runs a risk of generating hybrids, not pure descendants.

Outbreeding Depression in Slimy Sculpin

Fish are frequently the target of reintroduction programs that rely on individuals from different sources. When these sources are geographically isolated—as is often the case for freshwater species—outbreeding depression can result. In Minnesota, Huff and colleagues (Huff et al. (2011)) examined the consequences of mixing of different parental stock on fitness trates. They found pure populations were notably larger in length weight, body condition, and persistence, something they attribute to disrupted coadpated gene complexes.

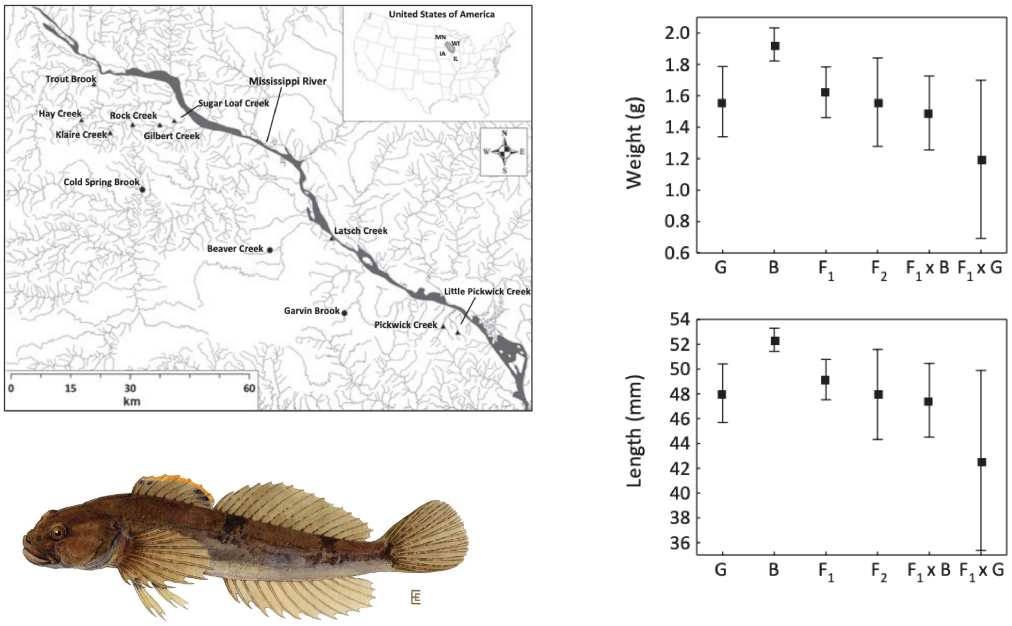


Figure 15.4: *Indicators of outbreeding depression in slimy sculpin* (Cottus cognatus). *Note weight and length decrease in later generation hybrids (indicated by abbreviations on the righthand side of boxplots).*

16 Hybridization

Hybridization is mating between "individuals from two populations, or groups of populations, which are distinguishable on the basis of one or more heritable characters" (Harrison (1990)). Studies of hybrid zones have become a major focus of evolutionary biology, as they provide a window into the speciation process before reproductive isolation is complete, permitting inference of mechanisms of divergence, premating and postmating barriers to gene flow, and the association of genes and traits. In conservation genetics, hybridization is of primarily of concern when it leads to the loss of the genetic material that made a conservation unit distinct, or when it leads to outbreeding depression. The former topic quickly becomes philosophical; the latter was discussed in the previous section. Here, we will discuss some practical consequences of hybridization on allele frequencies, and touch on the legal treatment of hybrid lineages and populations.

To do so, we will focus on models of frequencies at **diagnostic loci**: alleles that are fixed (i.e., $f(A_1) = 1$) in one population and absent in the other. In the event that individuals from these divergent populations interbreed, producing hybrids, there may be one of two outcomes, each with its own evolutionary and conservation implications. If hybrids are not themselves fertile, genetic material cannot cross back into pure parental lineages, a scenario described as **hybridization wtihout introgression**. Alternatively, if hybrids are fertile, they can interbreed among themselves and with pure parental indvidiuals, allowing genetic material to permeate population boundaries through *backcrosses* in the process of **introgressive hybridization**.

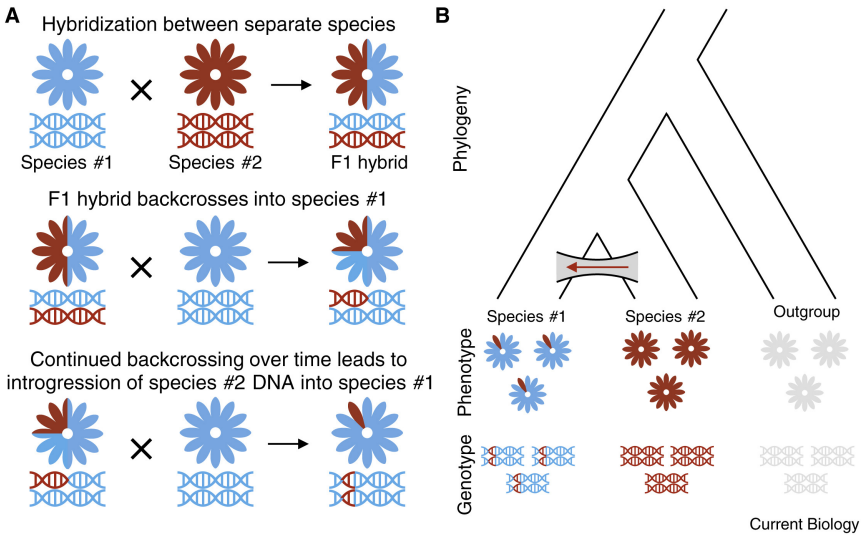
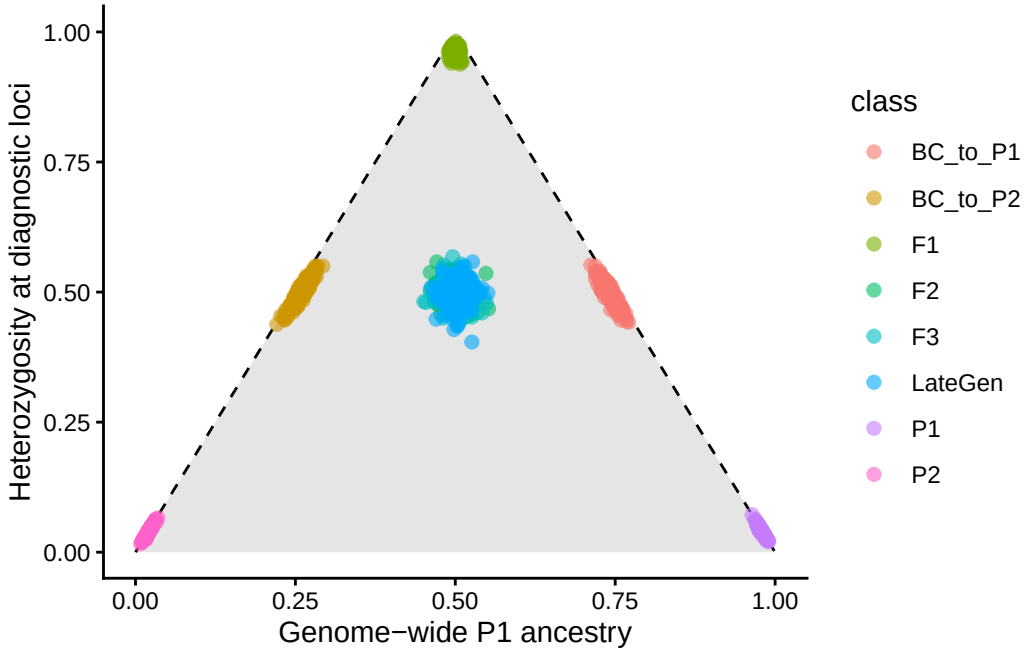


Figure 16.1: *Figure 1 from Aguillon et al. (2022), illustrating the difference between hybridization and introgressive hybridization*

First-generation hybrids between pure parental taxa are known as **F1s**. Interbreeding among hybrids then produces later generations: **F2s**, **F3s**. **late generation hybrids**. An F1 hybrid has 50% of its genome derived from each parent, with 100% of loci showing interpopulation ancestry. A later generation hybrid will still have roughly 50% of its genome derived from each parent, but will have fewer loci showing interpopulation ancestry, as hybrid x hybrid crosses will end up pairing alleles from the same parental taxon. Backcrosses will end up with greater than 50% of their genome from a single parental taxon, and an intermediate number of loci showing interpopulation ancestry (often referred to as their **admixture coefficient**). Individuals can be ploted on axes of diagnostic locus heterozygosity and interpopulation ancestry on ternary (triangle) plots, such as the one below (generated from a simple numberical simulation):



Crossing F1 hybrids produces a range of possible outcomes. For example, in an F2 hybrid, the probability an individual is heterozygous at 5 diagnostic loci is:

$$p(\text{H at 5 loci}) = 0.5^5 = 0.03125$$

This is because an F2 individual comes from an F1 x F1 cross. At diagnostic loci, F1 individuals are by definition hetoerozygous. An A_1A_2 x A_1A_2 cross will produce 2 heterozygous genotypes out of four possibilities (A_1A_2 and A_2A_1).

An F1 hybrid backcrossing with one parental taxon will lose 50% of the remainder of its ancestry derived from the other parent each generation. For example, an F1 hybrid formed by an intial cross between species A and species B followed by 5 generations of backcrossing into species A will end up with <2% of its genome derived from species B ($0.5^6 = 0.015625$).

Hybridization between genetically differentiated populations can cause linkage disequlibirum. Consider the following population formed by 5 individuals each of two parental taxa that differ at two diagnostic loci. Here, treating all individuals as a single population gives us the maximum possible value of D:

- **P**: AB, AB, AB, AB, AB, ab, ab, ab, ab, ab
- $f(A) = 0.5; f(a) = 0.5; f(B) = 0.5; f(b) = 0.5$
- $f(AB) = 0.5; f(ab) = 0.5$

- $D = p(A, B) - p(A)p(B) = 0.25$

A mix of F1 hybrids and pure parentals would continue to show strong linkage disequilibrium:

- **F1**: AB, ab, Ab, aB, AB, AB, ab, ab, aB, Ab
- $f(A) = 0.5; f(a) = 0.5; f(B) = 0.5; f(b) = 0.5$
- $f(AB) = 0.3; f(ab) = 0.3$
- $D = p(A, B) - p(A)p(B) = 0.1$

However, this decays with time:

- **F2**: aB, Ab, AB, ab, aB, Ab, AB, ab, Ab, aB
- $f(A) = 0.5; f(a) = 0.5; f(B) = 0.5; f(b) = 0.5$
- $f(AB) = 0.2; f(ab) = 0.2$
- $D = p(A, B) - p(A)p(B) = -0.05$

The consequences of hybridization on the intregrity of parental species is heavily influenced by the environmental and geographic context in which it occurs. If there are no barriers to gene flow—e.g., gradients of natural selection or obstacles to dispersal—hybridization can spread rampantly, creating a **hybrid swarm** where most individuals have some hybrid ancestry. If hybridization is instead restrained by the balance between natural selection and gene flow, it may create a **tension zone**, with genotypes and phenotypes showing clinal variation across a linear transect. A third possibility is that hybrids may have superior fitness to either parental taxon in a small region of space, something known as a **bounded hybrid superiority model**.

We previously discussed how hybridization (either natural or as an inadvertent consequence of well-intended management actions) may lead to outbreeding depression. Two other conservation concerns merit mention. First, hybridization at high enough rates can reduce population growth as reproductive output is put towards the "dead end" of hybrid individuals (**demographic swamping**). Second, introgressive hybridization without restraint can eliminate "pure" parental genomes, leading to **genomic extinction**. Whether this is something to be "passionately resisted" (to quote Gary Snyder) depends in part on species concepts, in part on management goals, and in part on values. Hybrid individuals may have (more or less) the same ecological role and phenotypic traits of their parent taxa; preserving "pure" genotypes may facilitate future adaptive change, or may be an end in its own right.

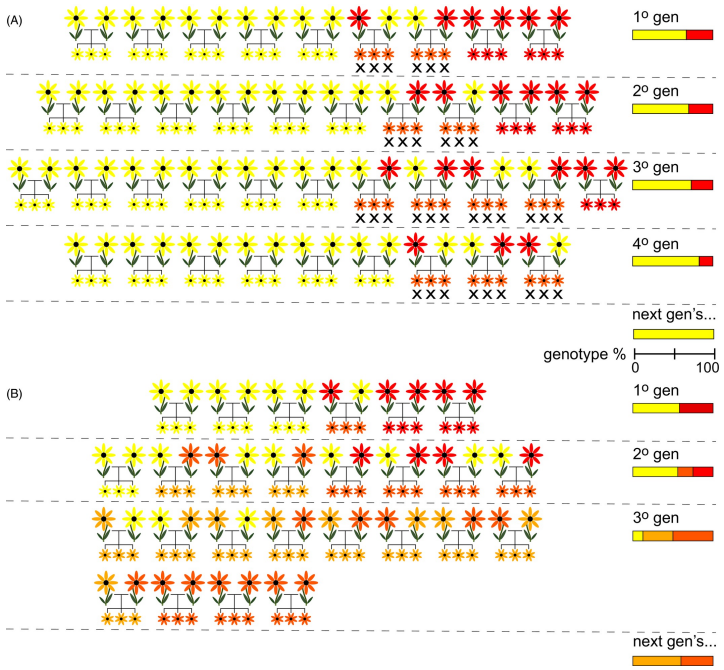


Figure 16.2: *Figure 1 from Todesco et al. (2016), illustrating demographic swamping and genomic extinction*

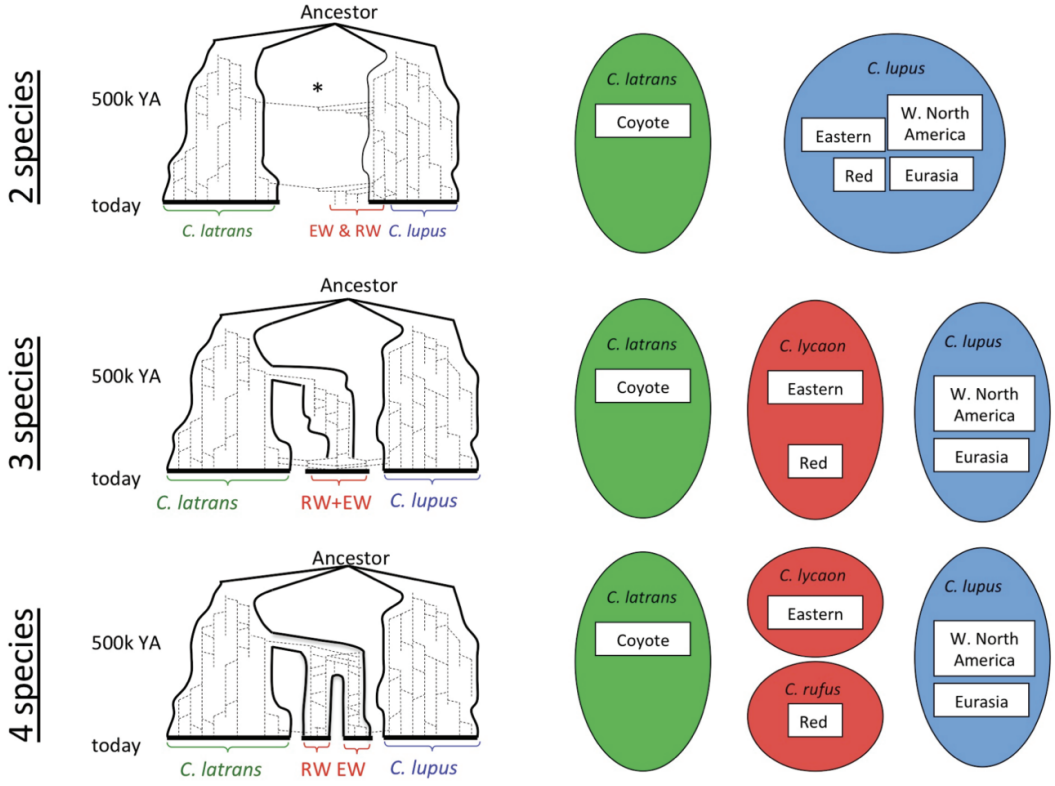
Unsurprisingly, hybrids pose problems for conservation policy. Different laws afford hybrids different rights under biodiversity protection policies. Under **Australia's Biodiversity Conservation Act (2016)**, hybrids are eligible for conservation listing if: 1) they are distinct entities, i.e. the progeny are consistent within the agreed taxonomic limits for that species group; 2) they are capable of self-perpetuating, and are not reliant on parent stock for replacement; 3) they are the product of a natural event. One implication of this policy is that hybrid-derived taxa—such as a stable hybrid swarm somewhere parental lineages have ceased to exist—**are eligible**, but hybrid zones are **not**.

Under the **US Endangered Species Act**, the legal status of hybrids is contentious and as of yet unresolved, with a long history. As early as May 1977—four years after the passage of the act—the U.S. Department of the Interior (DOI) stated that "because it defines 'fish or wildlife' to include any offspring without limitation, the Act's plain meaning dictates coverage of hybrids of listed animal species" (and also applies to plants). A few months later, the US Fish and Wildlife Service (USFWS) responded that "…since the Act was clearly passed to benefit endangered species, …it must have meant the offspring of two listed species and was not meant to protect a hybrid where that protection would in fact cause jeopardy to the continued existence of a species". One month later, the DOI concurs, claiming that "hybrids of listed species are not protected under the ESA" (because of potential for hybridization to cause harm). This "Hybrid Policy" held until December 1990, when it was summarily suspended by the USFWS for failing to protect taxa derived from hybridization (like the red wolf; see below),

or subspecies that interbreed (like Florida panther). Six years later, the USFWS & National Oceanic and Atmospheric Administration propose an "Intercross Policy" aimed at resolving remaining ambiguities. To date, it remains unpublished.

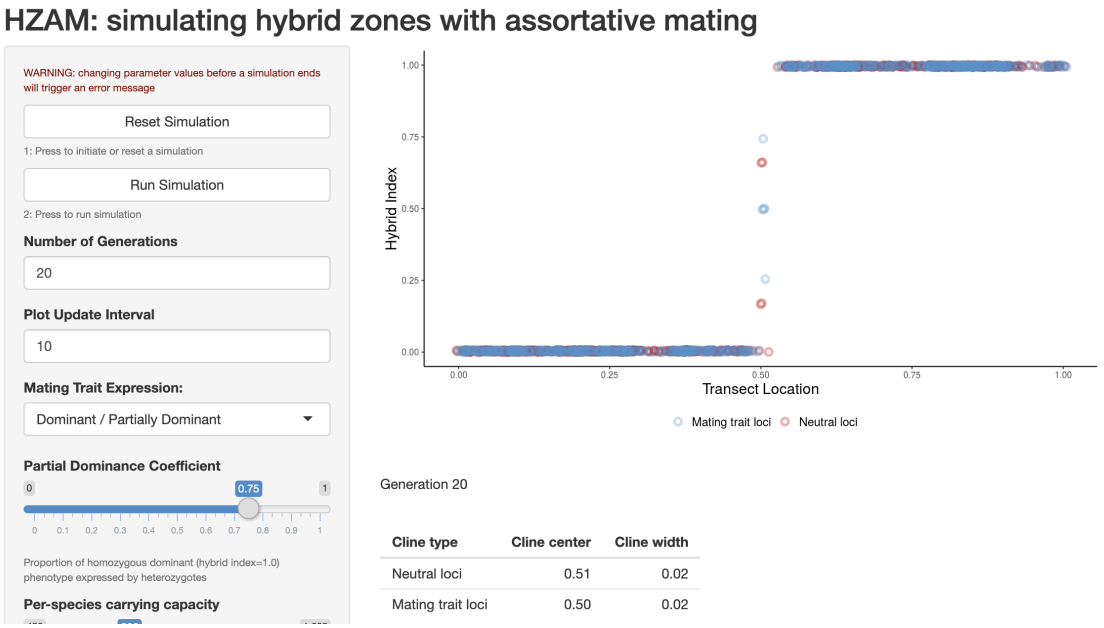
Are Red Wolves (*Canis rufus*) a DPS?

Red wolves (*Canis rufus*) are a critically endangered, taxonomically ambiguous population of canids in the southeastern US. Now restricted to 30 or so individuals at Alligator River National Wildlife Refuge (descended from a now-extinct population on the (Gulf Coast), conservation breeding efforts have been mandated by the taxon's status as a US Endangered Species. Yet recent and ongoing hybridization with coyotes and its uncertain evolutionary relationships has led to years of debate over whether *C. rufus* was better treated as an isolated population of Gray Wolves, a divergent population of Coyotoes with wolflike traits, the product of historic hybridization between Gray Wolves and Coyotoes, or the distinct species it was long assumed to be. Robin Waples and colleagues (Waples et al. (2018)) reviewed the literature on Red Wolf taxonomy, finding evidence for three primary taxonomic hypotheses, illustrated below. They argue that regardless of which is ultimately best supported by continuing research, *C. rufus* deserves treatment as a Distinct Population Segment (DPS; see previous section).



Shiny App: HZAM

A Shiny app I wrote adapting a program written by Darren Irwin (Irwin (2020)) called HZAM allows for simulations of hybrid zones under various degrees of assortative mating. It can be used to develop intuition about the behavior of tension zones under different levels of selection and gene flow.



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